

Opcenter Meta Link – User Manual

User Manual

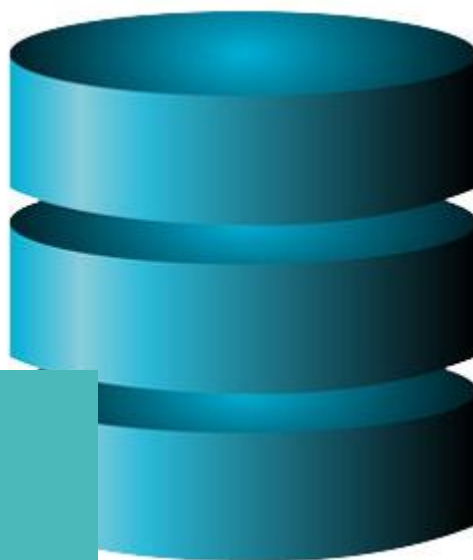


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1 Opcenter Meta Link

1.1 Introduction

Opcenter Meta Link is an application that reads the metadata from an MDB file and assigns referential integrity relationships for the Data Objects/Database Tables within the file. Opcenter Meta Link outputs its result in various outcomes like MSSQL data definition language, Oracle data definition language, Microsoft Access database, CSV file and Hierarchy view.

This guide will demonstrate the necessary steps and procedures to utilize Opcenter Meta Link.

1.2 Prerequisites

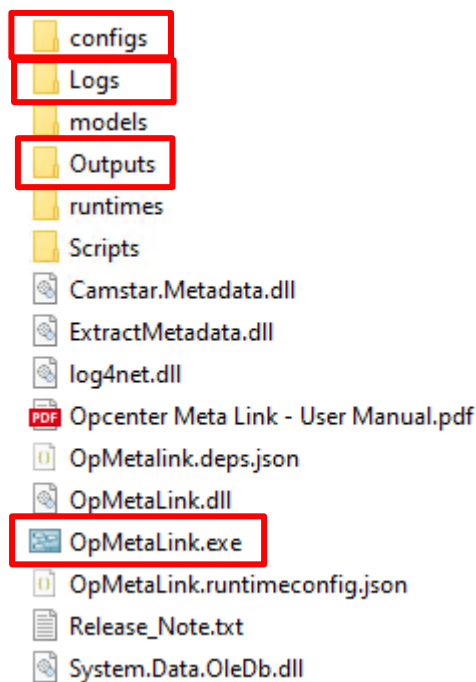
- Opcenter Meta Link application
- InSite.mdb
- SiteInfo.mdb

1.3 Disclaimers

- Outputs generated by Opcenter Meta Link are only for Entity Relationship Diagram (ERD) usage only.

1.4 How To Use

1.4.1 Folder Structure

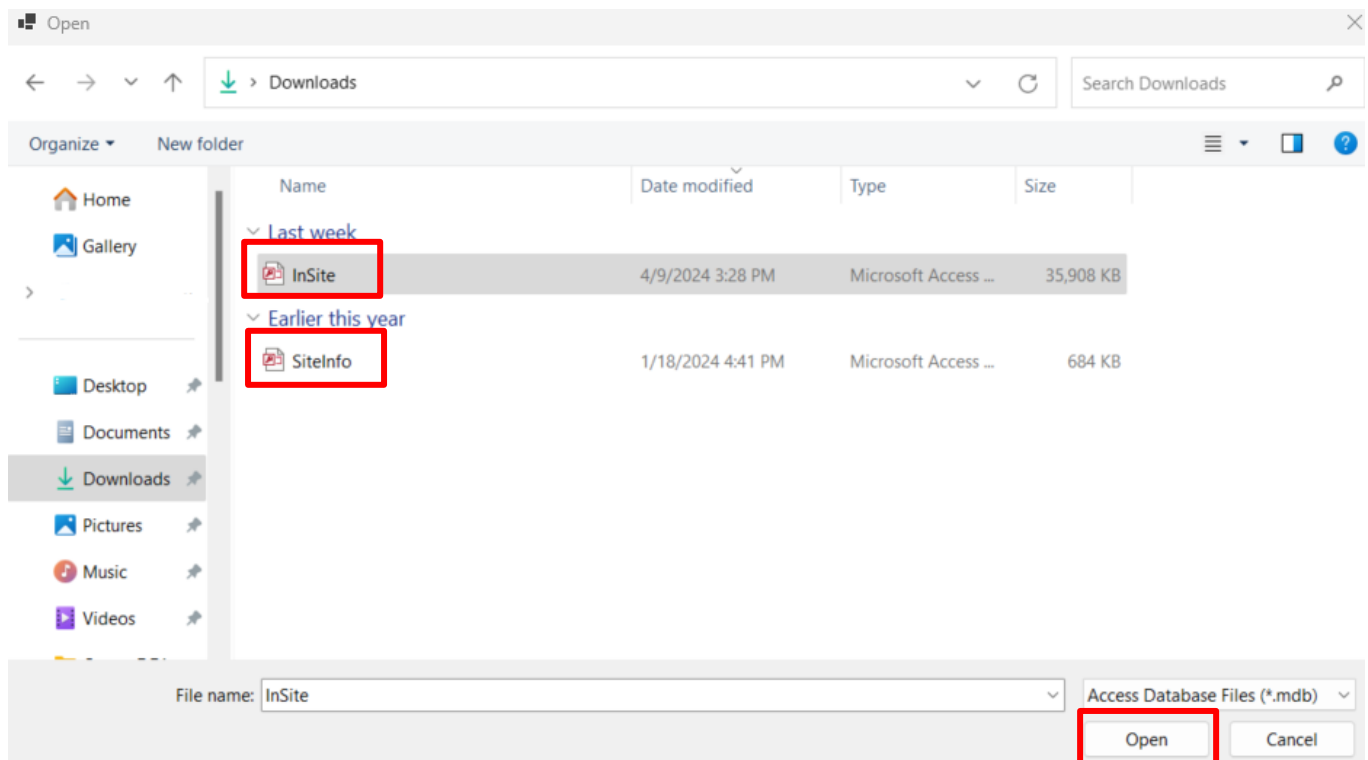
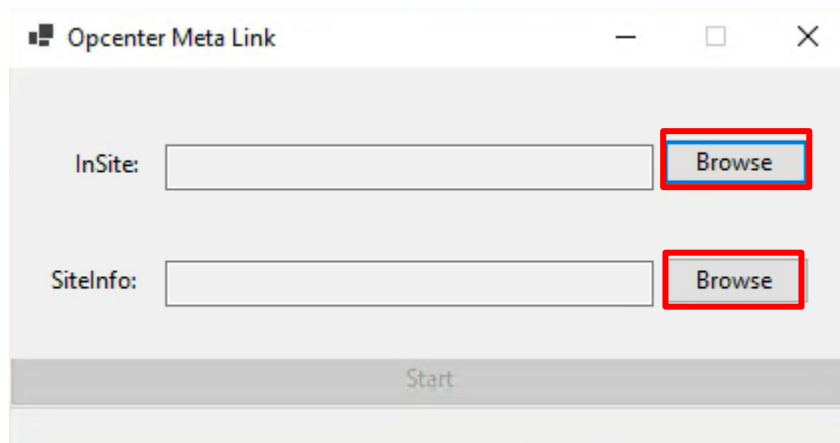


There are a few items to acknowledge.

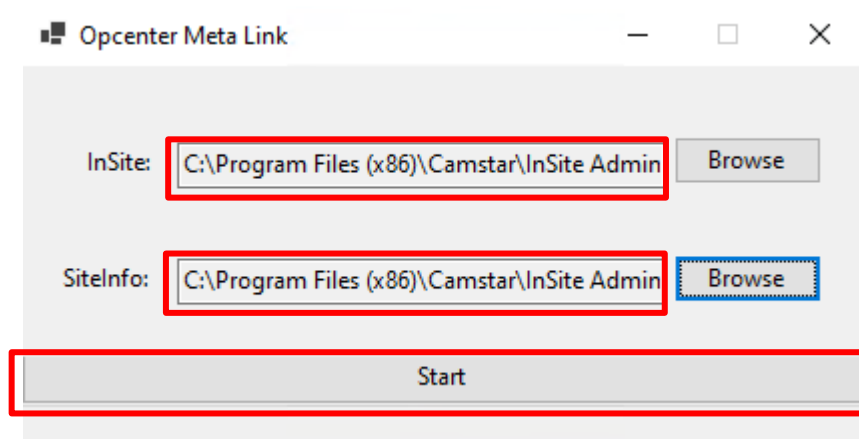
- “**Logs**” folder will consist of all transaction logs created by the application.
- “**Outputs**” folder will consist of all outcomes produced by the application.
- “**Configs**” folder will consist of all the application’s configurations.
- “**OpMetaLink**” is the application itself.

1.4.2 Load MDB File

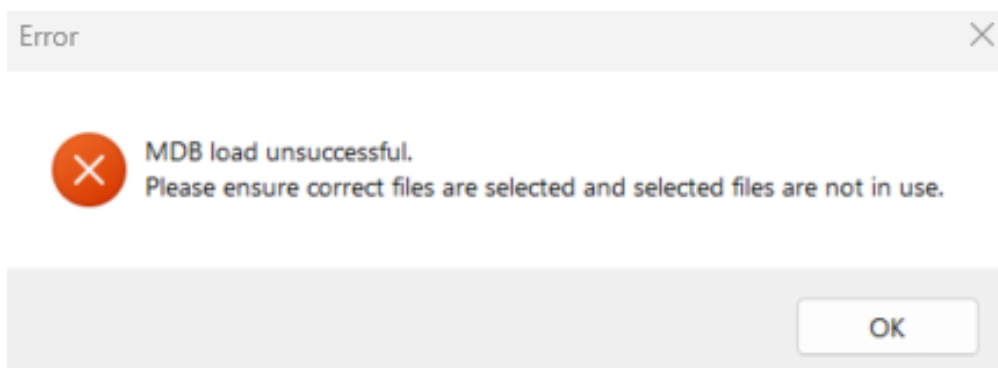
When the application is started up, the following will appear. An InSite.mdb file and a SiteInfo.mdb file is needed before the application can continue. Click on the “Browse” buttons to select the MDB files from our local environment.



Once both files have been selected, their file paths will be displayed on screen and the “Start” button will be enabled. Click on the “Start” button to proceed to the next screen.



If both file selections are correct, the next screen (Generation Page) will load successfully. Otherwise, an error will be prompted.



1.4.3 Generation Page

A: This part shows the **application name** and the **file path** of the InSite.mdb in used.

B: There are two relation options.

- When **Data Object** is selected, the “Checkbox List” in Box C will be populated with all the Configurable Data Objects retrieved from the MDB. The “Hierarchy” and “Purge” options in Box D will be disabled.
- When **Database** is selected, the “Checkbox List” in Box C will be populated with all the Database Tables retrieved from the MDB and the “Include Non-Persistent Fields” option in Box D will be disabled.

C: This is the item selection section.

- The “**Search Box**” allows for finding “Checkbox List” items based on their name.
- The “**Checkbox List**” provides selection of Configurable Data Objects/Database Tables to generate their referential integrity relationships.
- There is also a **label** below that indicates the number of items checked within “Checkbox List”.

D: This is the script settings section.

- The “**Script Type**” option indicates what type of output is generated (Sql Data Definition Language, Oracle Data Definition Language, Microsoft Access Database, CSV File and Hierarchy View). Take note that the “Hierarchy” option is only available when Database is selected in Box B and only 1 item is selected in the Checkbox list.
- The “**Include Non-PK/Non-FK Columns**” option indicates whether the output generated will either have only Primary Key and Foreign Keys columns (No selected) or all columns included (Yes selected).
- The “**Include Non-Persistent Fields**” option indicates whether the output generated will either consist of Persistent fields (No selected) or all fields included (Yes selected). If a field's CDO does not have its Persist to Table field defined, this will consider as non-persistent fields. Take note that this option is only available when the Data Object is selected in Box B.
- The “**Configuration File**” option indicates whether the output generated will be affected by the configuration file (Yes selected) or not affected at all (No selected). When Yes is selected, the “Save Configuration” button in Box E will appear while the “Configuration” tab-page in Box F will also appear. When No is selected, vice versa.
- The “**Table Level**” option indicates how far the output generation goes. If “1” is set, only the neighboring tables will be generated. If “2” is set, the neighbors’ neighboring tables will also be generated. This continues as the table level increases. The default table level is “0” which means all table levels are generated and the maximum input is “10”.

- The **“Auto Detect Relationship By”** indicates that Opcenter Meta Link will assist in finding relationships for the selected Data Objects/Database Tables. The options provided are FK, PK and Purge. When either FK, PK or Purge is selected, the “Table Level” setting in Box D appears. When “No” is selected, the “Table Level” setting disappears.
 - **“FK”: also known as Foreign Key.** Opcenter Meta Link will find relationships based on the selected Data Objects’/Database Tables’ foreign keys. The picture below shows the foreign keys for MfgOrder which will be used to find other Data Objects/Database Tables.

MfgOrder	
	WIPMsgDefMgrId
	ChangeStatusId
	UOMId
	UOM2Id
🔑	MfgOrderId
	ERPBOId
	OrderTypeId
	PriorityId
	ReportingFactoryId
	OrderStatusId
	PostProductionProcedureId
	PreProductionProcedureId
	ContainerNumberingRuleId
	BeginProductBaseId
	BeginProductId
	ProductBaseId
	ProductId
	ERPRouteBaseId
	ERPRouteId
	BillOfProcessBaseId
	BillOfProcessId

- **“PK”: also known as Primary Key.** Opcenter Meta Link will find relationships based on which Data Objects/Database Tables are referring to the primary key of the selected Data Objects/Database Tables. The picture below shows the primary key for MfgOrder which will be used to find other Data Objects/Database Tables.

MfgOrder	
	WIPMsgDefMgrId
	ChangeStatusId
	UOMId
	UOM2Id
🔑	MfgOrderId
	ERPBOId
	OrderTypeId
	PriorityId

- **“Purge”:** Opcenter Meta Link will find relationships based on which Database Tables will be affected if the selected Database Table is purged. Take note that the “Purge” option is only available when Database is selected in Box B and only 1 item is selected in the Checkbox list.

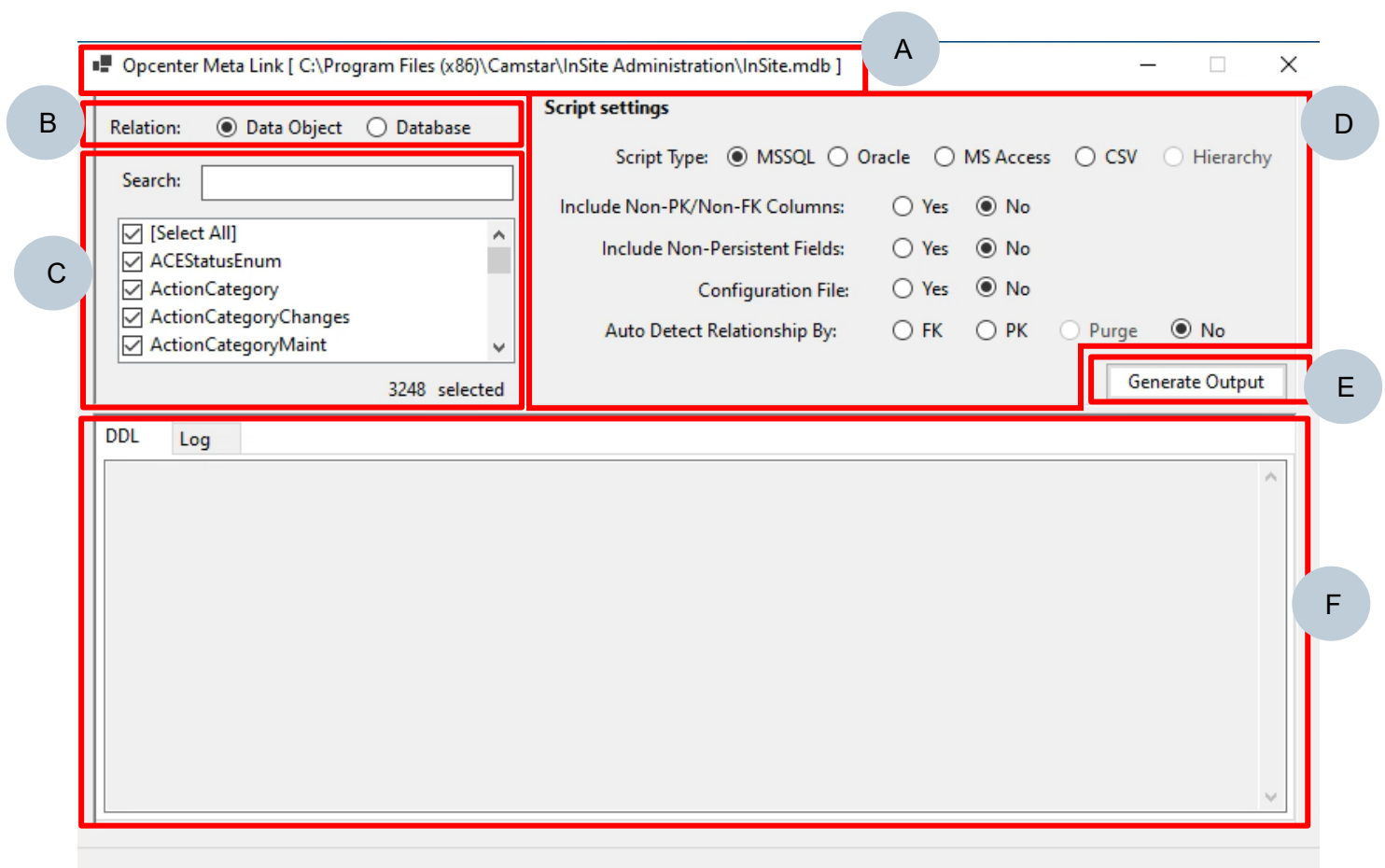
E: There are two buttons.

- The **“Save Configuration”** button saves changes made to the “Configuration” tab-page in Box F.
- The **“Generate”** button generates the output.

F: There are three tab-pages.

- The **“DDL”** tab will be populated by the output. This is only for Sql Data Definition Language and Oracle Data Definition Language. Microsoft Access Database and CSV file outputs will not display here.
- The **“Log”** tab will be populated by the transaction logs for each session.
- The **“Configuration”** tab will show xml formatted texts. For more information about this, refer to the next section.

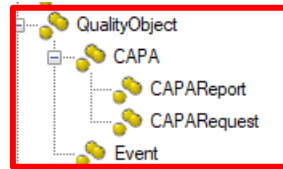
Here's the Generation Page.



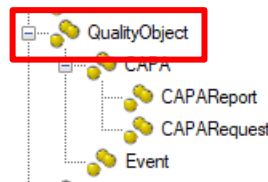
1.4.4 Configuration Format

A: These configurations apply to both Data Object and Database relations.

- `<ExcludeSuperClass> ..... </ExcludeSuperClass>`. This segment is used to exclude a Data Object/Database Table together with all its child from the output generation. For example, we added QualityObject in this segment. Hence, QualityObject, CAPA, CAPAReport, CAPARequest and Event will all be excluded from output generation.



- `< ExcludeSingleClass > ..... </ ExcludeSingleClass >`. This segment is used to exclude a singular Data Object/Database Table from the output generation. For example, we added QualityObject in this segment. Hence, only QualityObject will be excluded from output generation.



B: This configuration only applies to Database relation with “Purge” option selected at “Auto Detect Relationship By” settings. This configuration is used to handle special tables in which their relationships cannot be detected by Opcenter Meta Link.

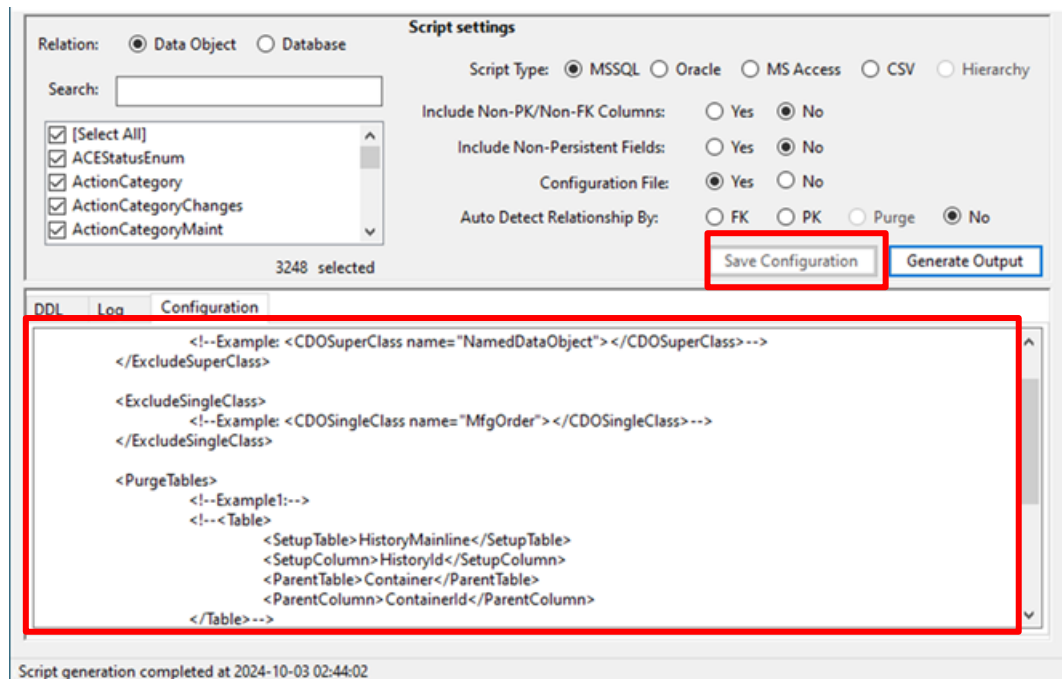
- `<PurgeTables> ..... </PurgeTables>`. This segment is used to include Database Tables into the output generation.
 - For the first example, we want to include HistoryMainline table into the output generation of Container table. Hence, our Setup Table will be HistoryMainline, Setup Column will be HistoryId, Parent Table will be Container and Parent Column will be ContainerId. The Setup Column and Parent Column are the columns used to link both tables together.
 - For the second example, we want to include CurrentStatus table into the output generation of Container table. Hence, our Setup Table will be CurrentStatus, Setup Column will be CurrentStatusId, Parent Table will be Container and Parent Column will be CurrentStatusId. Take note that there is an extra configuration here which is Parent Column Is Primary Key set to false. This extra configuration is needed when the Parent Column is not the primary key of the Parent Table.

Here's the Configuration Format. Highlighted are examples on how to add Data Objects/Database Tables into each segment. The name must match its Data Object/Database Table name (Case-Insensitive) else it will not work.

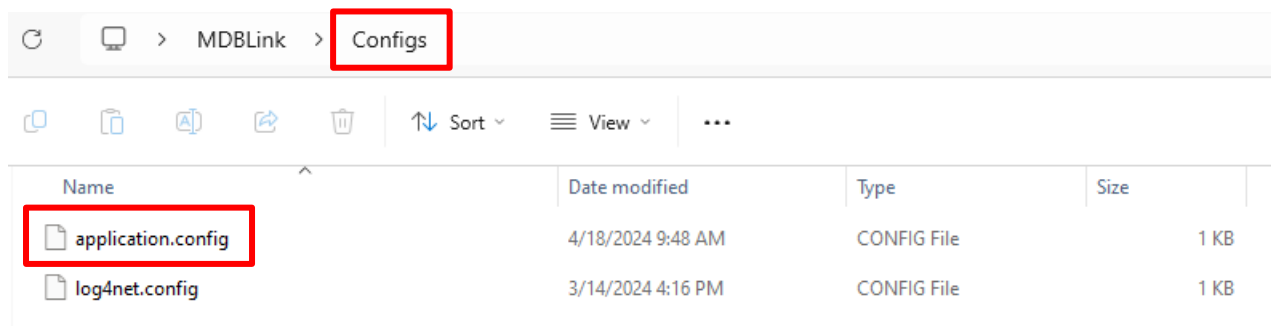
```
<?xml version="1.0" encoding="utf-8" ?>
<configuration>
  <ExcludeSuperClass>
    <!--Example: <CDOSuperClass name="QualityObject"></CDOSuperClass>-->
  </ExcludeSuperClass>
  <ExcludeSingleClass>
    <!--Example: <CDOSingleClass name=" QualityObject "></CDOSingleClass>-->
  </ExcludeSingleClass>
  <PurgeTables>
    <!--Example1:-->
    <!--<Table>
      <SetupTable>HistoryMainline</SetupTable>
      <SetupColumn>HistoryId</SetupColumn>
      <ParentTable>Container</ParentTable>
      <ParentColumn>ContainerId</ParentColumn>
    </Table>-->
    <!--Example2:-->
    <!--<Table>
      <SetupTable>CurrentStatus</SetupTable>
      <SetupColumn>CurrentStatusId</SetupColumn>
      <ParentTable>Container</ParentTable>
      <ParentColumn>CurrentStatusId</ParentColumn>
      <ParentColumnIsPrimaryKey>false</ParentColumnIsPrimaryKey>
    </Table>-->
  </PurgeTables>
</configuration>
```

Users will have two ways of editing the Configuration.

- (Recommended) The first way is through the Generation Page. There is a “Configuration” tab that allow users to edit its content and a “Save Configuration” button to save its changes. This is the recommended way as xml validation will occur during the save process to reduce human errors.



- (Advanced) The second way is through the Configuration file (application.config). This file can be found at the “Configs” folder. Users can directly make changes here if they are familiar with xml format.



1.4.5 Generate Output Based on Data Object

Step 1:

Select Data Object in the “Relation” option.

Relation: ☒ Data Object ☐ Database

Step 2:

Select the Data Objects from the “Checkbox List” that require their relationships generated. Deselect all checkboxes by clicking on the [Select All] and use the “Search Box” to find Container then select it.

Search:

- ☐ ConciergeTypeEnum
- ☐ ConditionTypeEnum
- ☐ Constants
- ☐ ContactCustomerDetail
- ☒ Container

1 selected

Step 3:

Select the script settings based on user’s needs. For this example, we are generating a Sql Data Definition Language script with only Primary Key and Foreign Keys columns. We also only include Persistent fields. We activate the “Configuration File” by clicking “Yes” and select the “FK” option for “Auto Detect Relationship By” setting. After clicking on the “FK” option, a “Table Level” setting should appear. We shall leave it as it is and move on.

Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level:

Step 4:

After clicking on Yes for the “Configuration File” option, a “Save Configuration” button and “Configuration” tab-page will appear. The “Save Configuration” button is currently disabled as no changes has been made to the “Configuration” tab. Click on the “Configuration” tab-page to edit it.

Relation: ☒ Data Object ☐ Database

Search:

- ☐ ConciergeTypeEnum
- ☐ ConditionTypeEnum
- ☐ Constants
- ☐ ContactCustomerDetail
- ☒ Container

1 selected

Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level:

DDL Log **Configuration**

Step 5:

Add the following xml texts to the “Configuration” tab then the “Save Configuration” button will be enabled. Click on it. Do take note that the “Configuration” tab has a xml validation during save, so incorrect xml format can cause an error to prompt.

☒ Container 1 selected Table Level:

DDL Log **Configuration**

```
<?xml version="1.0" encoding="utf-8" ?>
<configuration>
  <ExcludeSuperClass>
    <!--Example: <CDOSuperClass name="NamedDataObject"> </CDOSuperClass>-->
    <CDOSuperClass name="NamedDataObject"> </CDOSuperClass>
    <CDOSuperClass name="RevisionedObject"> </CDOSuperClass>
    <CDOSuperClass name="RevisionBase"> </CDOSuperClass>
  </ExcludeSuperClass>
  <ExcludeSingleClass>
    <!--Example: <CDOSingleClass name="MfgOrder"> </CDOSingleClass>-->
  </ExcludeSingleClass>
  <PurgeTables>
    <!--Example1:-->
  </PurgeTables>
</configuration>
```

The following text will appear if the save was successful.

Relation: ☒ Data Object ☐ Database

Search:

☐ ConnectAdapterParams
☐ ConnectAdapterParamsChanges
☐ Constants
☐ ContactCustomerDetail
☒ Container

1 selected

Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level: 0

DDL Log Configuration

```
<?xml version="1.0" encoding="utf-8" ?>
<configuration>
  <ExcludeSuperClass>
    <!--Example: <CDOSuperClass name="NamedDataObject"> </CDOSuperClass> -->
  </ExcludeSuperClass>
  <CDOSuperClass name="NamedDataObject"> </CDOSuperClass>
  <CDOSuperClass name="RevisionedObject"> </CDOSuperClass>
  <CDOSuperClass name="RevisionBase"> </CDOSuperClass>
  </ExcludeSuperClass>
  <ExcludeSingleClass>
    <!--Example: <CDOSingleClass name="MfgOrder"> </CDOSingleClass> -->
  </ExcludeSingleClass>
  <PurgeTables>
    <!--Example1!-->
  </PurgeTables>
</configuration>
```

Configuration Saved at 2024-10-03 03:57:56

Step 6:

Time to generate the output. Click on the "Generate Output" button. During output generation, the UI will be disabled and a loading bar will appear.

Relation: ☒ Data Object ☐ Database

Search:

☐ ConciergeTypeEnum
☐ ConditionTypeEnum
☐ Constants
☐ ContactCustomerDetail
☒ Container

1 selected

Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level: 0

Once generation is complete, the following text will appear.

Relation: ☒ Data Object ☐ Database

Search:

☐ ConnectAdapterParams
☐ ConnectAdapterParamsChanges
☐ Constants
☐ ConnectCustomerDetail
☒ Container

1 selected

Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

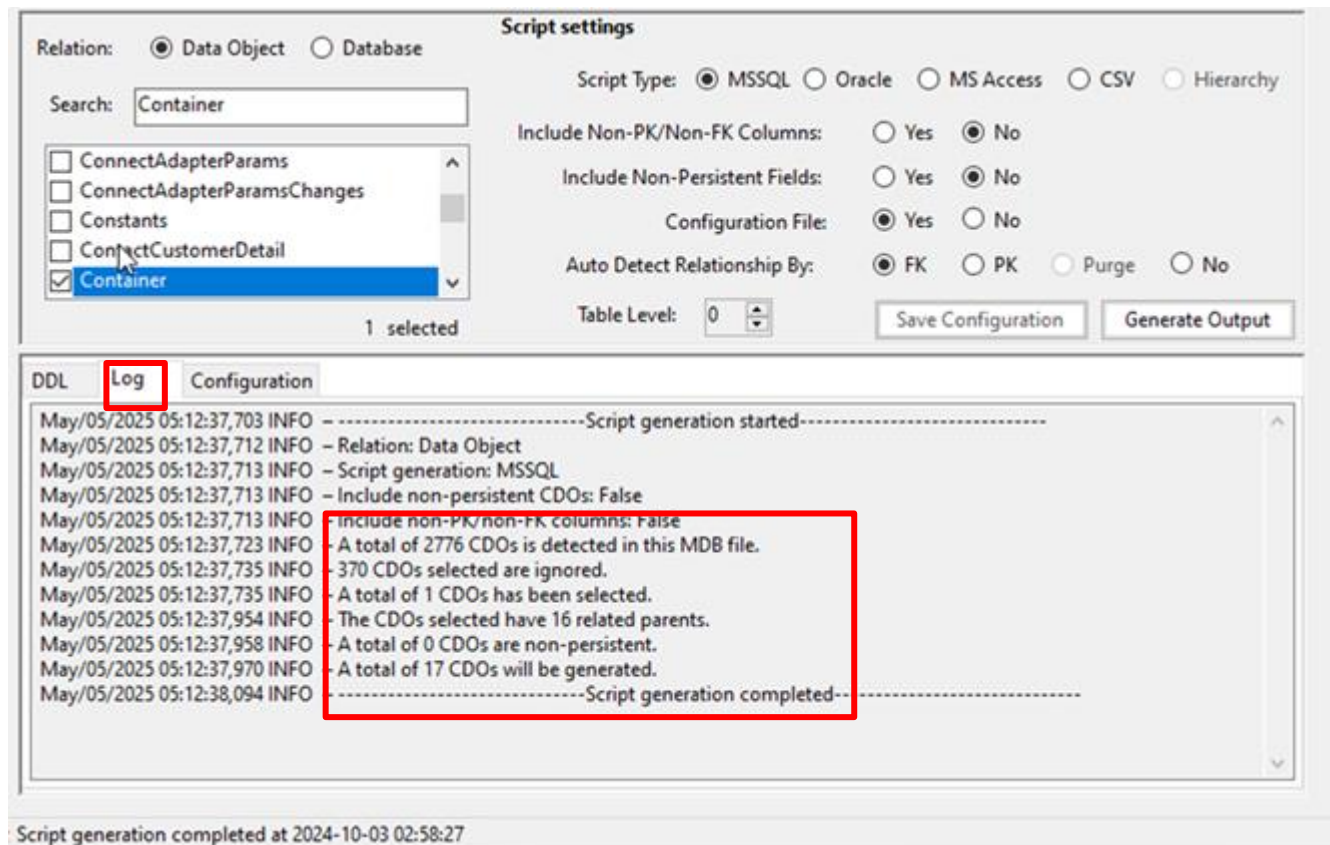
Table Level:

DDL Log Configuration

May/05/2025 05:12:37,703 INFO - -----Script generation started-----
May/05/2025 05:12:37,712 INFO - Relation: Data Object
May/05/2025 05:12:37,713 INFO - Script generation: MSSQL
May/05/2025 05:12:37,713 INFO - Include non-persistent CDOs: False
May/05/2025 05:12:37,713 INFO - Include non-PK/non-FK columns: False
May/05/2025 05:12:37,723 INFO - A total of 2776 CDOs is detected in this MDB file.
May/05/2025 05:12:37,735 INFO - 370 CDOs selected are ignored.
May/05/2025 05:12:37,735 INFO - A total of 1 CDOs has been selected.
May/05/2025 05:12:37,954 INFO - The CDOs selected have 16 related parents.
May/05/2025 05:12:37,958 INFO - A total of 0 CDOs are non-persistent.
May/05/2025 05:12:37,970 INFO - A total of 17 CDOs will be generated.
May/05/2025 05:12:38,094 INFO - -----Script generation completed-----

Script generation completed at 2024-10-03 02:58:27

The “Log” tab will be populated with the transaction logs while the “DDL” tab will be populated with the result.



Script settings

Relation: ☒ Data Object ☐ Database

Search:

☐ ConnectAdapterParams
☐ ConnectAdapterParamsChanges
☐ Constants
☐ ContactCustomerDetail
☒ Container

1 selected

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

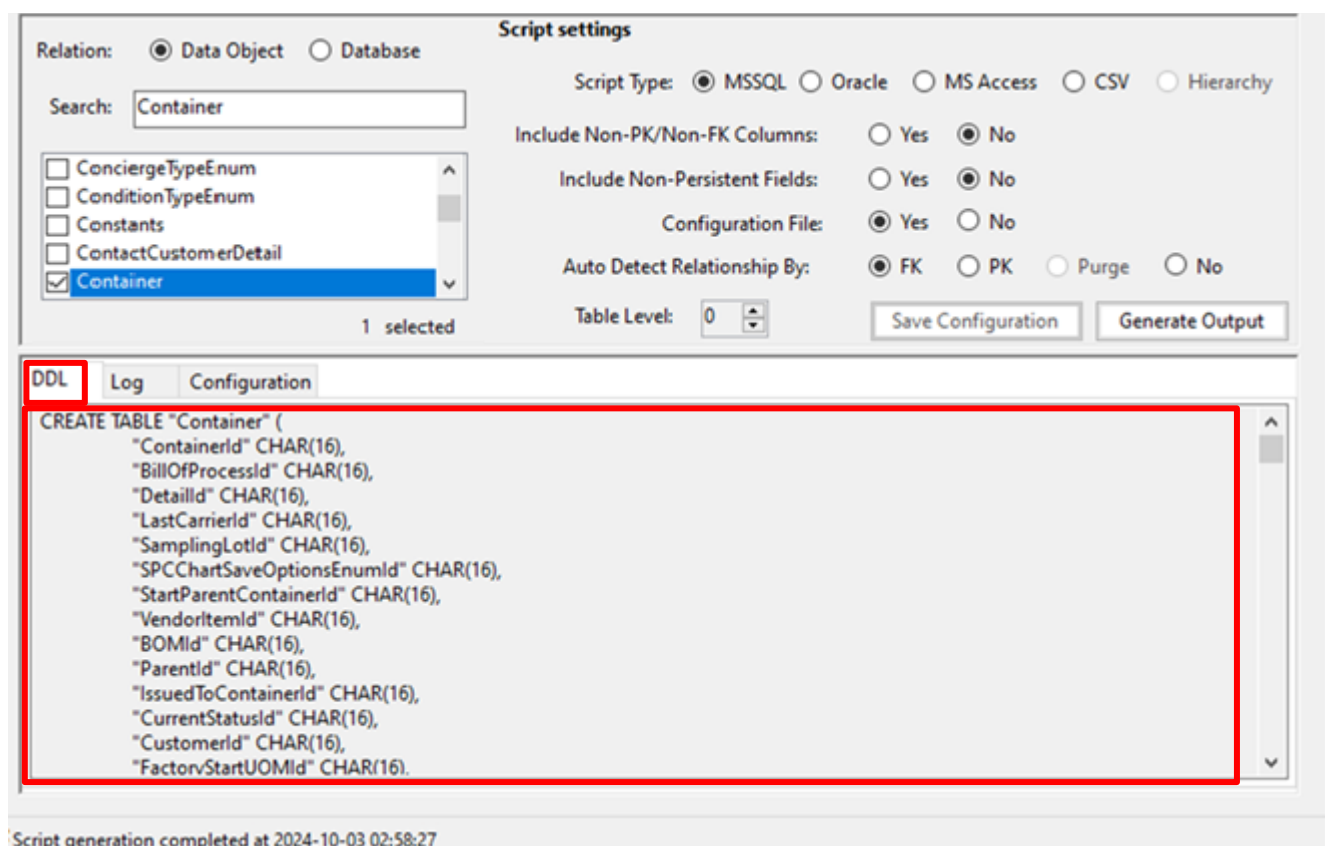
Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level: 0

DDL **Log** Configuration

May/05/2025 05:12:37,703 INFO -----Script generation started-----
May/05/2025 05:12:37,712 INFO - Relation: Data Object
May/05/2025 05:12:37,713 INFO - Script generation: MSSQL
May/05/2025 05:12:37,713 INFO - Include non-persistent CDOs: False
May/05/2025 05:12:37,713 INFO - Include non-PK/non-FK columns: False
May/05/2025 05:12:37,723 INFO - A total of 2776 CDOs is detected in this MDB file.
May/05/2025 05:12:37,735 INFO - 370 CDOs selected are ignored.
May/05/2025 05:12:37,735 INFO - A total of 1 CDOs has been selected.
May/05/2025 05:12:37,954 INFO - The CDOs selected have 16 related parents.
May/05/2025 05:12:37,958 INFO - A total of 0 CDOs are non-persistent.
May/05/2025 05:12:37,970 INFO - A total of 17 CDOs will be generated.
May/05/2025 05:12:38,094 INFO -----Script generation completed-----

Script generation completed at 2024-10-03 02:58:27



Script settings

Relation: ☒ Data Object ☐ Database

Search:

☐ ConciergeTypeEnum
☐ ConditionTypeEnum
☐ Constants
☐ ContactCustomerDetail
☒ Container

1 selected

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level: 0

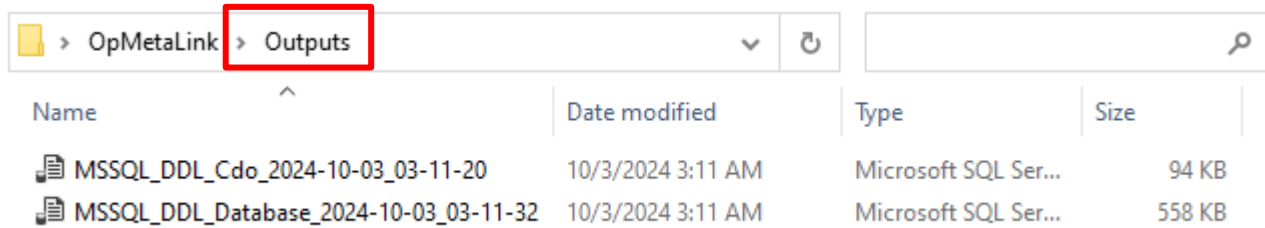
DDL Log Configuration

CREATE TABLE "Container" (
"ContainerId" CHAR(16),
"BillOfProcessId" CHAR(16),
"DetailId" CHAR(16),
"LastCarrierId" CHAR(16),
"SamplingLotId" CHAR(16),
"SPCChartSaveOptionsEnumId" CHAR(16),
"StartParentContainerId" CHAR(16),
"VendorItemId" CHAR(16),
"BOMId" CHAR(16),
"ParentId" CHAR(16),
"IssuedToContainerId" CHAR(16),
"CurrentStatusId" CHAR(16),
"CustomerId" CHAR(16),
"FactoryStartUOMId" CHAR(16).

Script generation completed at 2024-10-03 02:58:27

Step 7:

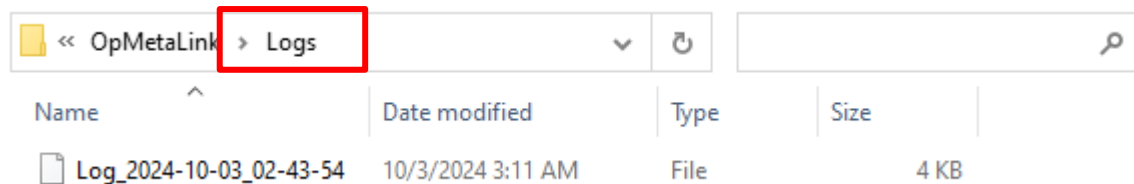
If we generate multiple outputs, we can always retrieve our previous outputs in the “Outputs” folder.



Name	Date modified	Type	Size
MSSQL_DDL_Cdo_2024-10-03_03-11-20	10/3/2024 3:11 AM	Microsoft SQL Ser...	94 KB
MSSQL_DDL_Database_2024-10-03_03-11-32	10/3/2024 3:11 AM	Microsoft SQL Ser...	558 KB

Step 8:

We can also retrieve our log files in the “Logs” folder. Each log file is session based.



Name	Date modified	Type	Size
Log_2024-10-03_02-43-54	10/3/2024 3:11 AM	File	4 KB

Note: The name of the column in DDL is based on the field’s name with append of “Id”.

1.4.6 Generate Output Based on Database

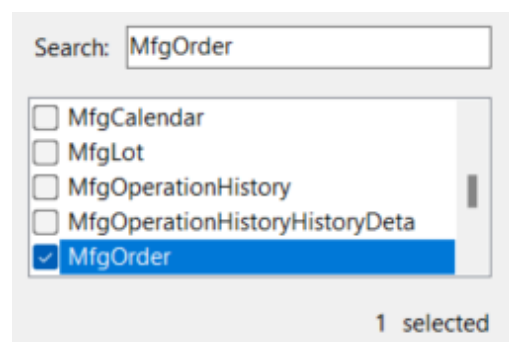
Step 1:

Select Database in the “Relation” option.

Relation: ☐ Data Object ☒ Database

Step 2:

Select the Database Tables from the “Checkbox List” that require their relationships generated. Deselect all checkboxes by clicking on the [Select All] and use the “Search Box” to find MfgOrder then select it.



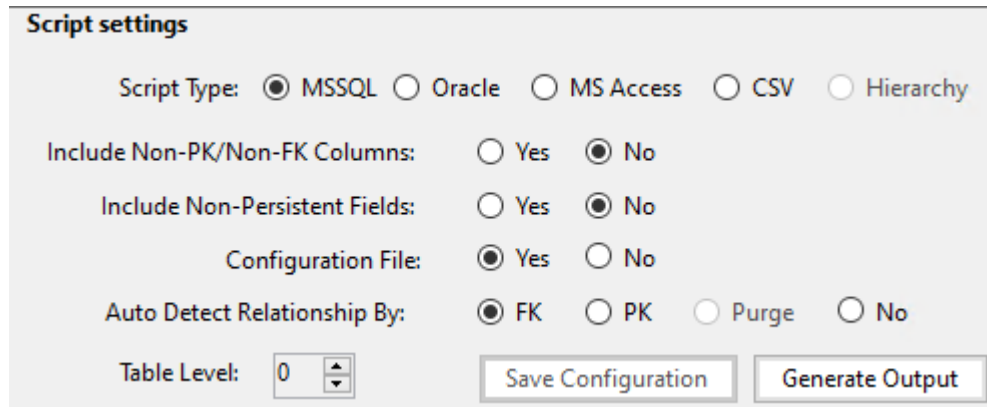
Search:

- ☐ MfgCalendar
- ☐ MfgLot
- ☐ MfgOperationHistory
- ☐ MfgOperationHistoryHistoryData
- ☒ MfgOrder

1 selected

Step 3:

Select the script settings based on user's needs. For this example, we are generating a Sql Data Definition Language script with only Primary Key and Foreign Keys columns. We also include Configuration file and "FK" for "Auto Detect Relationship By" setting. The "Include Non-Persistent Fields" option will be disabled for Database. After clicking on the "FK" option, a "Table Level" setting should appear. We shall leave it as it is and move on.



Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

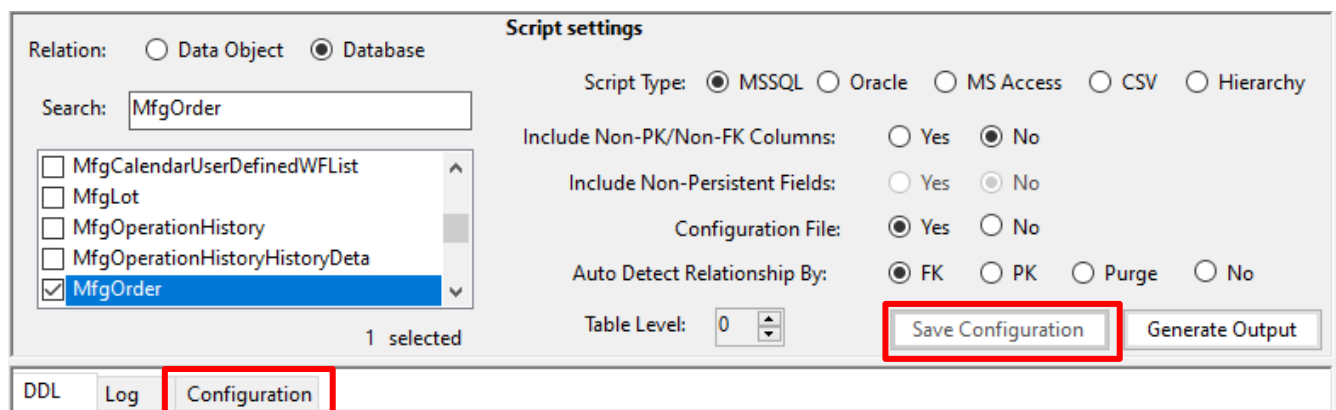
Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level: 0

Save Configuration Generate Output

Step 4:

After clicking on Yes for the "Configuration File" option, a "Save Configuration" button and "Configuration" tab-page will appear. The "Save Configuration" button is currently disabled as no changes has been made to the "Configuration" tab. Click on the "Configuration" tab-page to edit it.



Script settings

Relation: ☐ Data Object ☒ Database

Search: MfgOrder

- ☐ MfgCalendarUserDefinedWFList
- ☐ MfgLot
- ☐ MfgOperationHistory
- ☐ MfgOperationHistoryHistoryData
- ☒ MfgOrder

1 selected

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

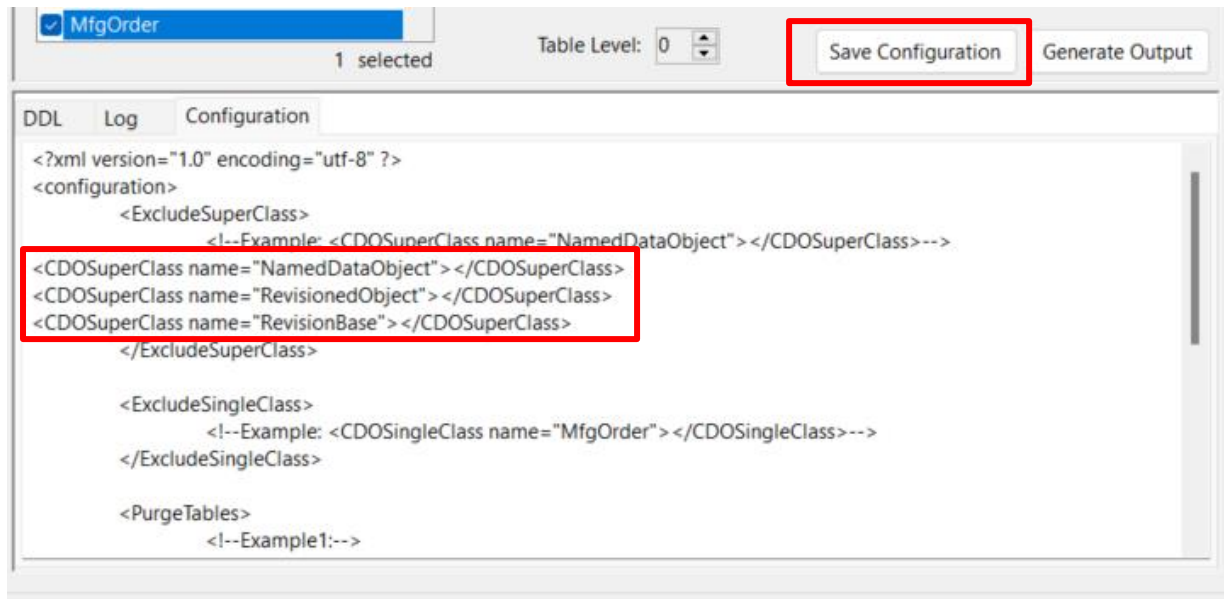
Table Level: 0

Save Configuration Generate Output

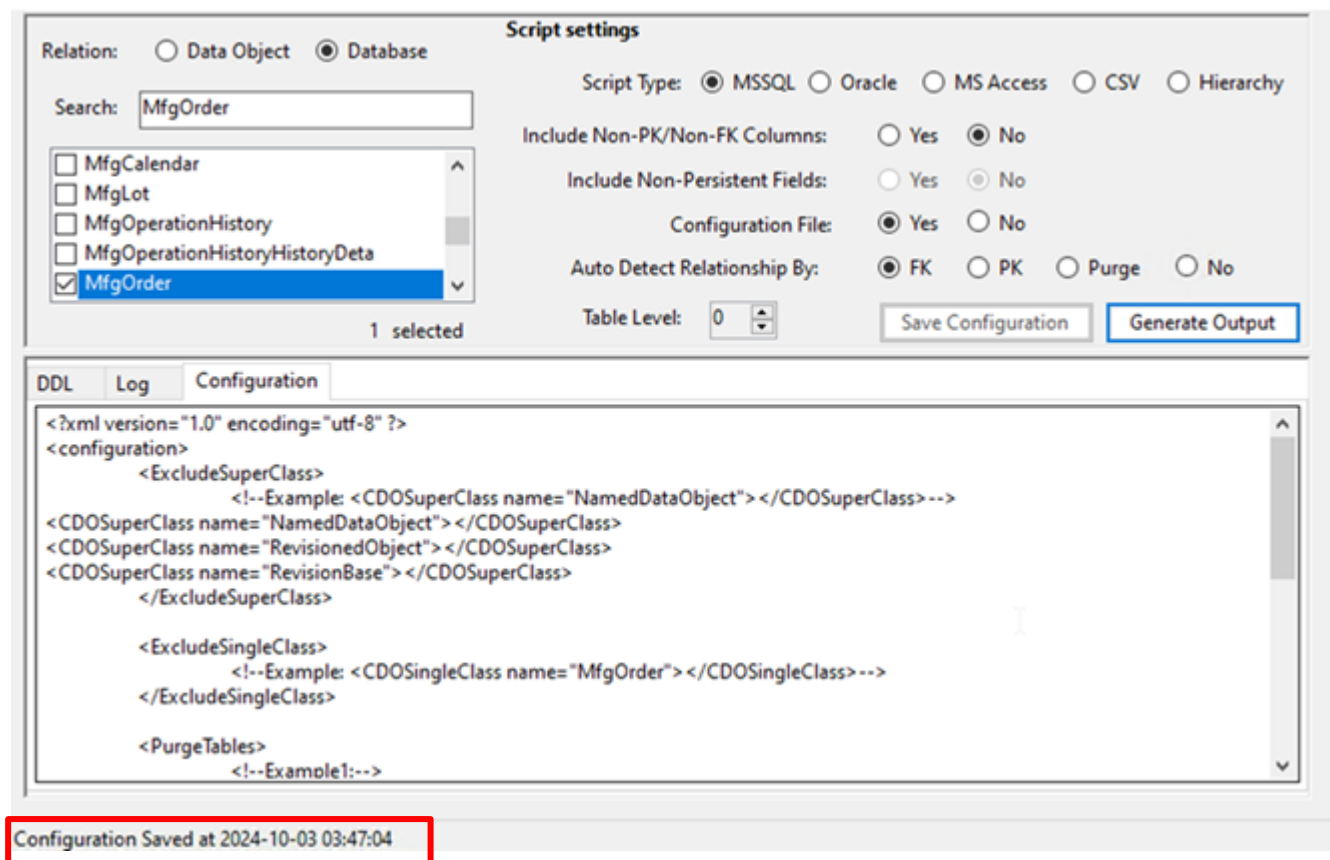
DDL Log **Configuration**

Step 5:

Add the following xml texts to the “Configuration” tab then the “Save Configuration” button will be enabled. Click on it. Do take note that the “Configuration” tab has a xml validation during save, so incorrect xml format can cause an error to prompt.



The following text will appear if the save was successful.



Step 6:

Time to generate the output. Click on the “Generate Output” button. During output generation, the UI will be disabled and a loading bar will appear.

Relation: ☐ Data Object ☒ Database

Search:

☐ MfgCalendar
☐ MfgLot
☐ MfgOperationHistory
☐ MfgOperationHistoryHistoryData
☒ MfgOrder

1 selected

Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level:

Once generation is complete, the following text will appear.

Relation: ☐ Data Object ☒ Database

Search:

☐ MfgCalendar
☐ MfgLot
☐ MfgOperationHistory
☐ MfgOperationHistoryHistoryData
☒ MfgOrder

1 selected

Script settings

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level:

DDL Log Configuration

May/05/2025 05:17:09,726 INFO - A total of 1 DBTables will be generated.
May/05/2025 05:17:09,769 INFO - -----Script generation completed-----
May/05/2025 05:20:30,553 INFO - -----Script generation started-----
May/05/2025 05:20:30,554 INFO - Relation: Database
May/05/2025 05:20:30,554 INFO - Script generation: MSSQL
May/05/2025 05:20:30,554 INFO - Include non-persistent CDOs: False
May/05/2025 05:20:30,554 INFO - Include non-PK/non-FK columns: False
May/05/2025 05:20:30,554 INFO - A total of 1008 DBTables is detected in this MDB file.
May/05/2025 05:20:30,693 INFO - 352 DBTables selected are ignored.
May/05/2025 05:20:30,694 INFO - A total of 1 DBTables has been selected.
May/05/2025 05:20:30,739 INFO - The DBTables selected have 4 related parents.
May/05/2025 05:20:30,739 INFO - A total of 5 DBTables will be generated.
May/05/2025 05:20:30,821 INFO - -----Script generation completed-----

Script generation completed at 2024-10-03 03:48:45

The “Log” tab will be populated with the transaction logs while the “DDL” tab will be populated with the result.

Script settings

Relation: ☐ Data Object ☒ Database

Search:

☐ MfgCalendar
☐ MfgLot
☐ MfgOperationHistory
☐ MfgOperationHistoryHistoryData
☒ MfgOrder

1 selected

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level:

DDL **Log** Configuration

```

May/05/2025 05:17:09,726 INFO - A total of 1 DBTables will be generated.
May/05/2025 05:17:09,769 INFO - -----Script generation completed-----
May/05/2025 05:20:30,553 INFO - -----Script generation started-----
May/05/2025 05:20:30,554 INFO - Relation: Database
May/05/2025 05:20:30,554 INFO - Script generation: MSSQL
May/05/2025 05:20:30,554 INFO - Include non-persistent CDOs: False
May/05/2025 05:20:30,554 INFO - Include non-PK/non-FK columns: False
May/05/2025 05:20:30,554 INFO - A total of 1008 DBTables is detected in this MDB file.
May/05/2025 05:20:30,693 INFO - 352 DBTables selected are ignored.
May/05/2025 05:20:30,694 INFO - A total of 1 DBTables has been selected.
May/05/2025 05:20:30,739 INFO - The DBTables selected have 4 related parents.
May/05/2025 05:20:30,739 INFO - A total of 5 DBTables will be generated.
May/05/2025 05:20:30,821 INFO - -----Script generation completed-----
  
```

Script generation completed at 2024-10-03 03:48:45

Script settings

Relation: ☐ Data Object ☒ Database

Search:

☐ MfgCalendar
☐ MfgLot
☐ MfgOperationHistory
☐ MfgOperationHistoryHistoryData
☒ MfgOrder

1 selected

Script Type: ☒ MSSQL ☐ Oracle ☐ MS Access ☐ CSV ☐ Hierarchy

Include Non-PK/Non-FK Columns: ☐ Yes ☒ No

Include Non-Persistent Fields: ☐ Yes ☒ No

Configuration File: ☒ Yes ☐ No

Auto Detect Relationship By: ☒ FK ☐ PK ☐ Purge ☐ No

Table Level:

DDL Log Configuration

```

CREATE TABLE "WIPMsgKey" (
  "WIPMsgKeyld" CHAR(16),
  "WIPMsgDefMgrld" CHAR(16)
);
CREATE TABLE "WIPMsgDefMgr" (
  "WIPMsgDefMgrld" CHAR(16),
  "Parentld" CHAR(16),
  "WIPMsgEntityld" CHAR(16),
  "AllKeyld" CHAR(16),
  "ChangeHistoryld" CHAR(16)
);
CREATE TABLE "MfgOrder" (
  "MfgOrderld" CHAR(16),
  "ChangeStatusld" CHAR(16),
  "BeginProductld" CHAR(16)
  
```

Script generation completed at 2024-10-03 03:48:45

Step 7:

If we generate multiple outputs, we can always retrieve our previous outputs in the “Outputs” folder.

OpMetaLink > Outputs		Search Outputs	
Name	Date modified	Type	Size
MSSQL_DDL_Cdo_2024-10-03_03-51-39	10/3/2024 3:51 AM	Microsoft SQL Ser...	502 KB
MSSQL_DDL_Database_2024-10-03_03-48-45	10/3/2024 3:48 AM	Microsoft SQL Ser...	3 KB

Step 8:

We can also retrieve our log files in the “Logs” folder. Each log file is session based.

OpMetaLink > Logs		Search Logs	
Name	Date modified	Type	Size
Log_2024-10-03_02-43-54	10/3/2024 3:51 AM	File	5 KB

1.4.7 File Naming Convention

Output files:

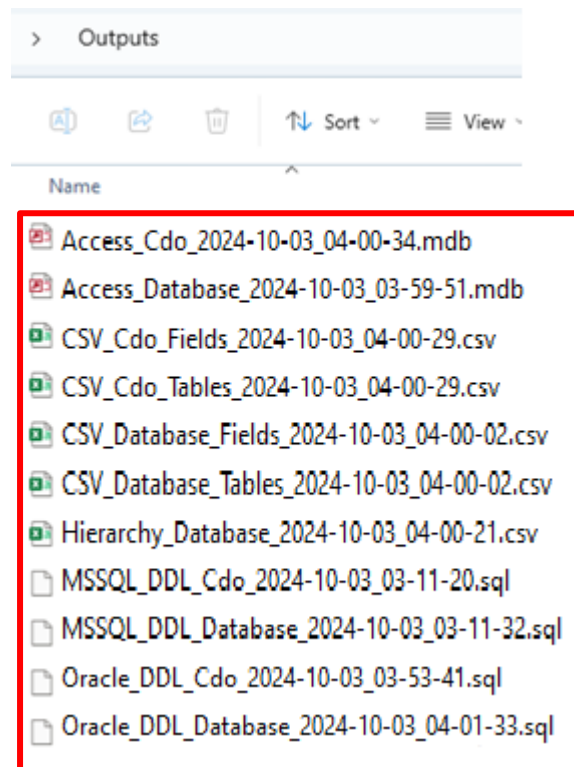
- MSSQL DDL, Oracle DDL, Microsoft Access Database & Hierarchy View

<Script Type>_<Relation>_<Date Time Created>

- Csv File

<Script Type>_<Relation>_Tables/Fields_<Date Time Created>

- Example:



Log files:

- **Log_<Date Time Created>**

- Example:

OpMetaLink > Logs					Search Logs	
Name	Date modified	Type	Size			
Log_2024-10-03_02-43-54	10/3/2024 3:51 AM	File	5 KB			
Log_2024-10-03_03-43-44	10/3/2024 3:51 AM	File	5 KB			

2 Generate ERD by Microsoft SQL Server Management Studio (SSMS)

2.1 Introduction

Microsoft SQL Server Management Studio (SSMS) is a software application developed by Microsoft, which is used for configuring, managing, and administering all components within Microsoft SQL Server. One of its functionalities includes generating an Entity Relationship Diagram (ERD) based on the Microsoft SQL Server database.

This guide would demonstrate the necessary steps and procedures to generate the Entity Relationship Diagram (ERD) with SSMS and export it to Portable Document Format (PDF).

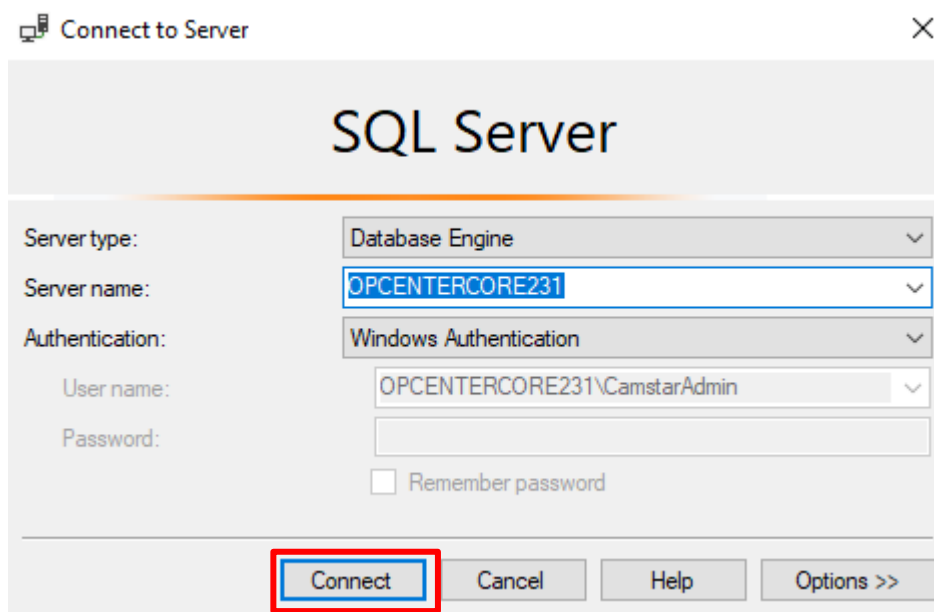
2.2 Prerequisites

- Microsoft SQL Server
- Microsoft SQL Server Management Studio (SSMS)

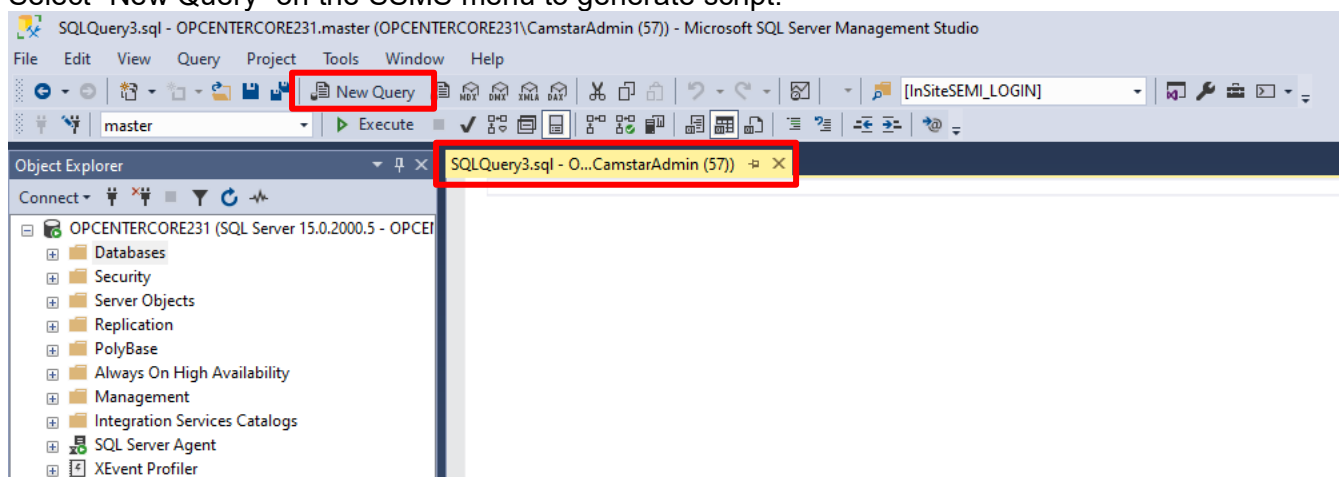
2.3 Demonstration

2.3.1 Database Setup

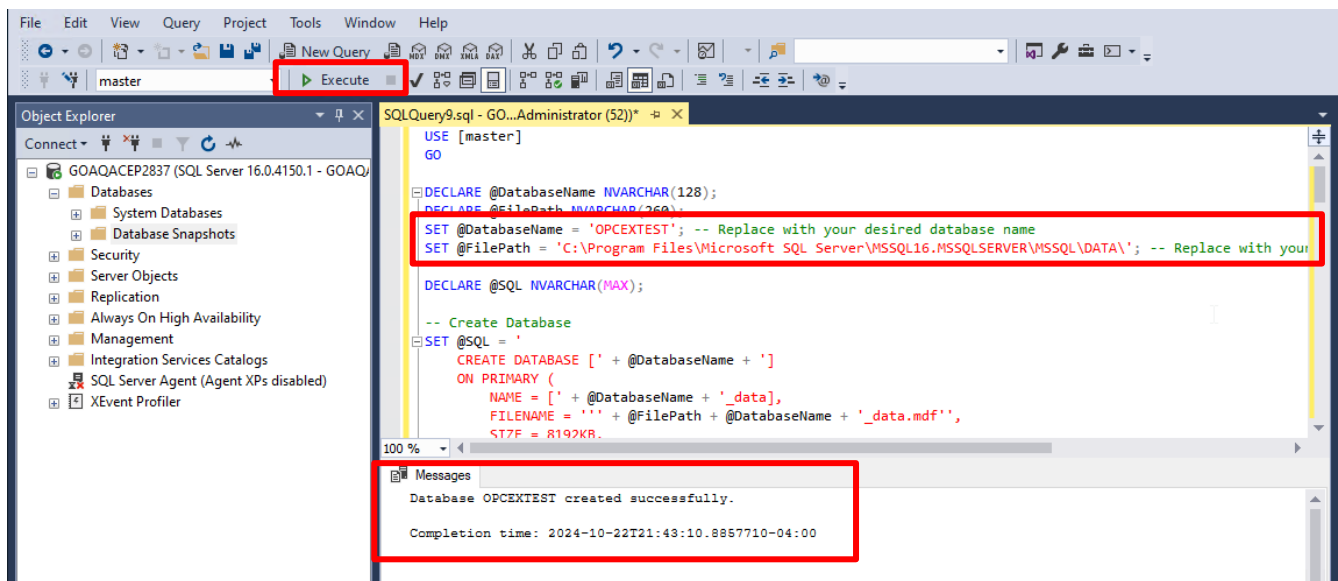
Connect and authenticate into the Microsoft SQL Server.



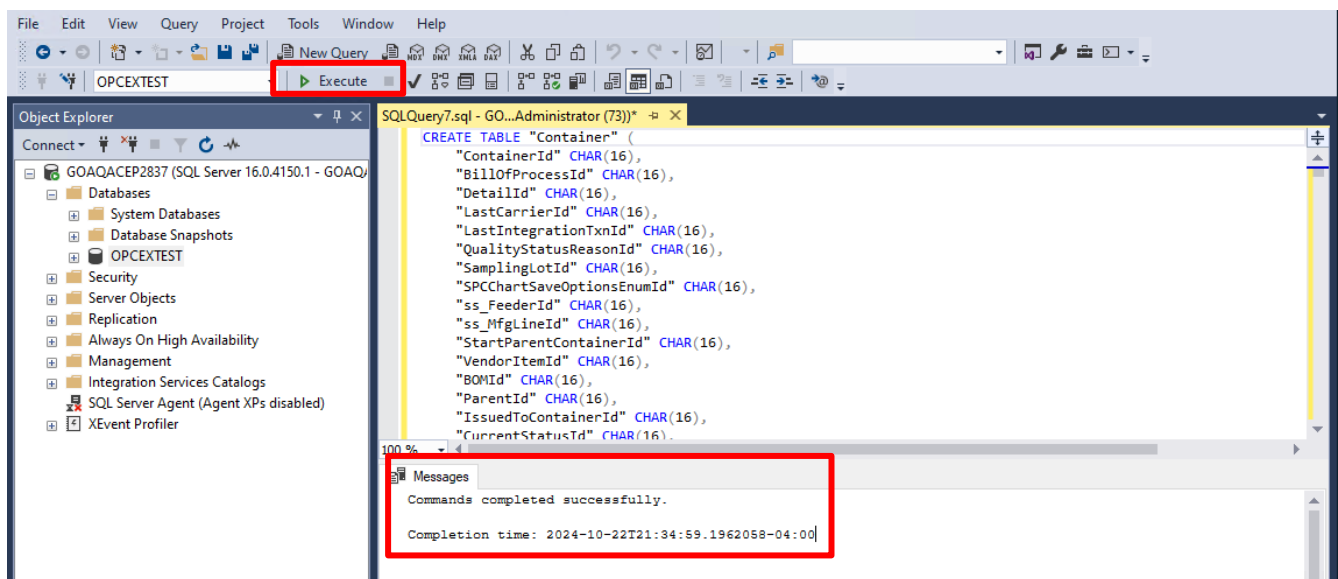
Select "New Query" on the SSMS menu to generate script.



Copy the script from “**CreateDB.sql**” from path: “C:\Program Files (x86)\Camstar\InSite Administration\OpMetaLink\scripts\Mssql”, update Database Name and File Path Name for backup files, then execute to create the database.

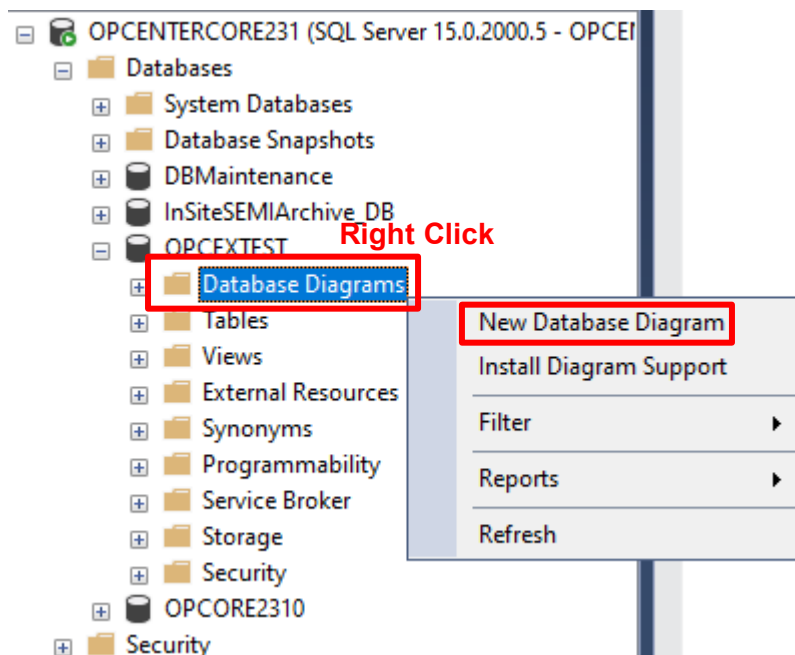


Copy the script from output file from path: “C:\Program Files (x86)\Camstar\InSite Administration\OpMetaLink\Outputs”, execute to create the database with relationship.

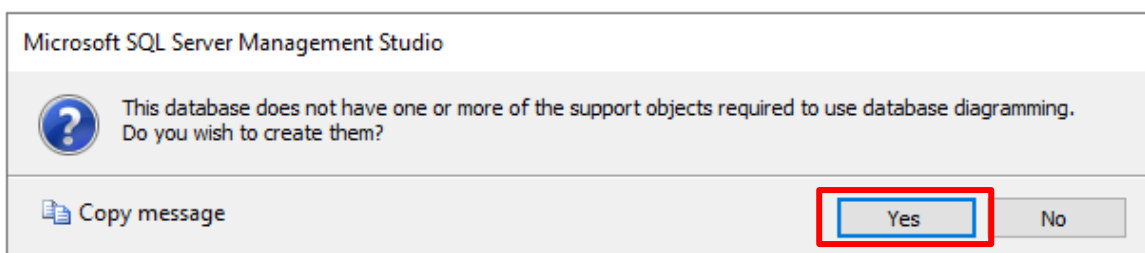


2.3.2 Generate ERD

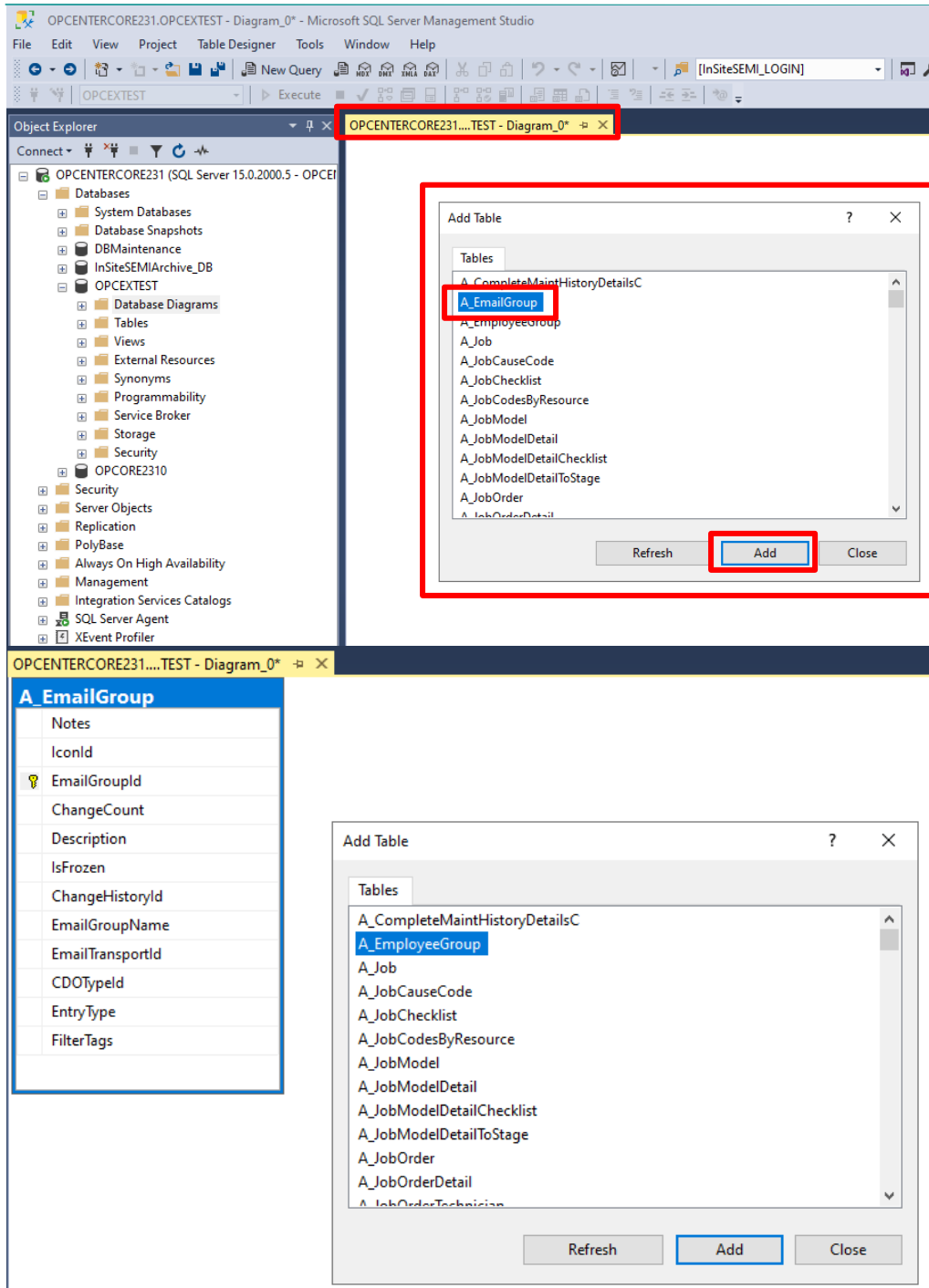
Then, expand **OPCEXTEST** database and right click on “**Database Diagrams**” and select “**New Database Diagram**”.



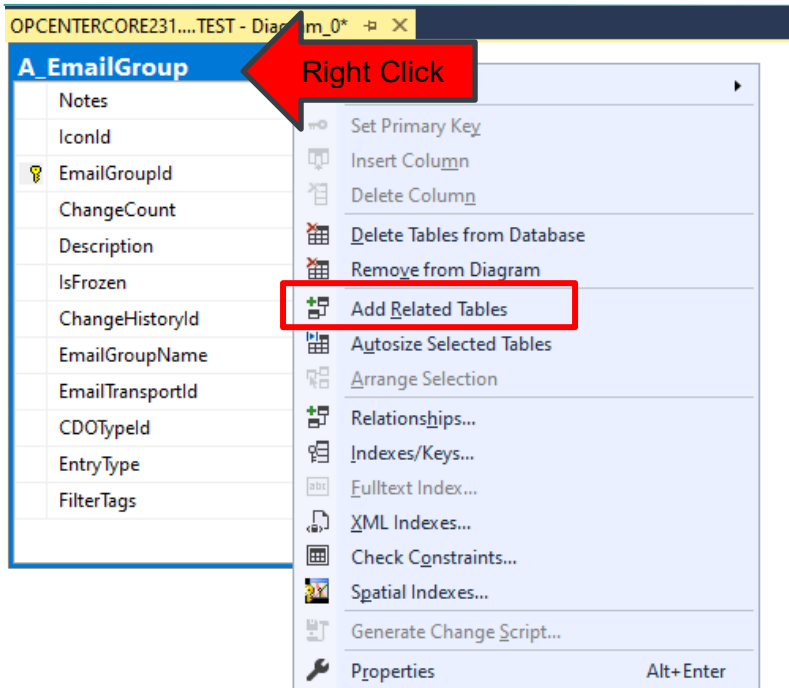
Select “Yes” on the following pop-up window to allow ERD diagramming on SSMS.



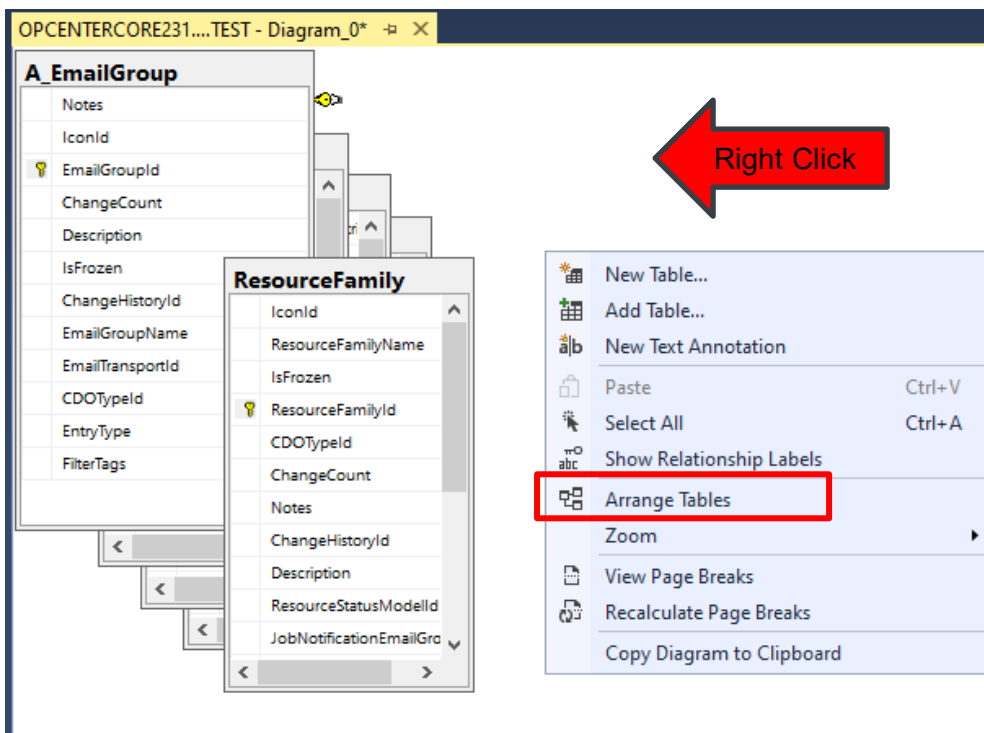
The following window and pop-up will appear. Then, select and choose which table to display the ERD. After selecting the table, click “**Add**”. (In this example, **A_EmailGroup** is the targeted table.)

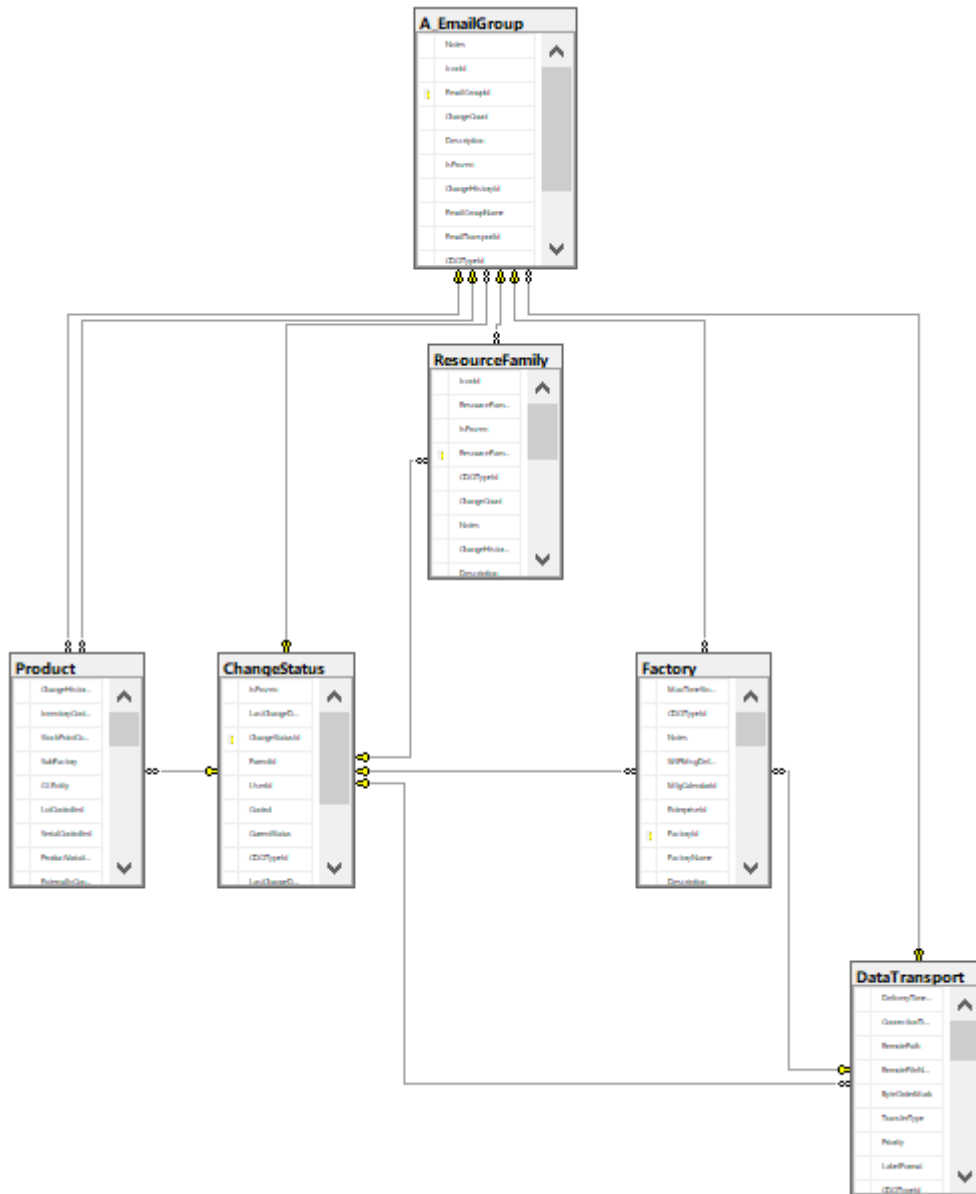


To display its related relationship and table entities, **right-click** on the table and select “**Add Related Tables**”.



Then, **right-click** on the blank space and select “**Arrange Tables**”.

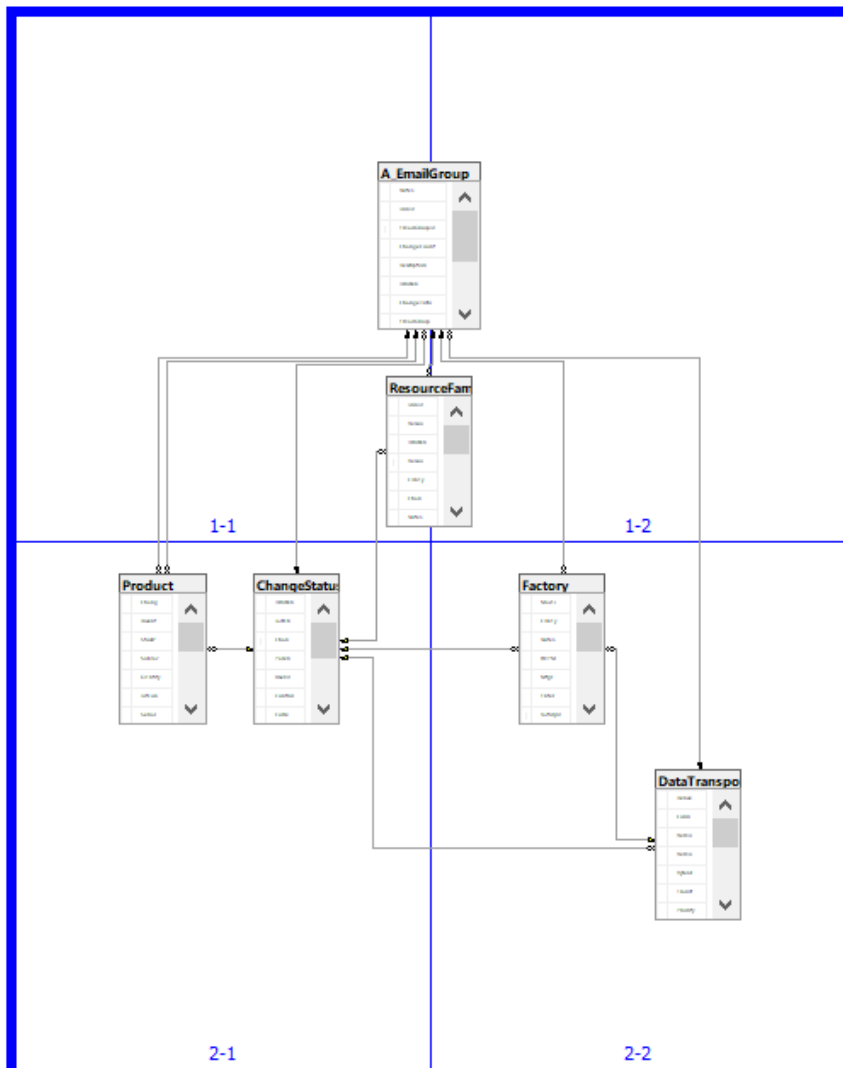
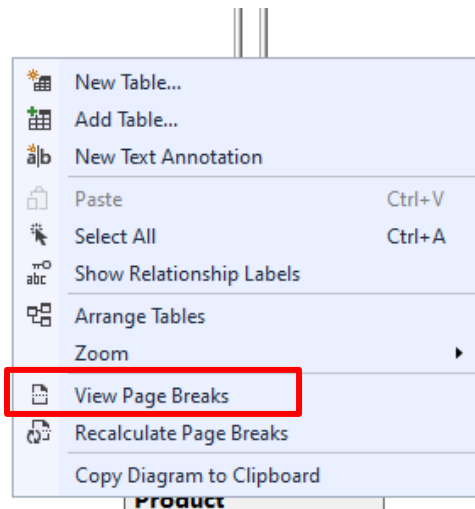




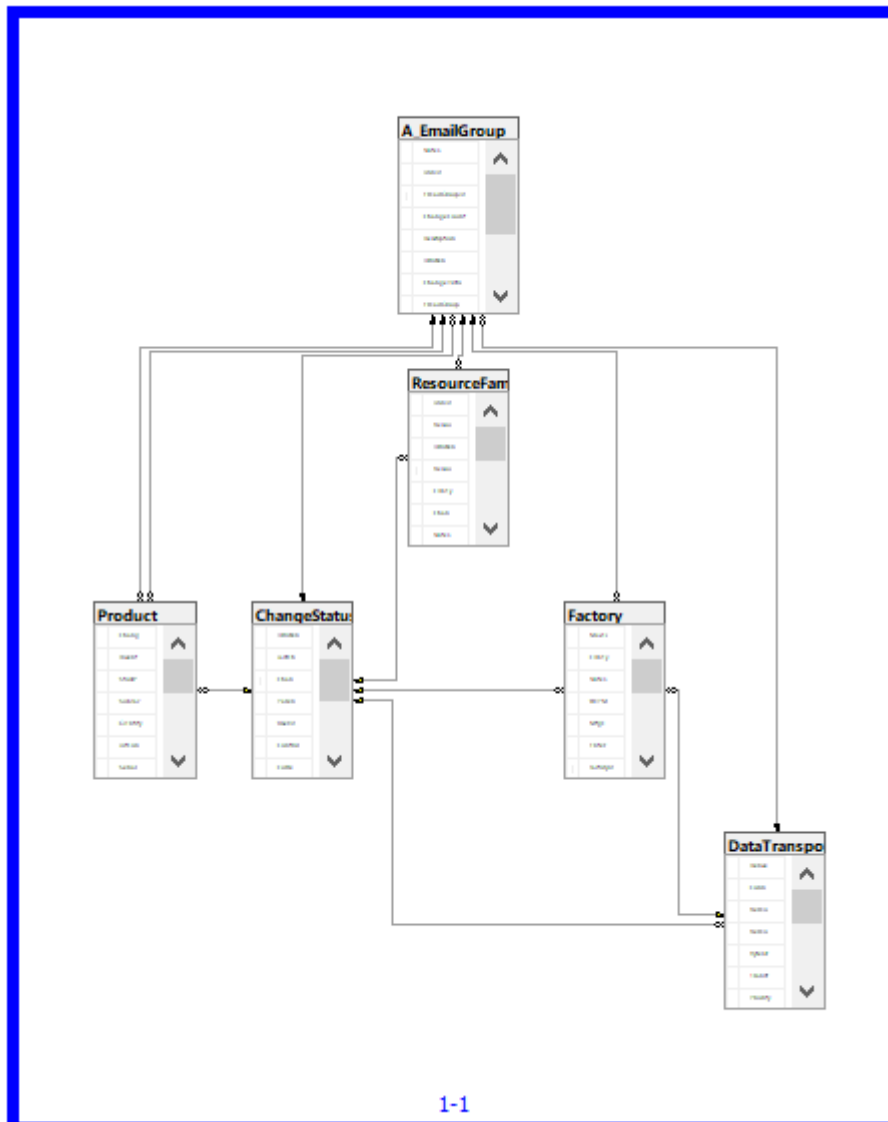
2.3.3 Export ERD

2.3.3.1 Export to PDF

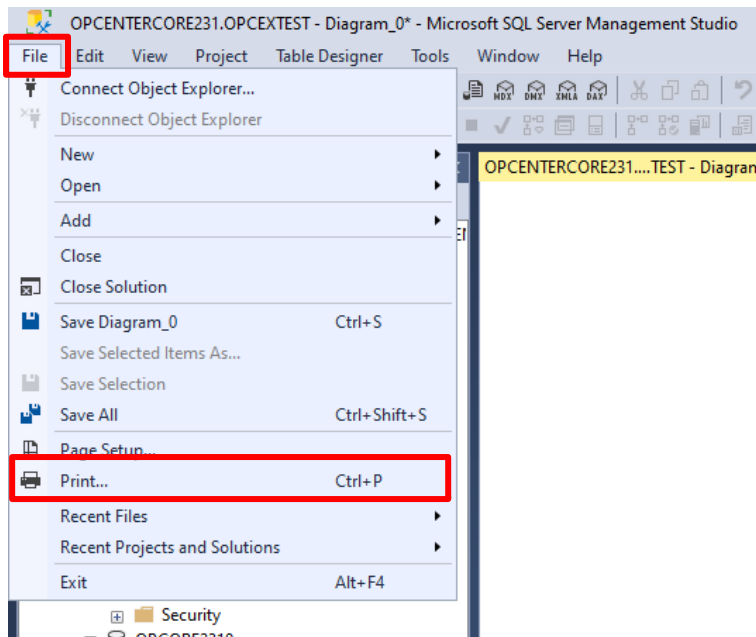
Before exporting the ERD, right-click on the blank space and select “View Page Breaks” to view how the diagram would be displayed in PDF. In the following example, the ERD would be divided into 4 pages.



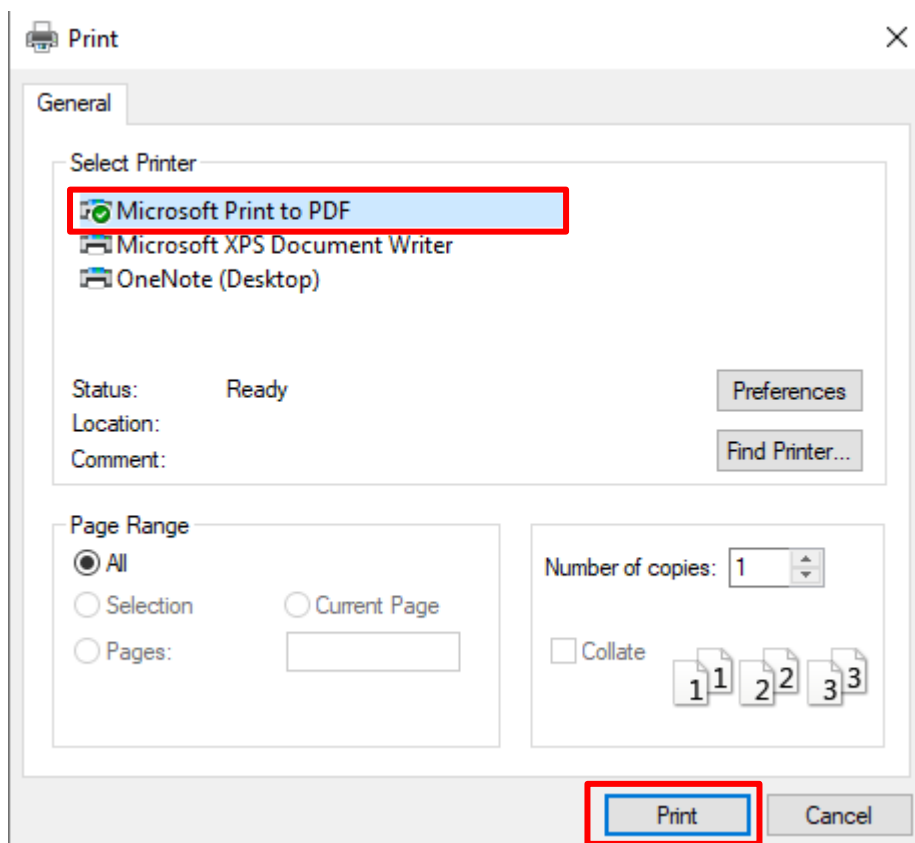
The user can drag the **blue-lined box** to adjust the printing position of the ERD document.



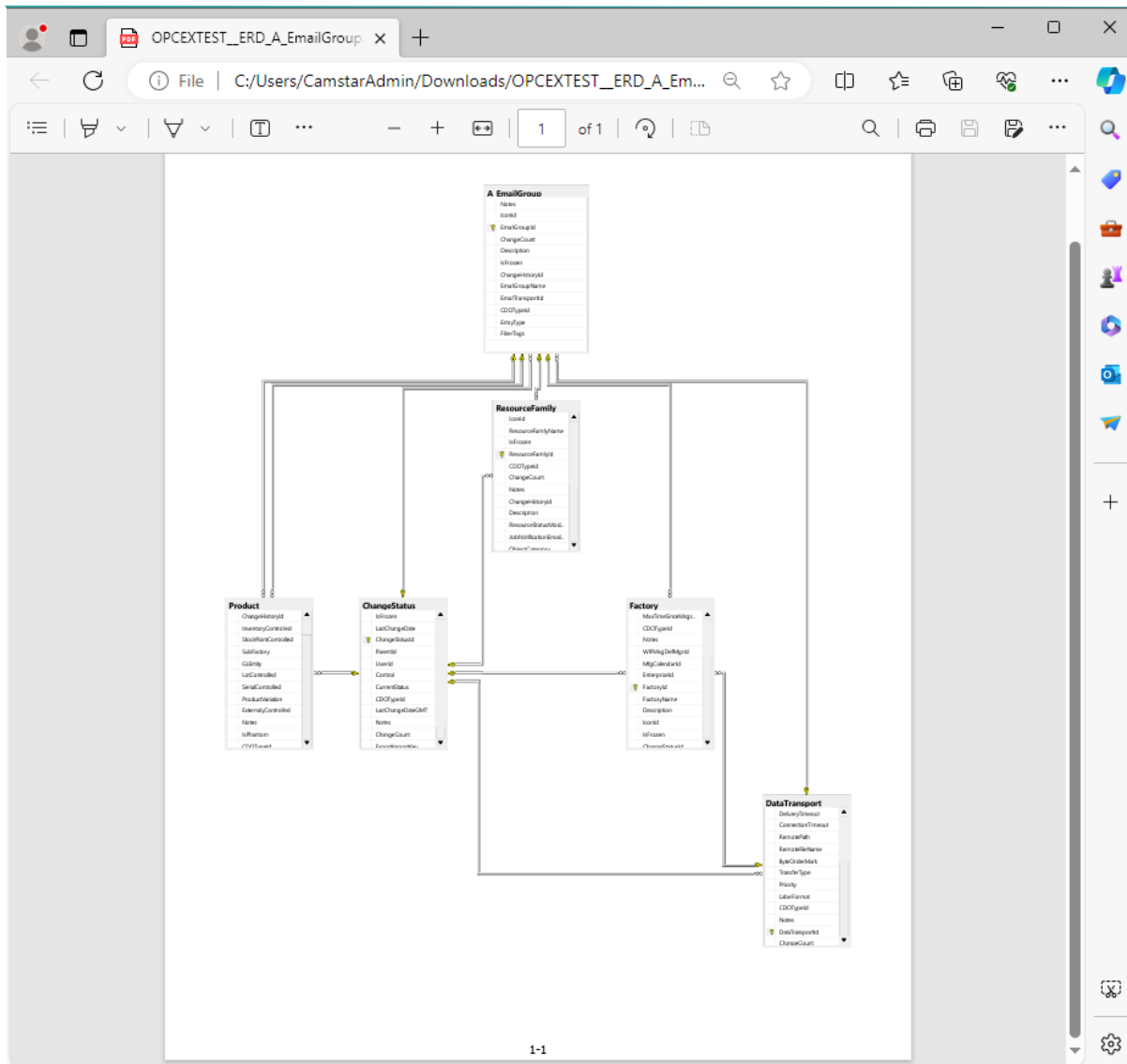
To export the ERD, select **"File"** and select **"Print..."** from the SSMS menu.



Then, select **"Microsoft Print to PDF"** under the Select Printer tab, then click on **"Print"** and proceed to save the PDF document on your destination directory.

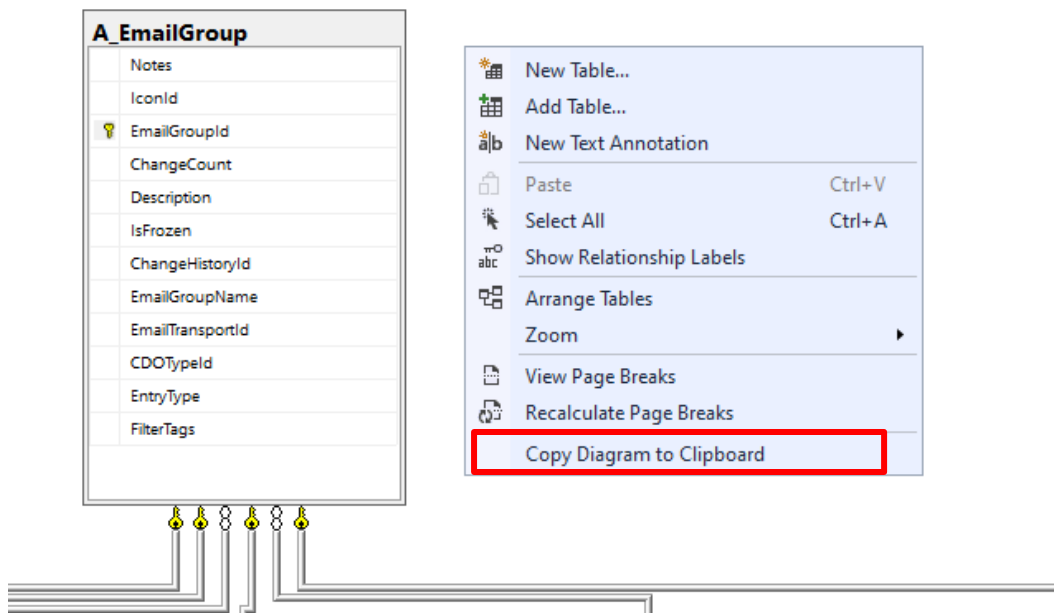


Name	Date modified	Type	Size
Today (1)			
 OPCEXTEST_ERD_A_EmailGroup	26/2/2024 3:07 PM	Microsoft Edge P...	307 KB

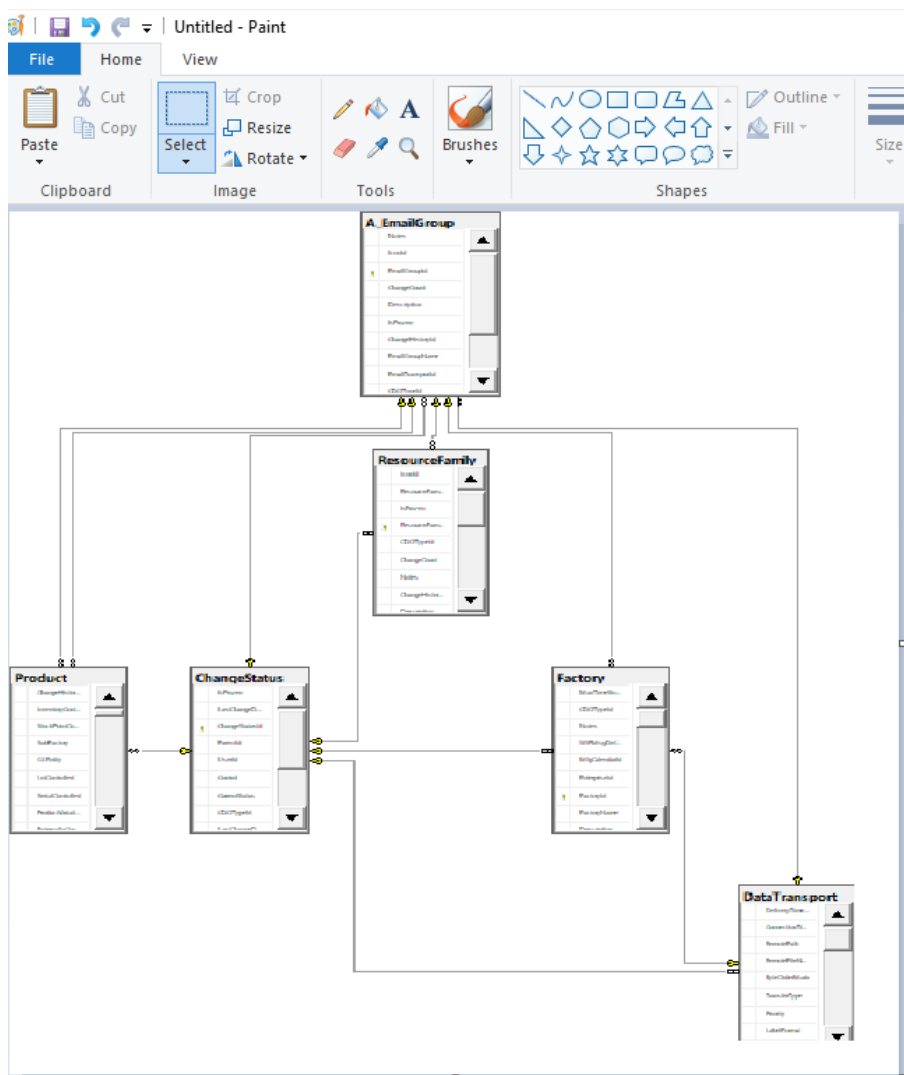


2.3.3.2 Export to Clipboard

Besides exporting to PDF, SSMS supports exporting the ERD to clipboard. **Right-click** on any blank space and select “**Copy Diagram to Clipboard**”.



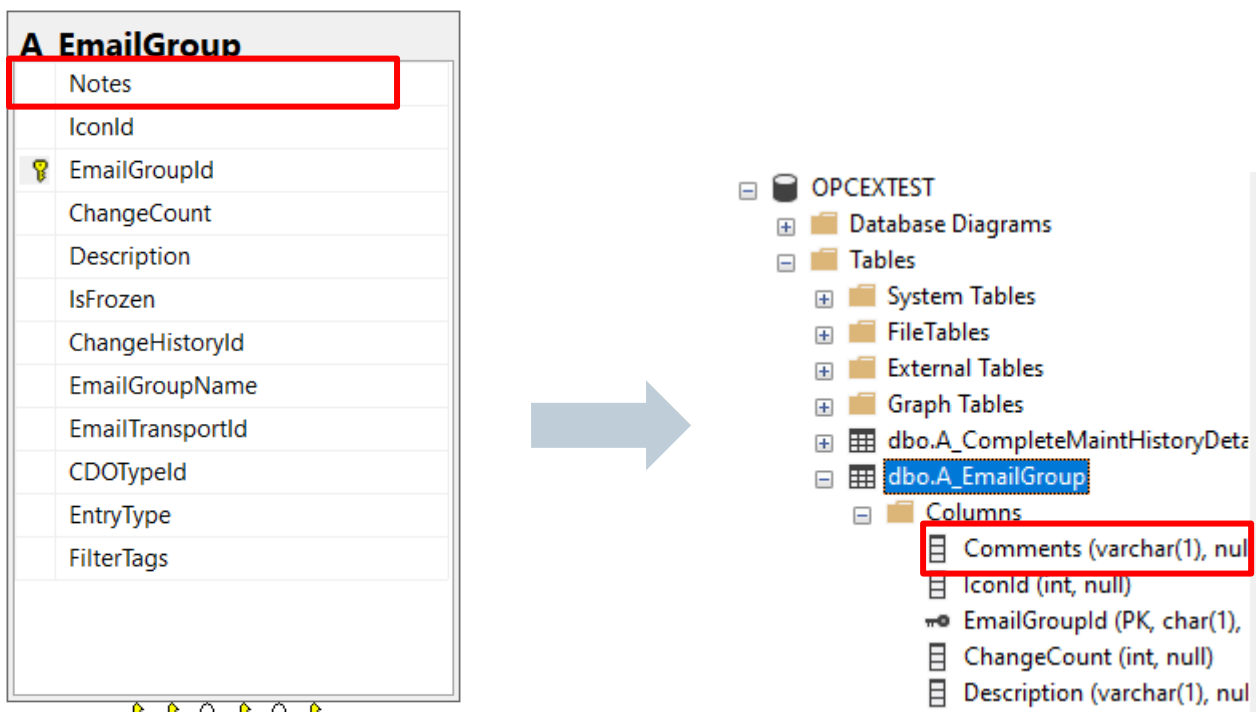
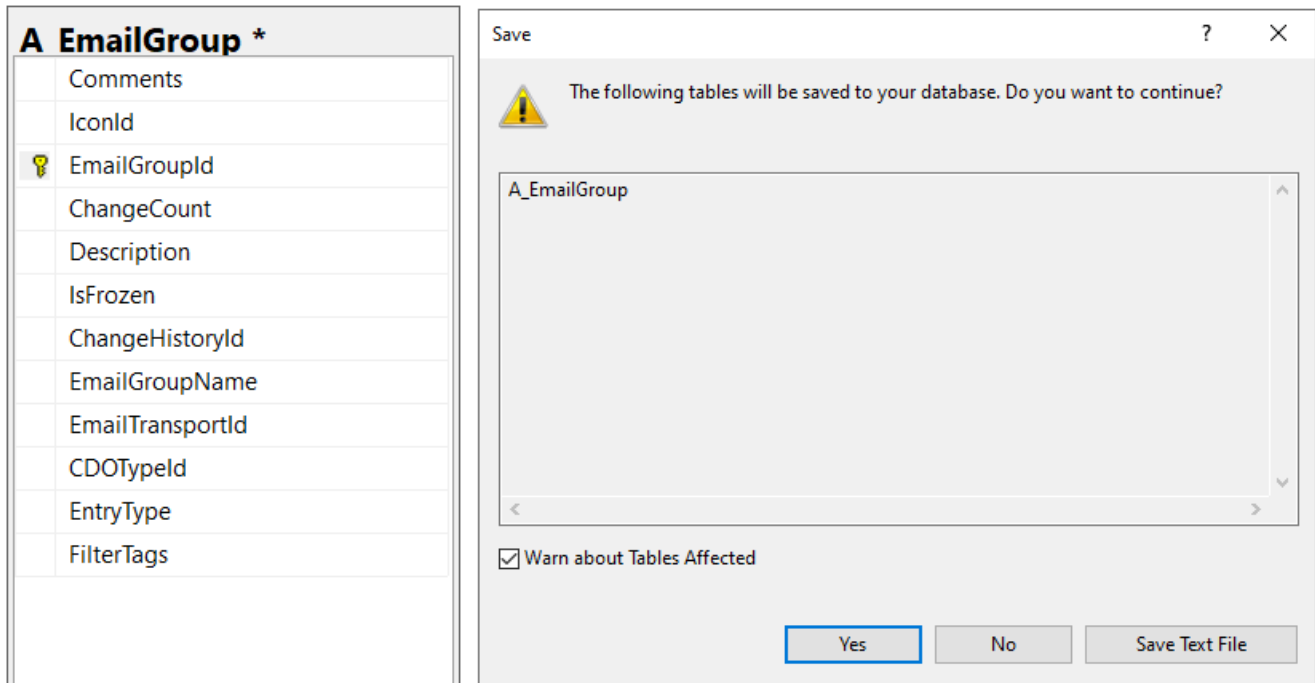
This allows users to **paste** the ERD with **CTRL + V** on applications such as Paint or Microsoft Word.



2.3.4 Disclaimers

After displaying the Entity Relationship Diagram (ERD) on SSMS, be aware that the entities and columns are reflected directly on the Microsoft SQL Server database. Any changes made to the ERD (e.g. changing column name) would result in changes in the actual database itself.

For example, changing the column name from “**Notes**” to “**Comments**” on the ERD would reflect the changes to the actual database.



3 Generate ERD by Microsoft Access Database

3.1 Introduction

Microsoft Access is a database management system from Microsoft that combines the relational Access Database Engine with a graphical user interface and software-development tools. It includes functionalities such as allowing users to organize and manipulate data in a structured manner. This documentation will provide an user guide on how to generate Entity Relationship Diagram (ERD) through Microsoft Access.

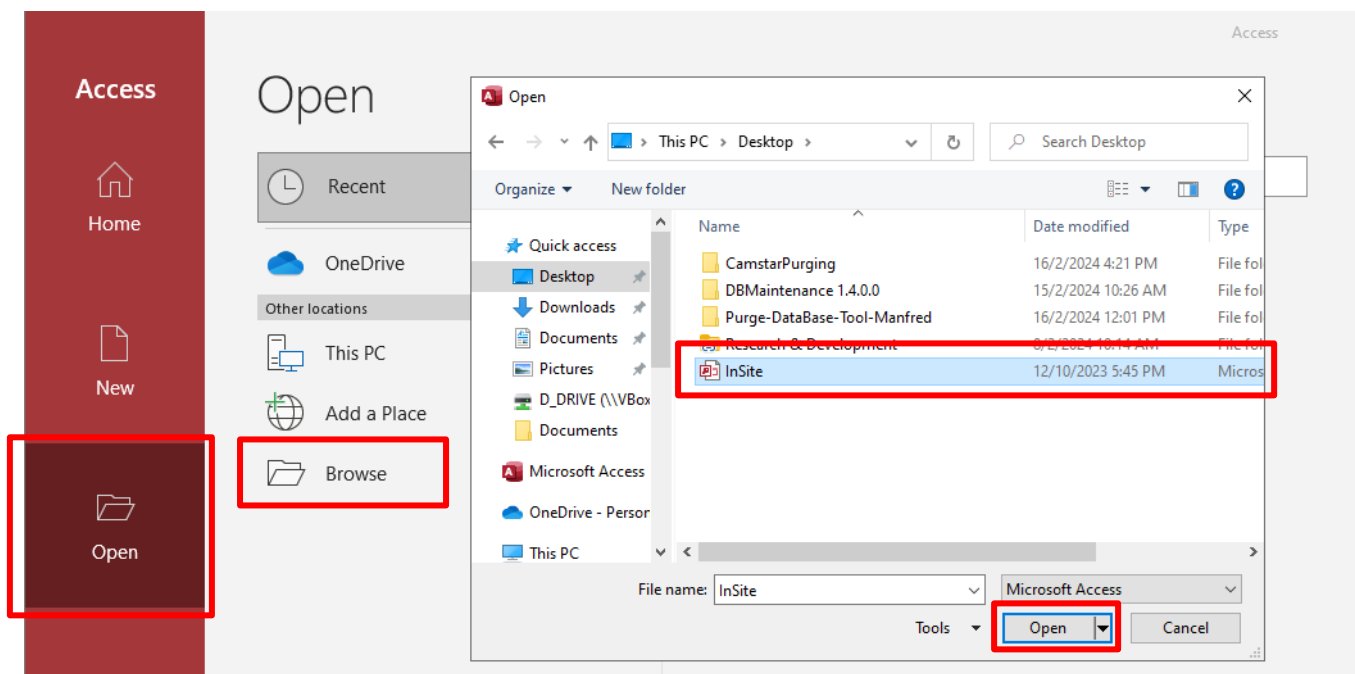
3.2 Prerequisites

- Microsoft Access Database Engine

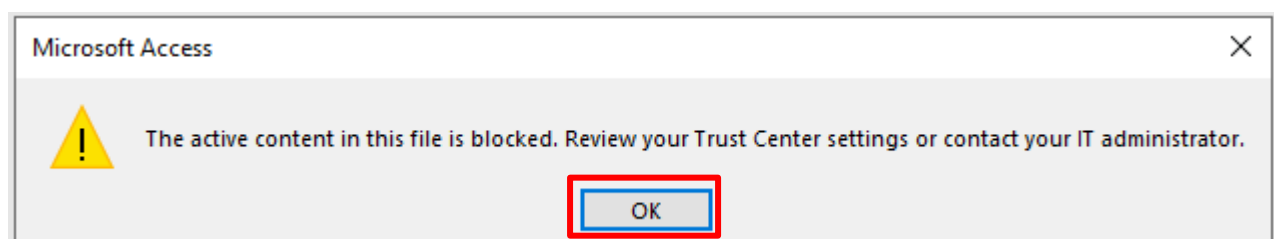
3.3 Demonstration

3.3.1 Database Setup

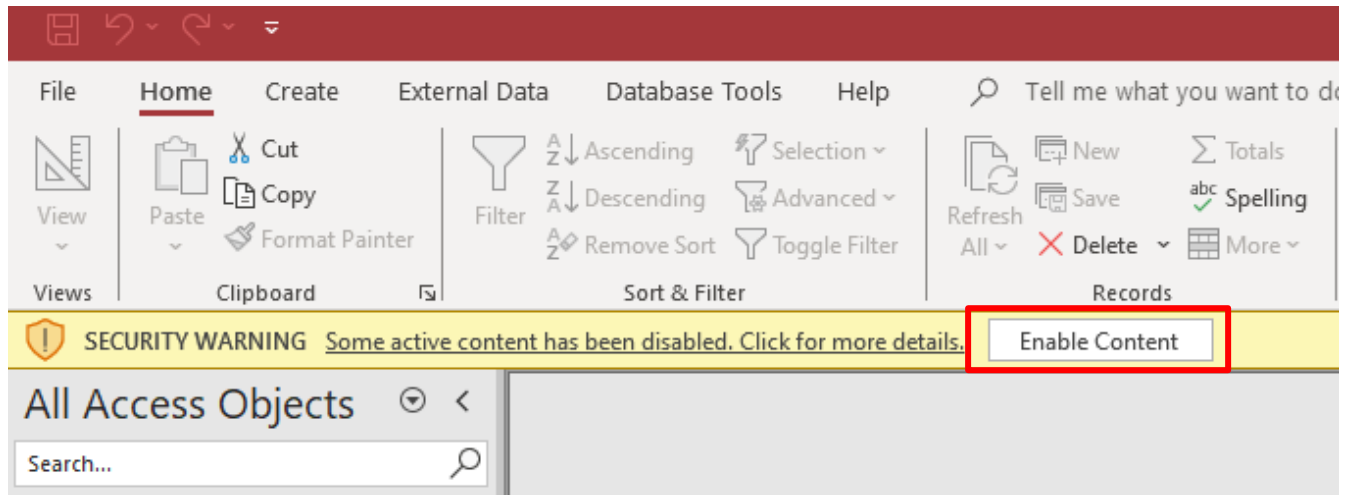
Open **Microsoft Access** and select “**Open**”, then select “**Browse**” to retrieve the database file (*.mdb).



Select “**OK**” on this pop-up window. The following step will demonstrate how to grant access to the database.

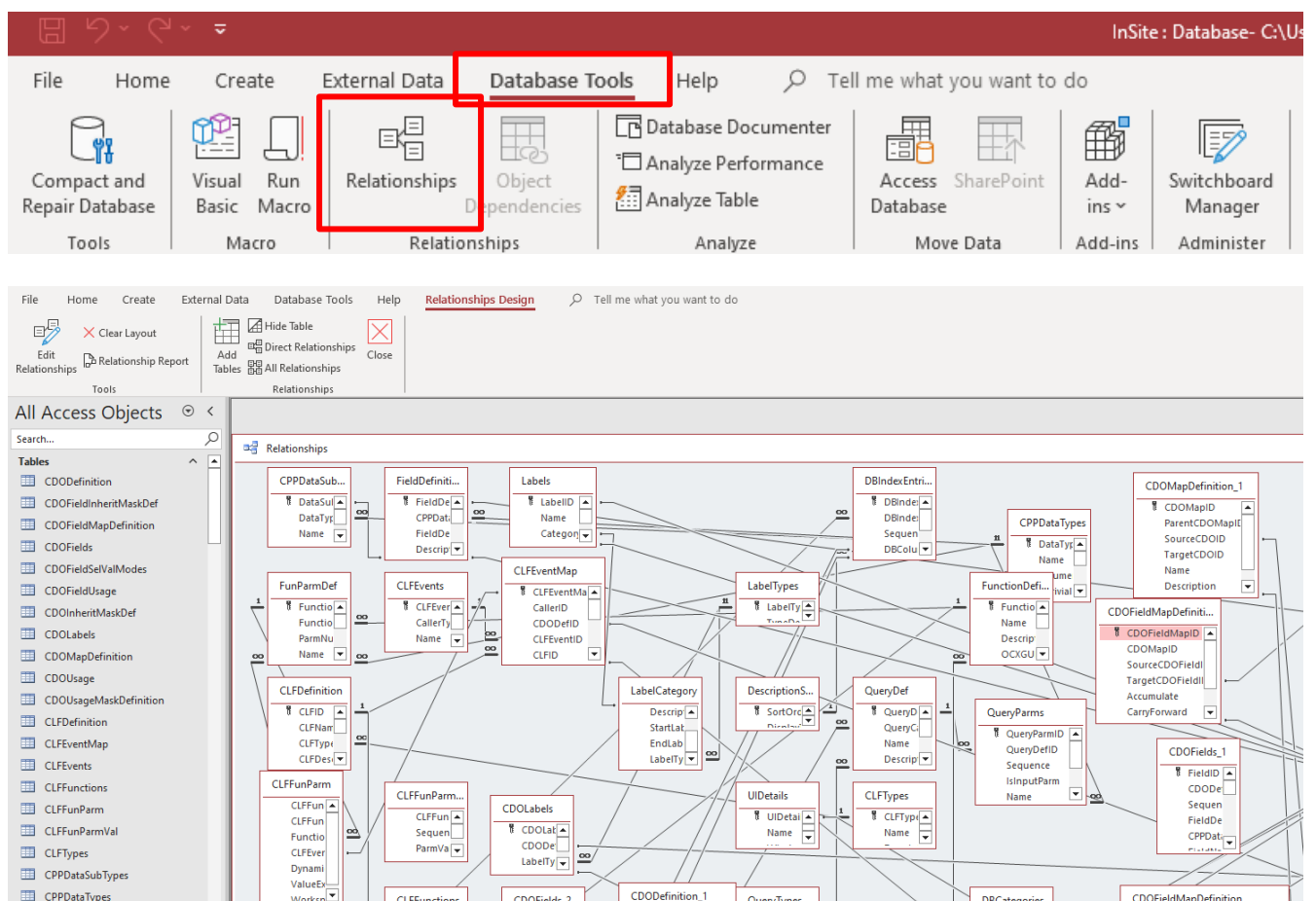


Select **“Enable Content”** below the menu bar.

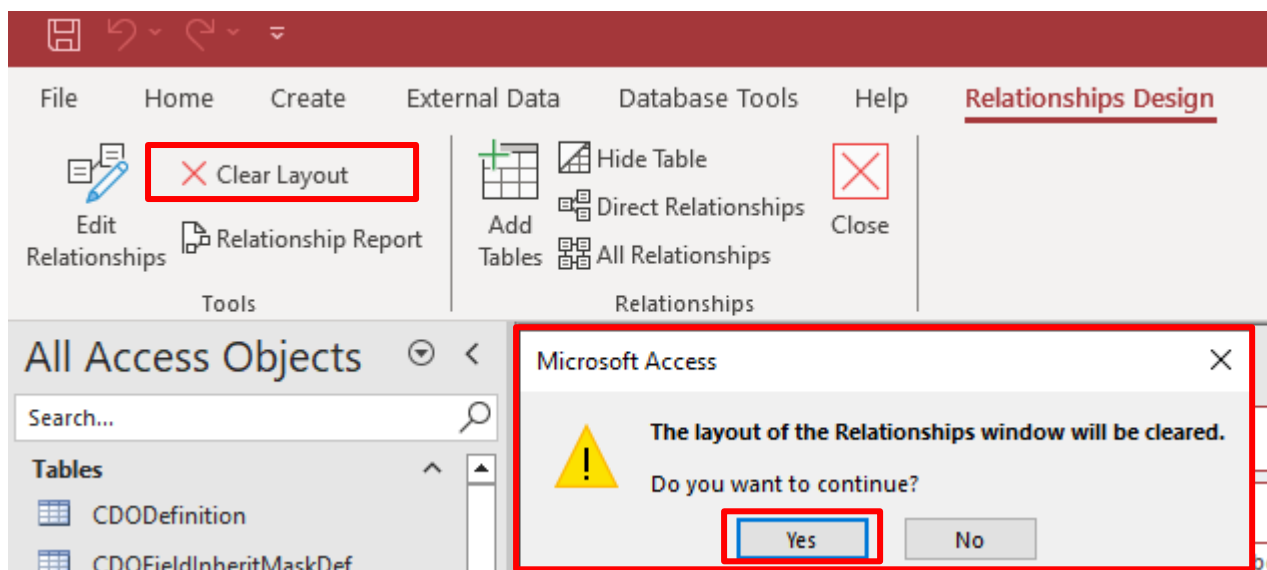


3.3.2 Generate Entity Relationship Diagram (ERD)

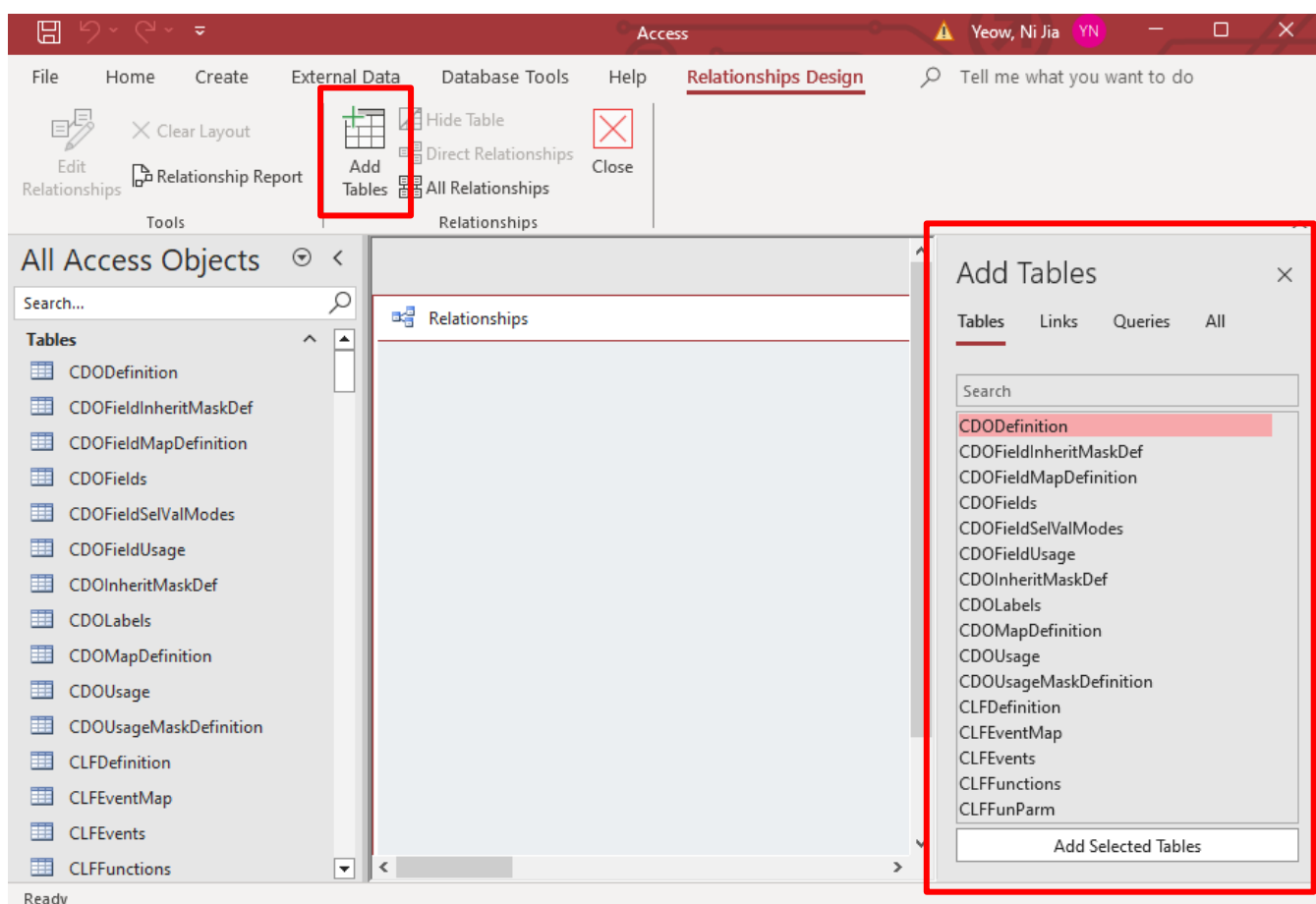
To display the ERD Diagram, select **“Database Tools”**, then select **“Relationships”**. It would display the ERD for the entire database including its associated relationships between tables.



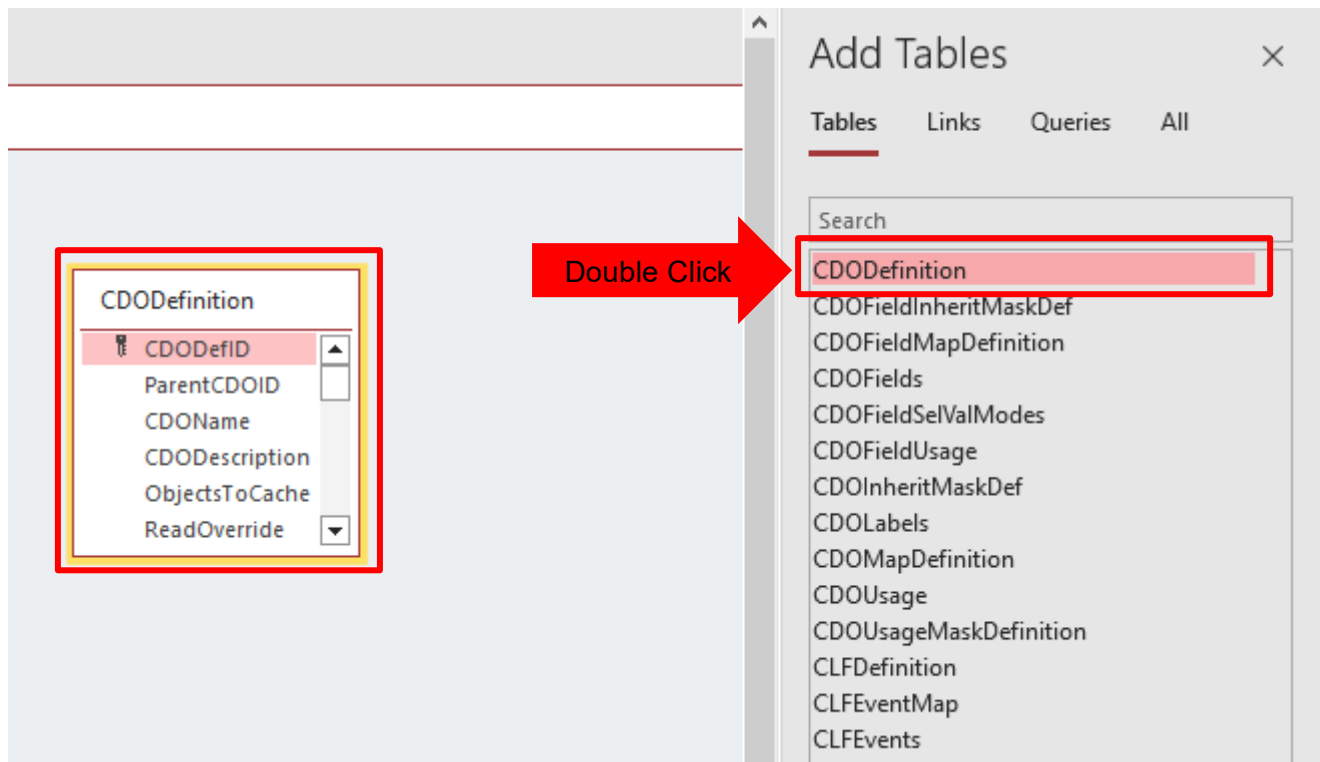
To view the ERD of any specific entities, select **“Clear Layout”** and select **“Yes”** on the following pop-up window.



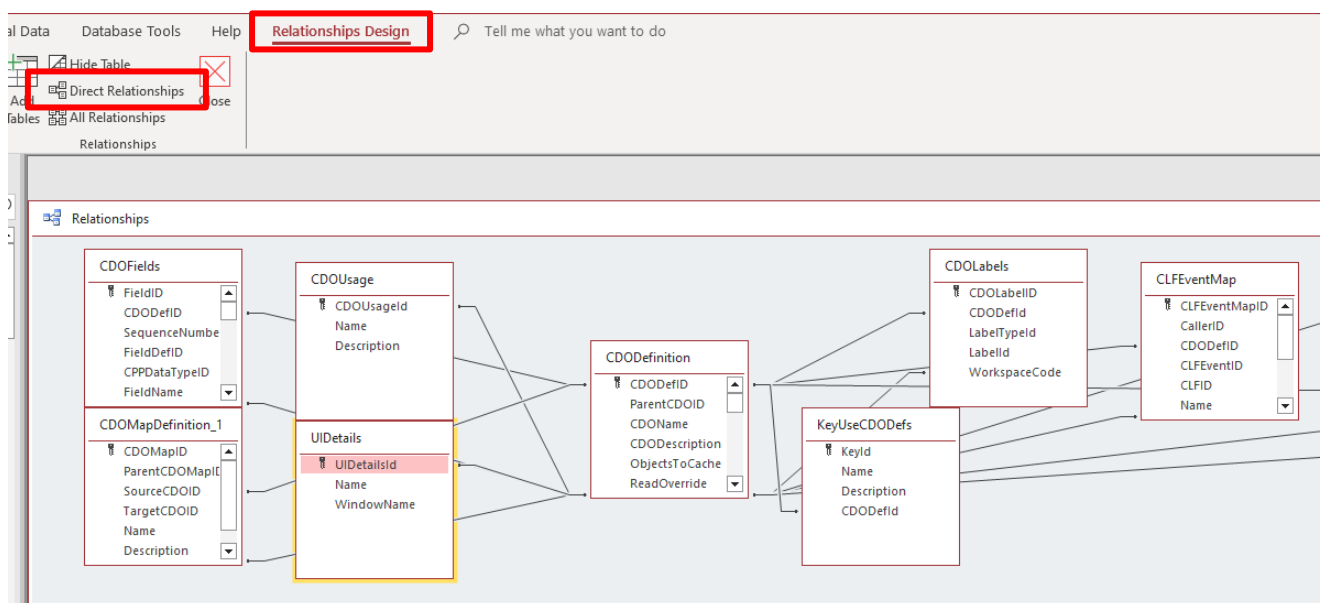
Then, select **“Add Tables”**. The following **Add Tables** menu will appear on the right side of Microsoft Access.



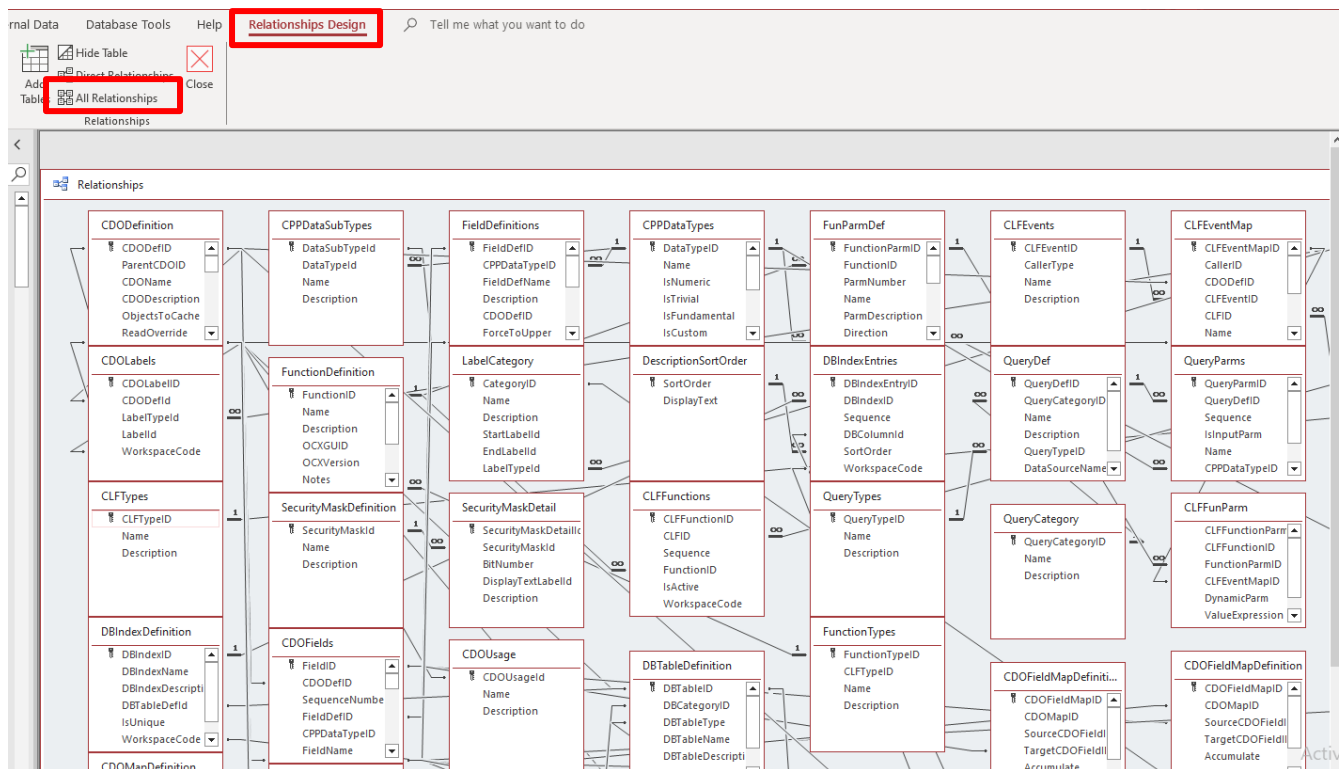
To view the ERD of the table, **double click** on the targeted table.



To view and display all the direct relationships associated with the table, select “**Direct Relationships**” on the menu bar.

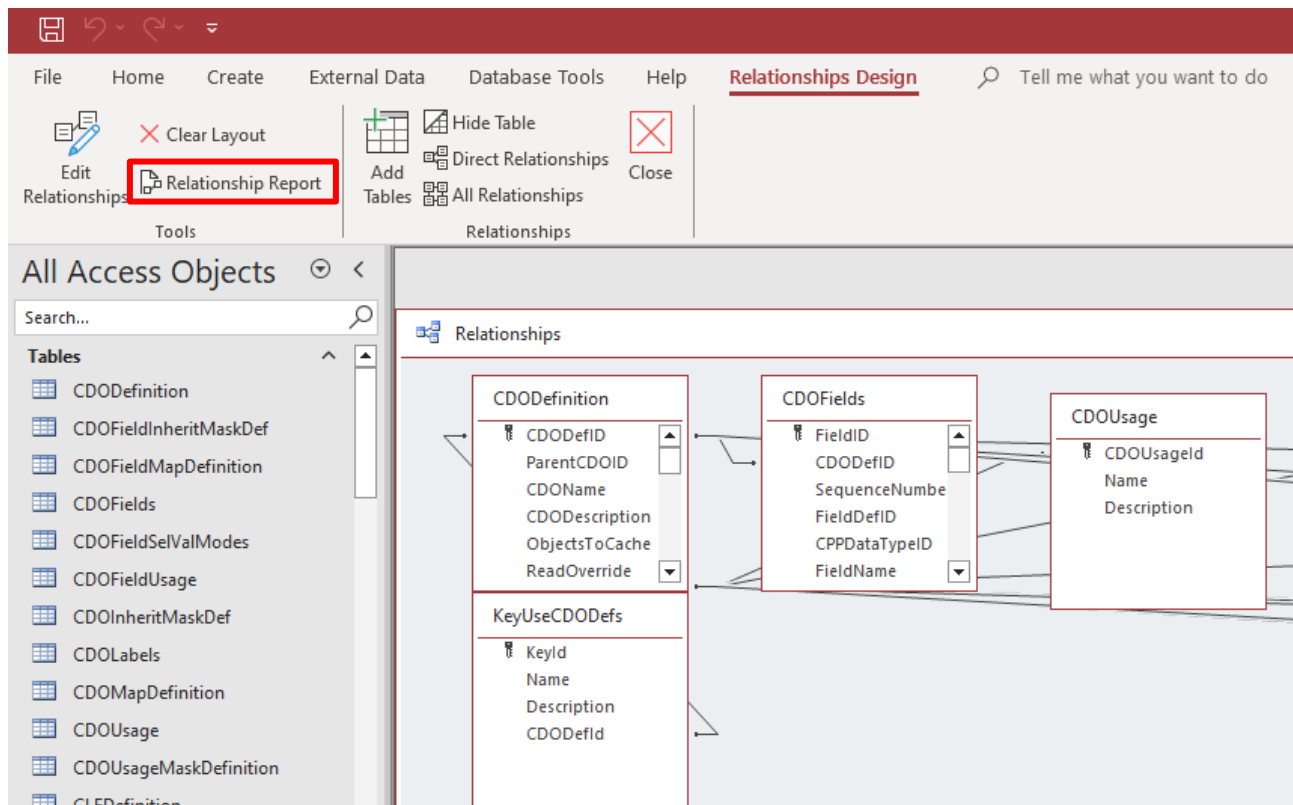


To view the Entity Relationship Diagram (ERD) of the database including its tables and associated relationships, select **“All Relationships”** on the menu bar under **“Relationships Design”**.

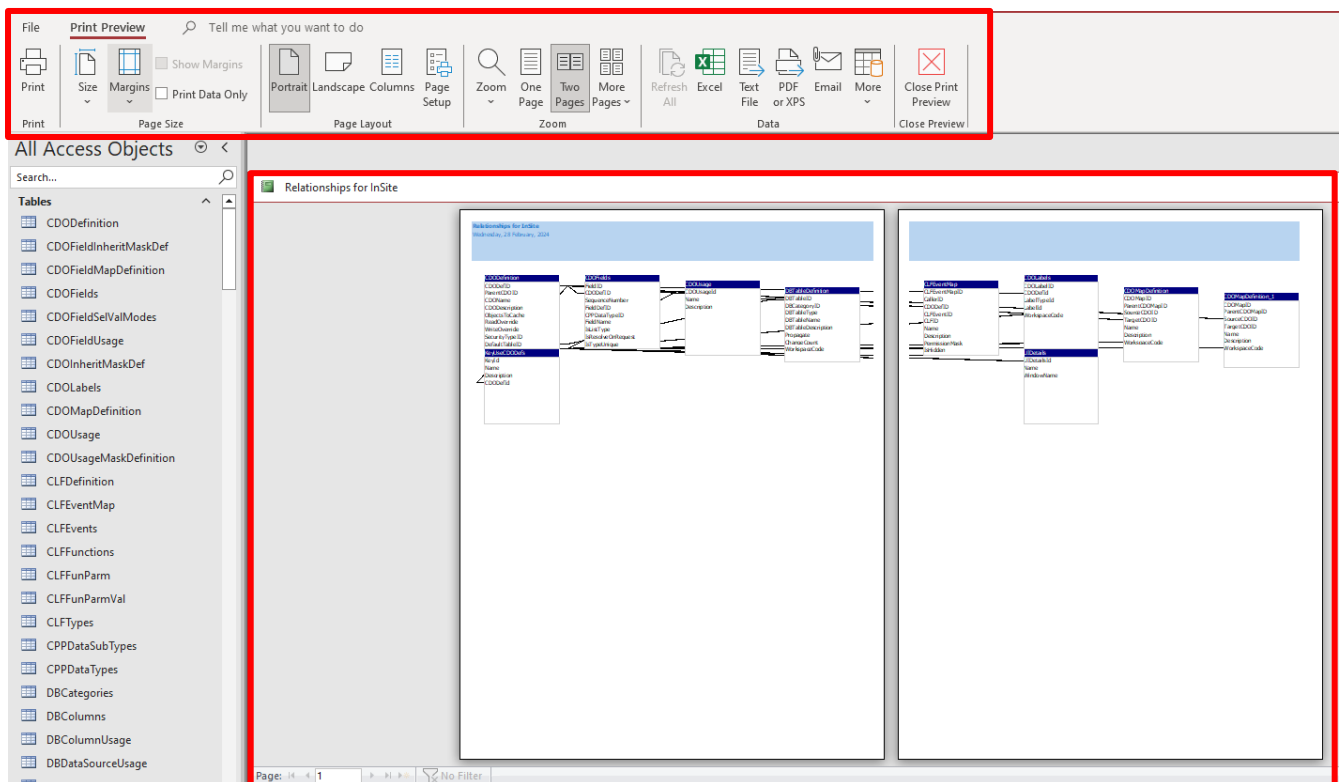


3.3.3 Export

To export the Entity Relationship Diagram (ERD), select **“Relationship Report”** on the menu bar under **“Relationships Design”** tab. It would generate an ERD report on the selected tables and its associated relationships.



In this section, users can specify the options for the report, including its size, margin, printing preferences (portrait or landscape) and viewing options. It also allows users to select the ERD export method, whether in Excel, text file, PDF and etcs.



4 Generate ERD by Microsoft Visio

4.1 Introduction

Microsoft Visio is a diagramming and vector graphics application and is part of the Microsoft 365 family. It provides a user-friendly interface with a wide range of shapes and templates to assist users in creating professional-looking visual representations of information.

One of the visual representations includes Entity Relationship Diagram (ERD), which this document will demonstrate on generating ERD through Microsoft Access and Microsoft SQL Server.

4.2 Visio with Microsoft SQL Server

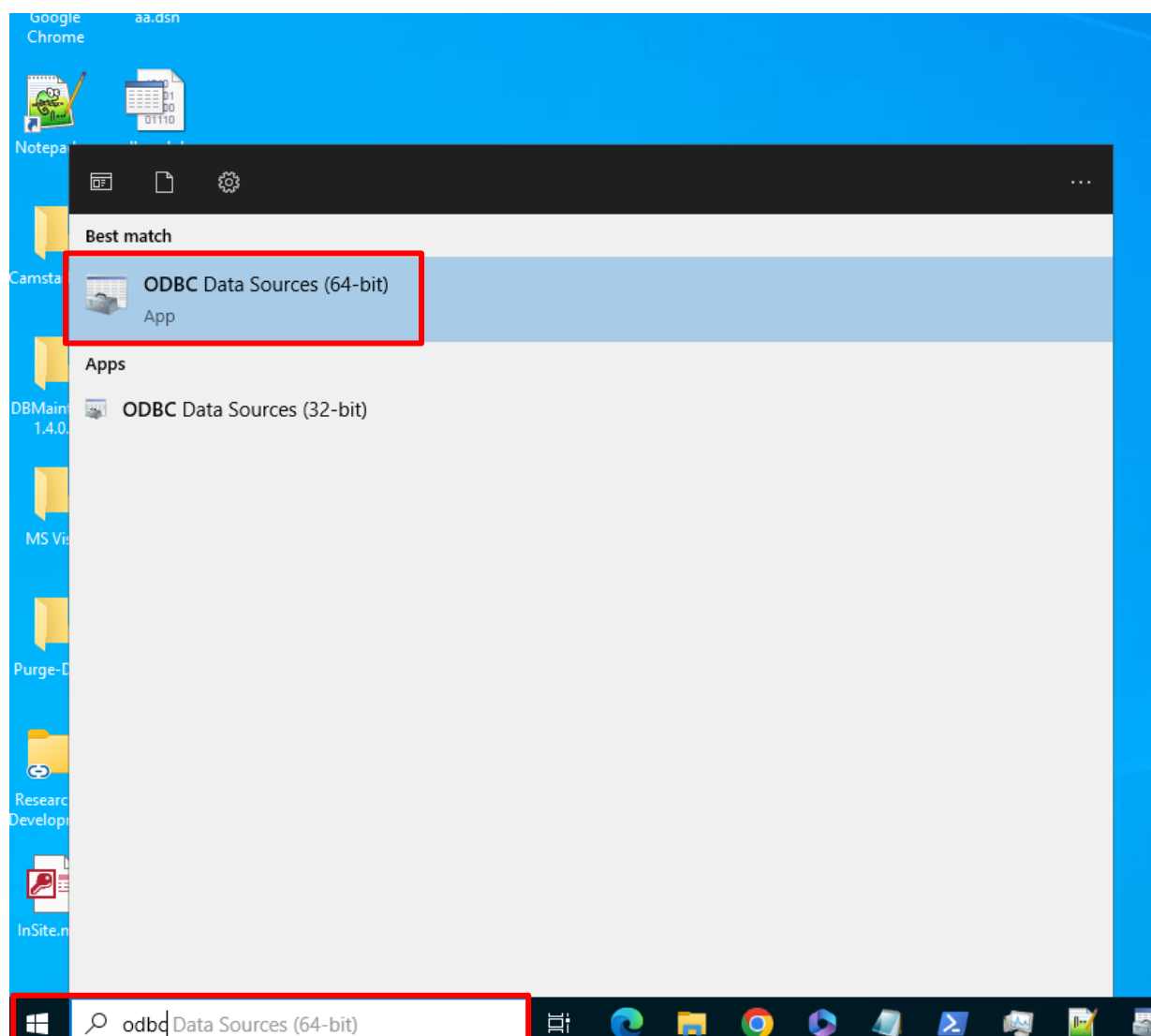
4.2.1 Prerequisites

- Microsoft Visio Professional 2019
- Microsoft ODBC Driver 17 for SQL Server
- Microsoft SQL Server 2019

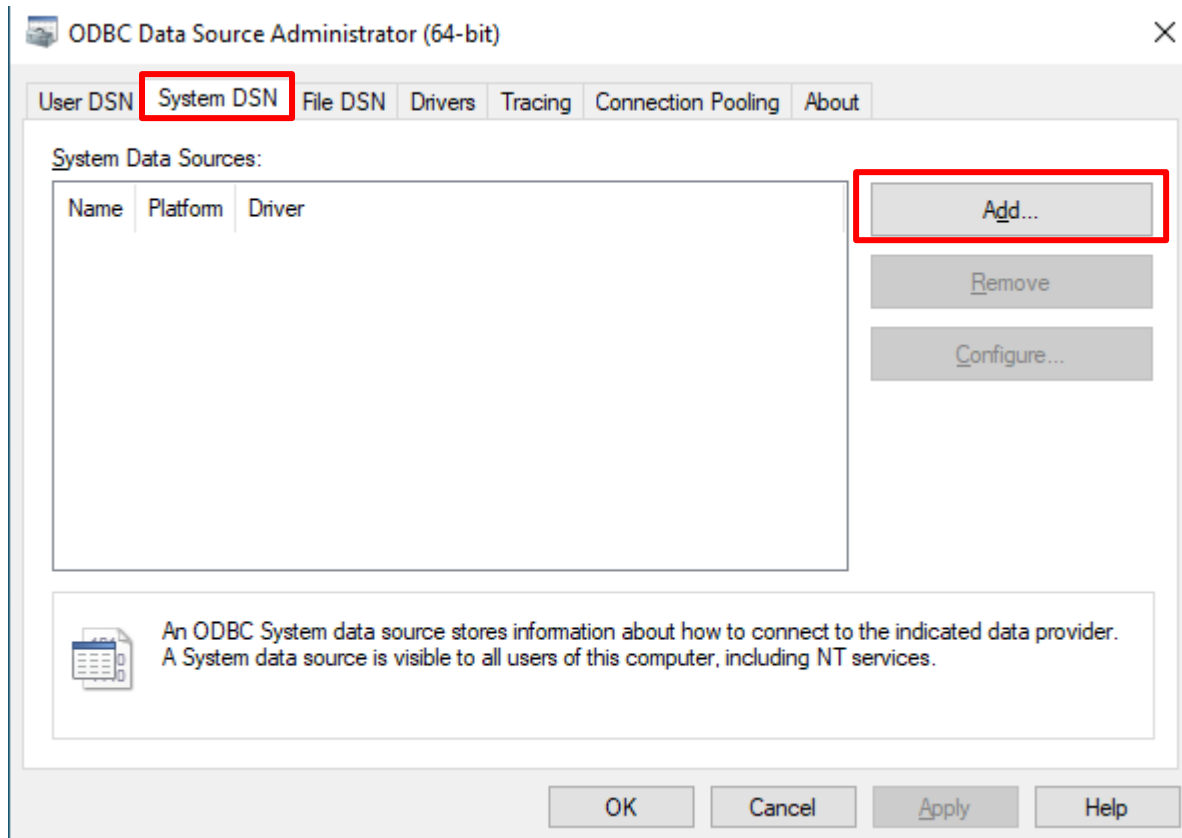
4.2.2 Demonstration

4.2.2.1 Data Source Setup

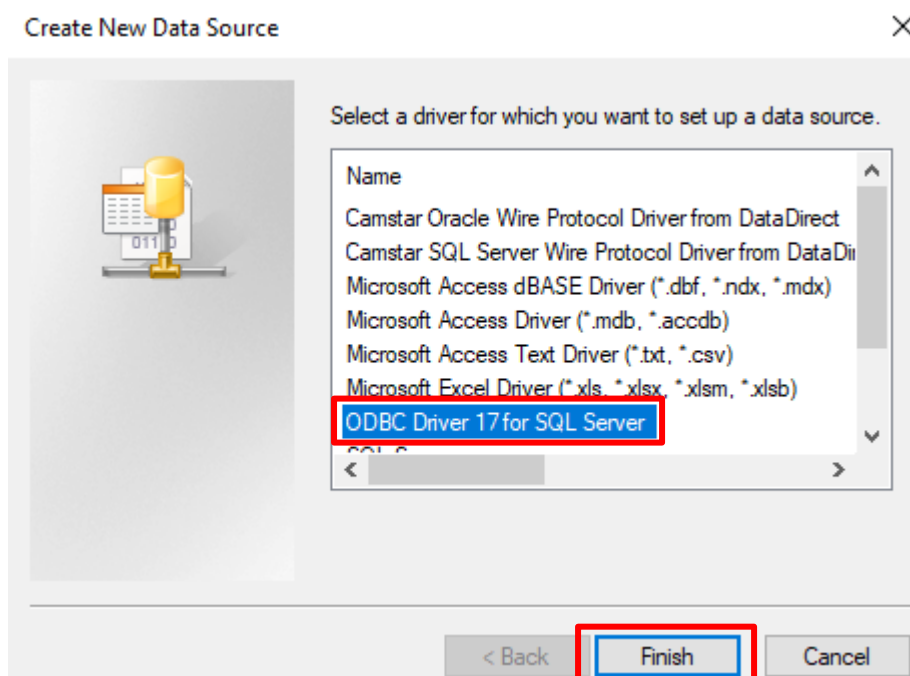
Go to the **Windows Search Bar**, and search “**ODBC**”, then select “**ODBC Data Sources (64-bit)**”.



Select **"System DSN"** and select **"Add..."**.



Select **"ODBC Driver 17 for SQL Server"**, then click **"Finish"**.



Enter the name of the New Data Source and the SQL Server name to connect, then click “**Next**”. (In this example, **OPCENTERCORE231** is the Name and the SQL Server.

Create a New Data Source to SQL Server

This wizard will help you create an ODBC data source that you can use to connect to SQL Server.

What name do you want to use to refer to the data source?

Name:

How do you want to describe the data source?

Description:

Which SQL Server do you want to connect to?

Server:

Click “**Next >**” for authentication with Integrated Windows authentication. Users can opt for other options for the authentication.

Create a New Data Source to SQL Server

How should SQL Server verify the authenticity of the login ID?

☒ With Integrated Windows authentication.

SPN (Optional):

☐ With Azure Active Directory Integrated authentication.

☐ With SQL Server authentication using a login ID and password entered by the user.

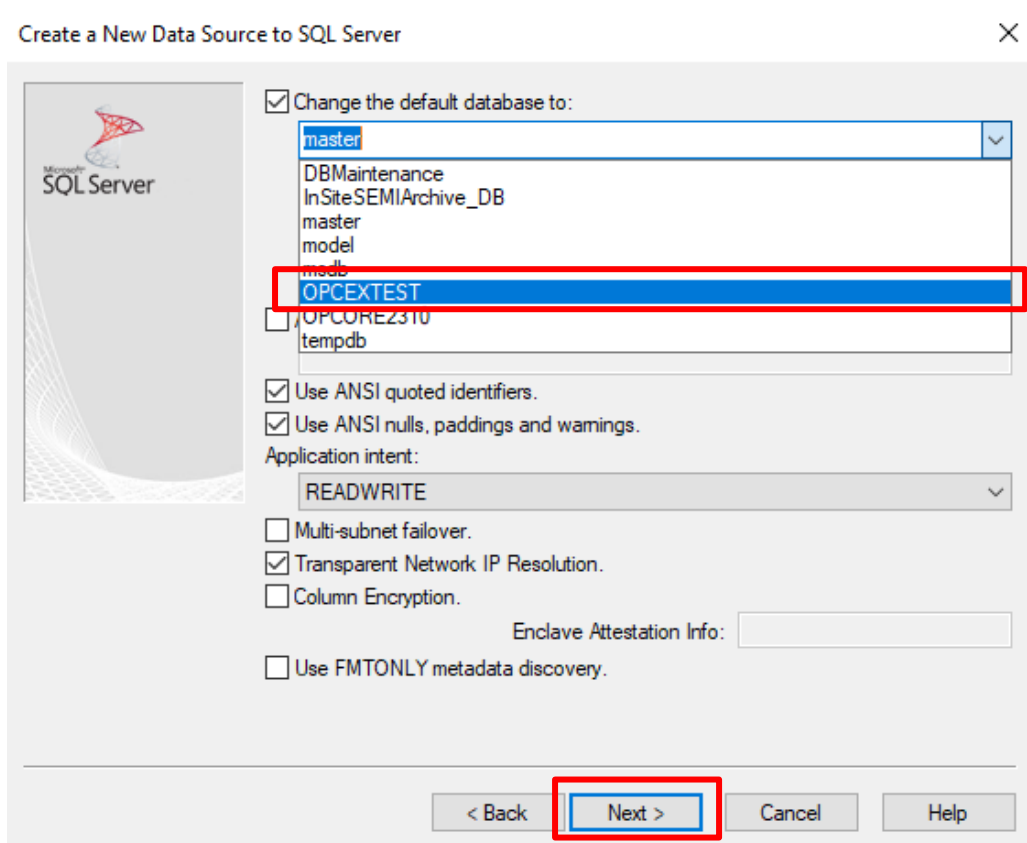
☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.

☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.

Login ID:

Password:

Select the checkbox for “**Change the default database to:**” and choose the target database to generate ERD, then click “**Next >**”. (In this example, **OPCEXTEST** is the target database.)



Microsoft SQL Server

☒ Change the default database to:

master
DBMaintenance
InSiteSEMIArchive_DB
master
model
model
OPCEXTEST
OPCOREZ310
tempdb

☐ Use ANSI quoted identifiers.
☒ Use ANSI nulls, paddings and warnings.

Application intent:
READWRITE

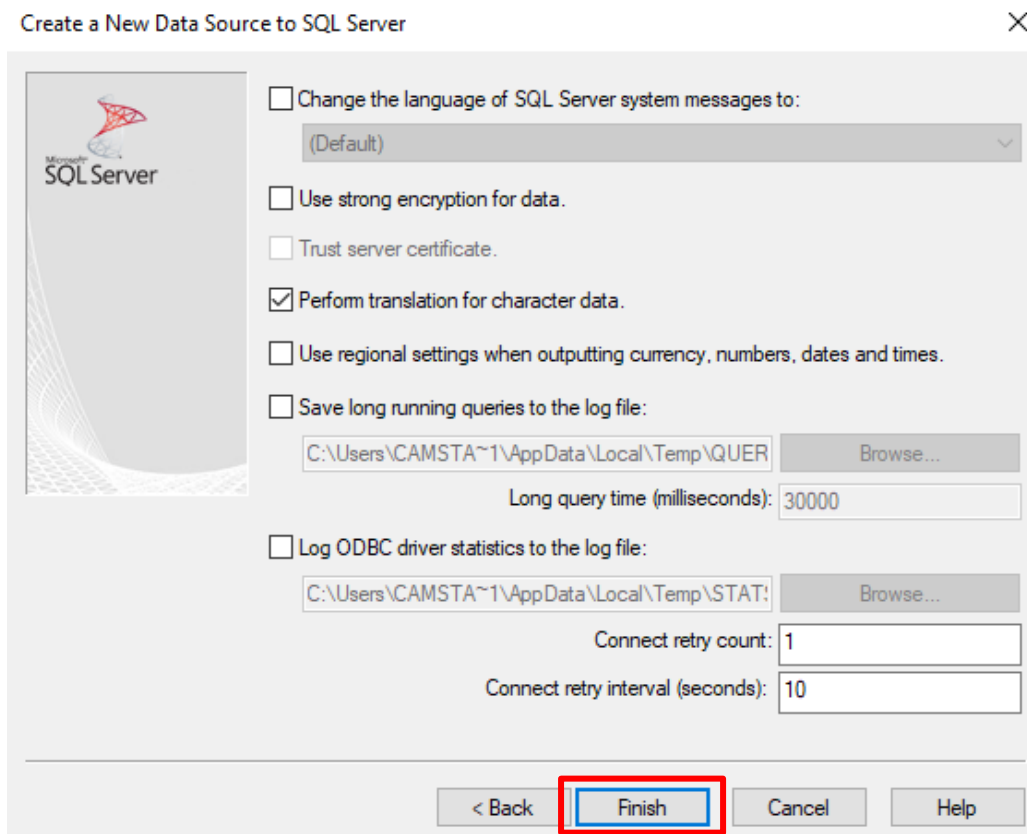
☐ Multi-subnet failover.
☒ Transparent Network IP Resolution.
☐ Column Encryption.

Enclave Attestation Info:

☐ Use FMONLY metadata discovery.

< Back **Next >** Cancel Help

Select “**Finish**” on this window.



Microsoft SQL Server

☐ Change the language of SQL Server system messages to:
(Default)

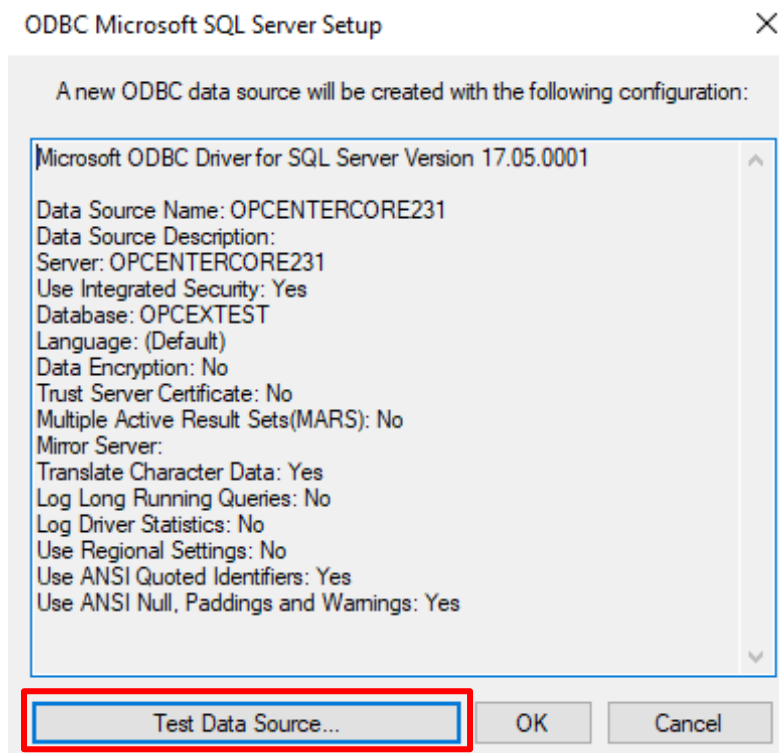
☐ Use strong encryption for data.
☐ Trust server certificate.
☒ Perform translation for character data.
☐ Use regional settings when outputting currency, numbers, dates and times.

☐ Save long running queries to the log file:
C:\Users\CAMSTA~1\AppData\Local\Temp\QUER Browse...
Long query time (milliseconds): 30000

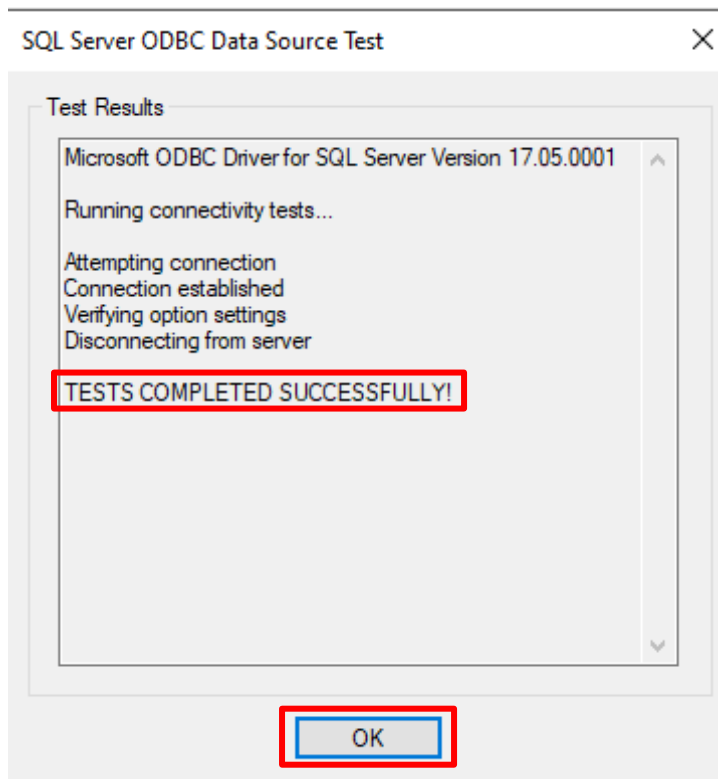
☐ Log ODBC driver statistics to the log file:
C:\Users\CAMSTA~1\AppData\Local\Temp\STAT: Browse...
Connect retry count: 1
Connect retry interval (seconds): 10

< Back **Finish** Cancel Help

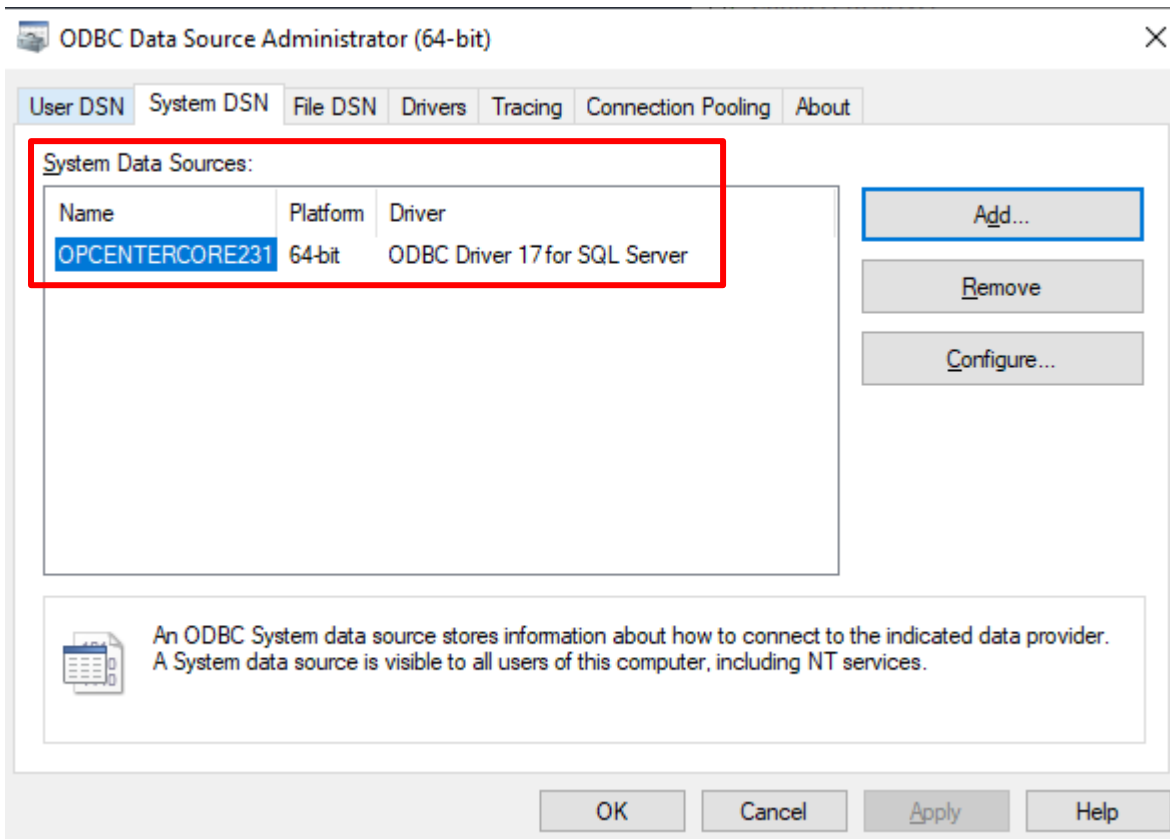
Then, the following pop-up windows will appear. Select “**Test Data Source...**” to validate the connection to the Microsoft SQL Server database.



If the connection is successfully established, it will show the following results. Then, click on “**OK**”.

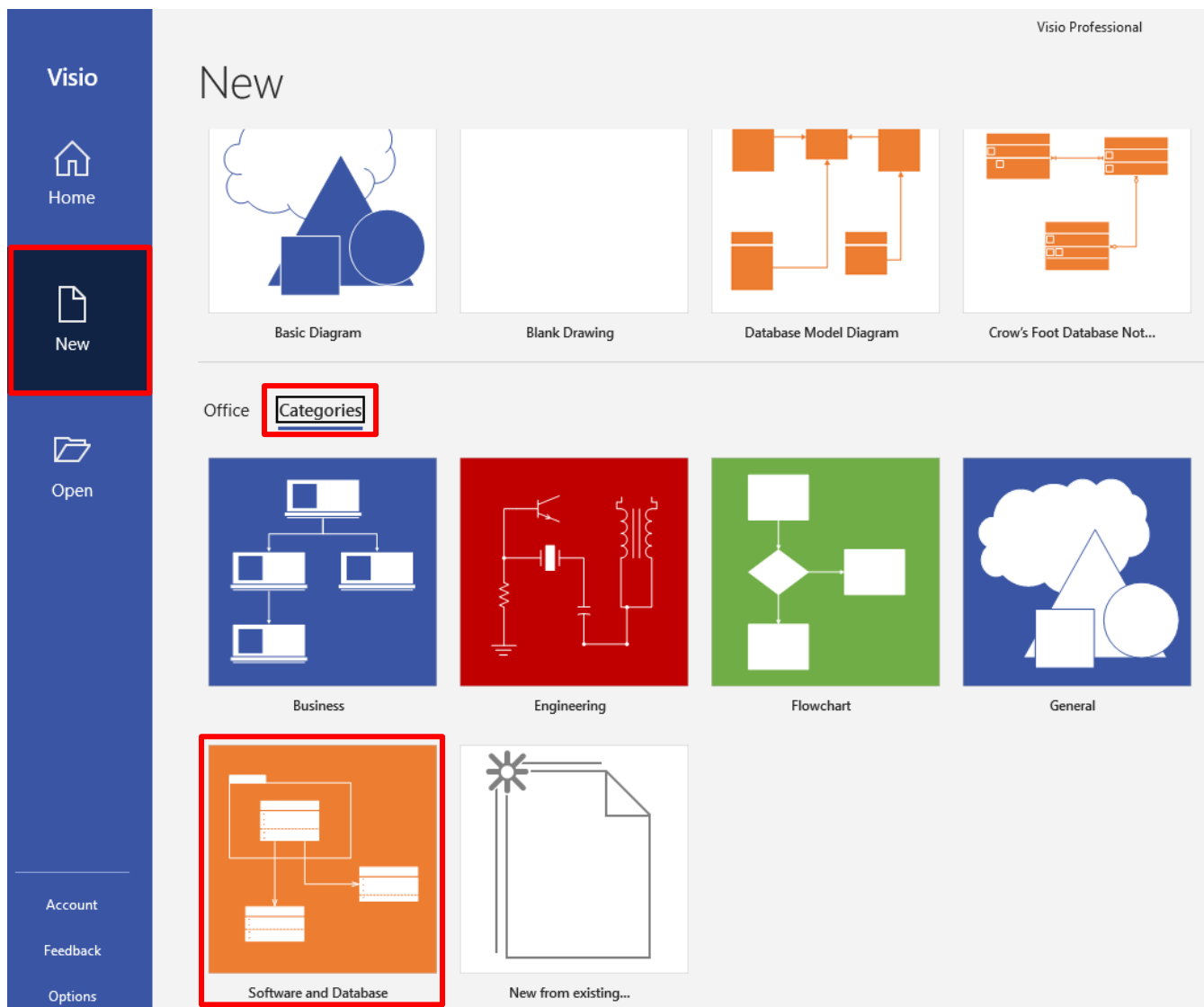


If the previous steps are done correctly, a new System Database Source will appear under “**System DSN**” named **OPCENTERCORE231**. Now we can proceed to generate the ERD with Microsoft Visio.

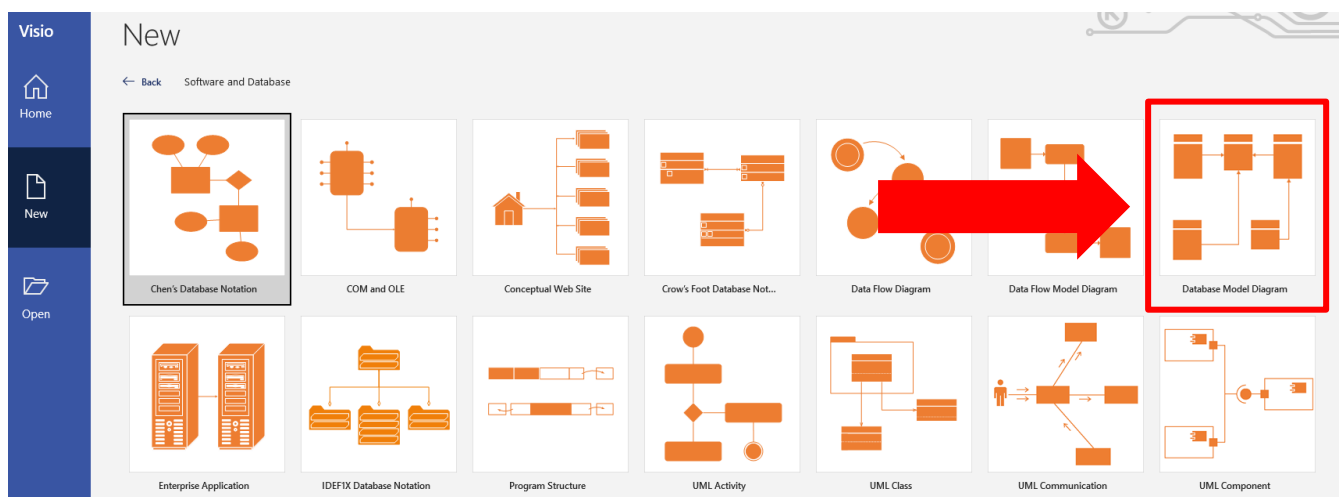


4.2.2.2 Reverse Engineering Setup

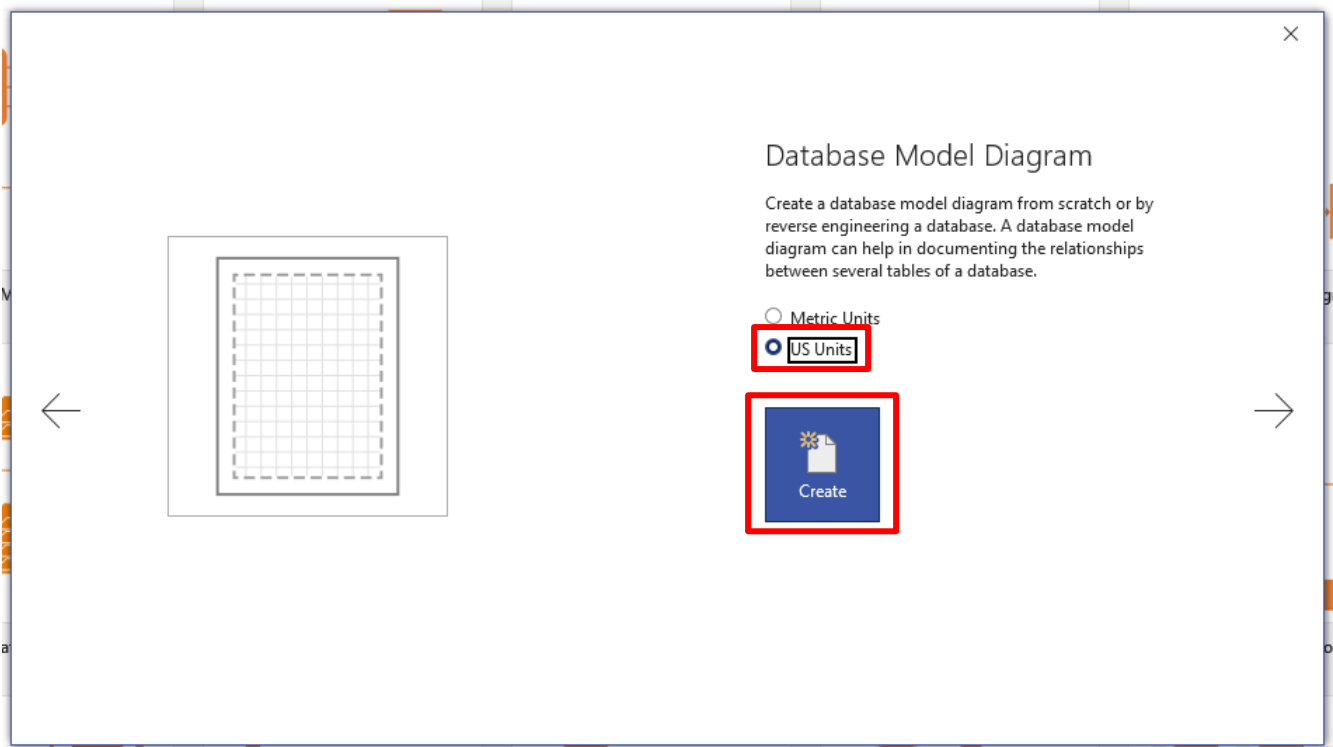
Open Microsoft Visio and select “**New**”, then select “**Categories**” and select “**Software and Database**”.



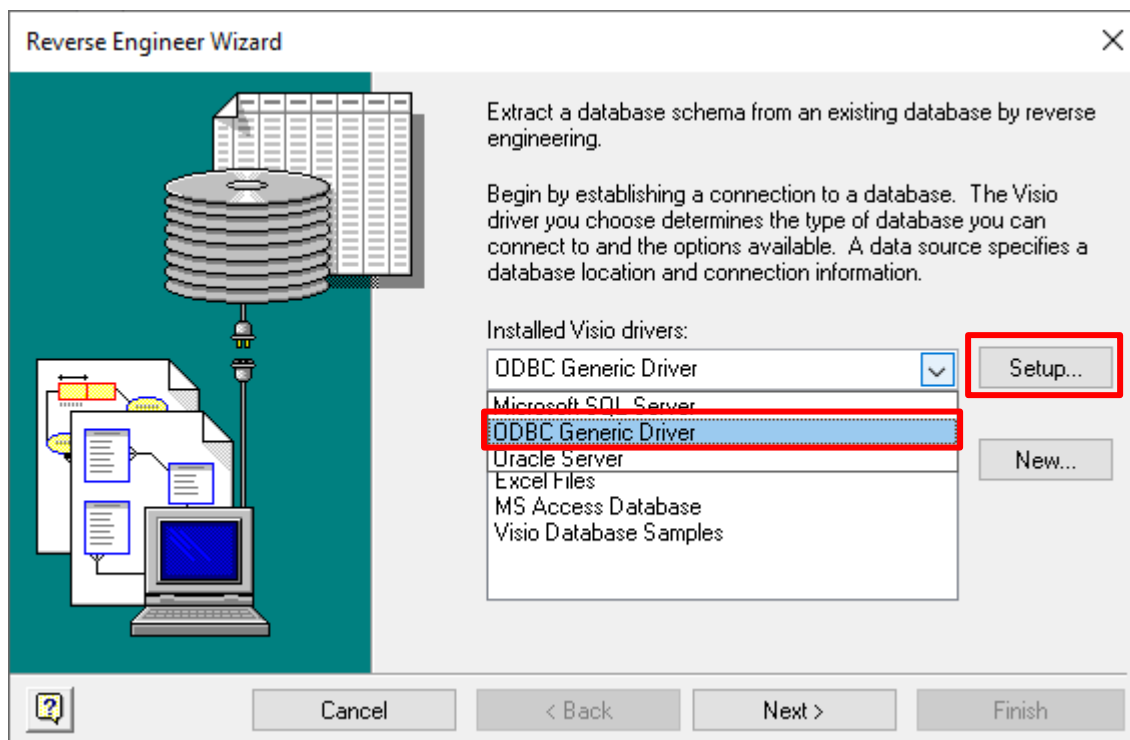
There are many Software and Database diagrams available in Microsoft Visio. To generate an Entity Relationship Diagram (ERD), select “Database Model Diagram”.



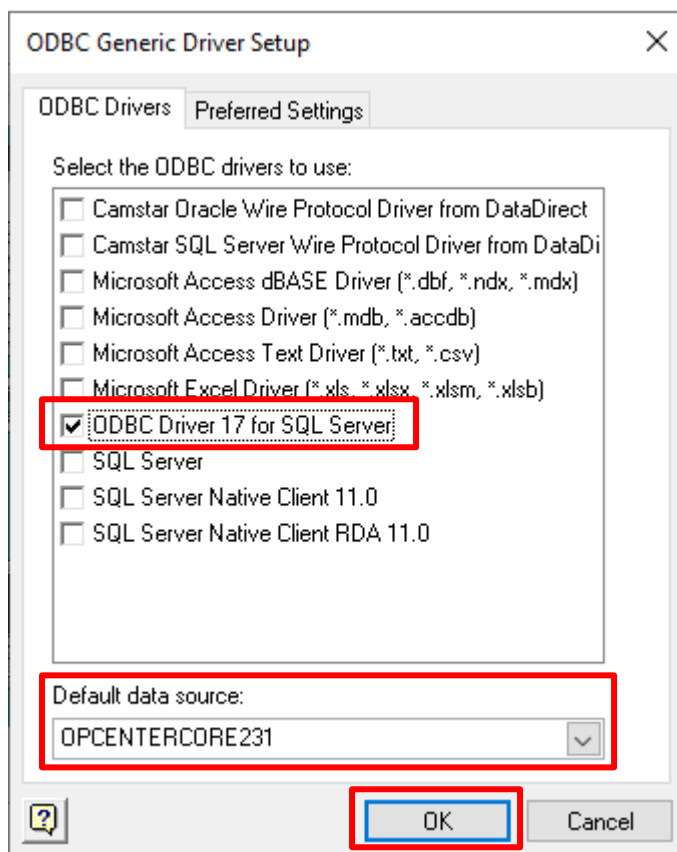
On the following pop-up window, select the “**US Units**” option and click “**Create**”.



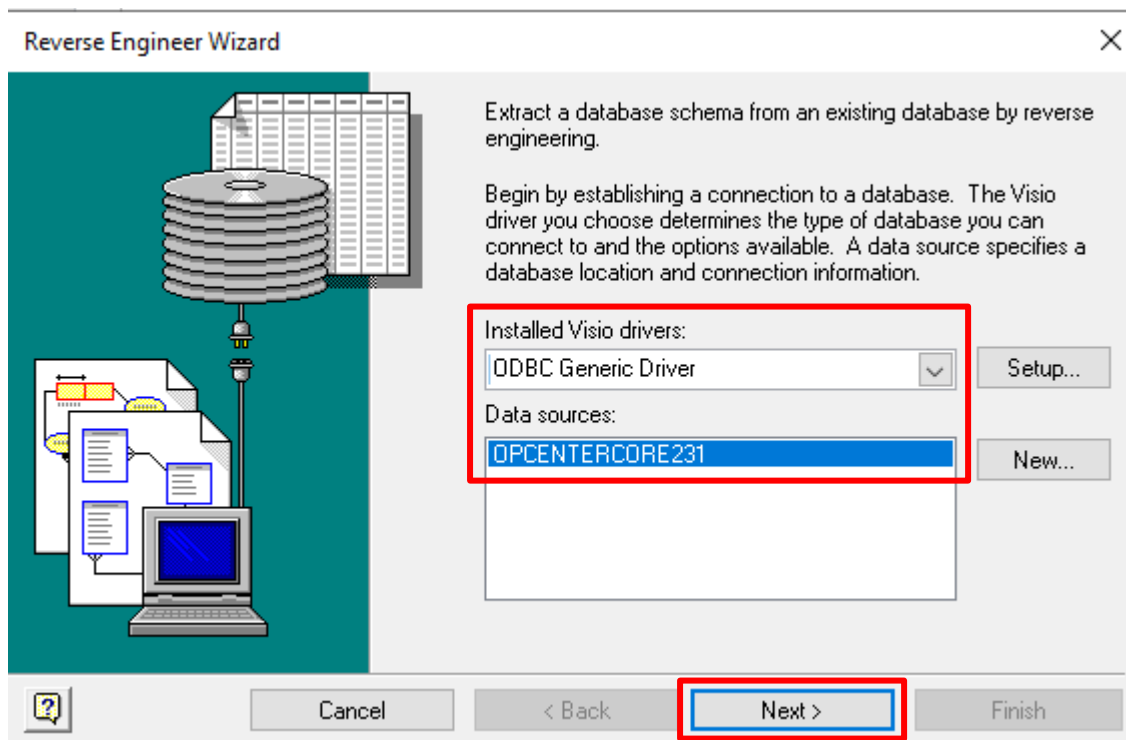
On the Reverse Engineer Wizard, select “**ODBC Generic Driver**” from the dropdown menu and select “**Setup...**”.



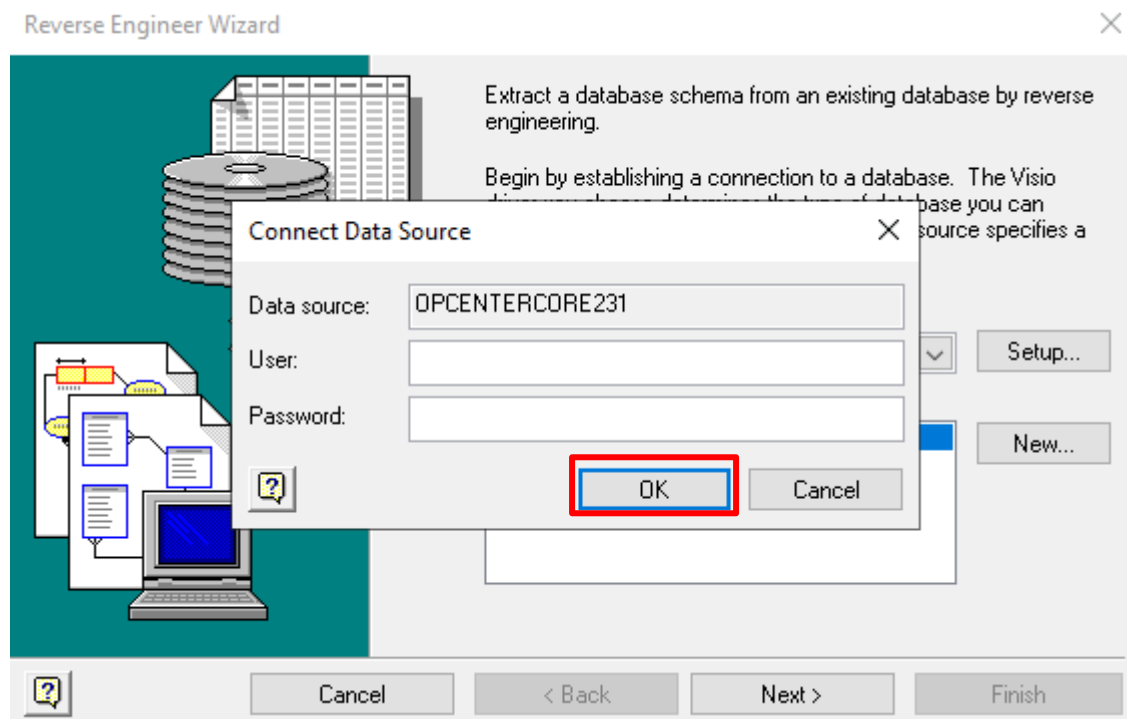
On the following pop-up window, **uncheck all other ODBC drivers** and **check only “ODBC Driver 17 for SQL Server”**. After selecting the checkbox, the default data source will automatically change to **OPCENTERCORE231** if the Data Source Setup is done correctly. Then, click **“OK”**.



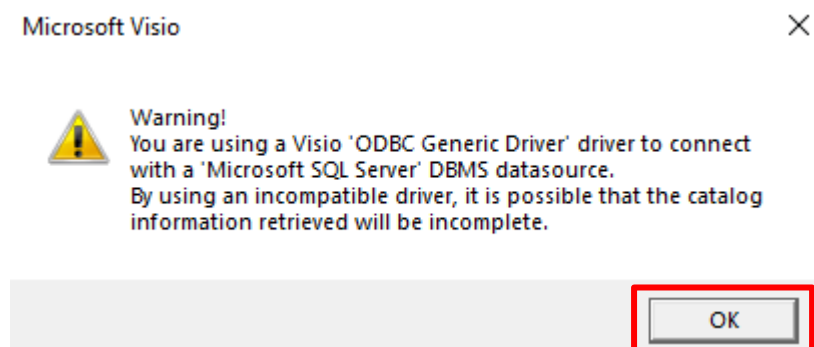
Then, the Reverse Engineer Wizard will be updated accordingly, and then click **“Next >”**.



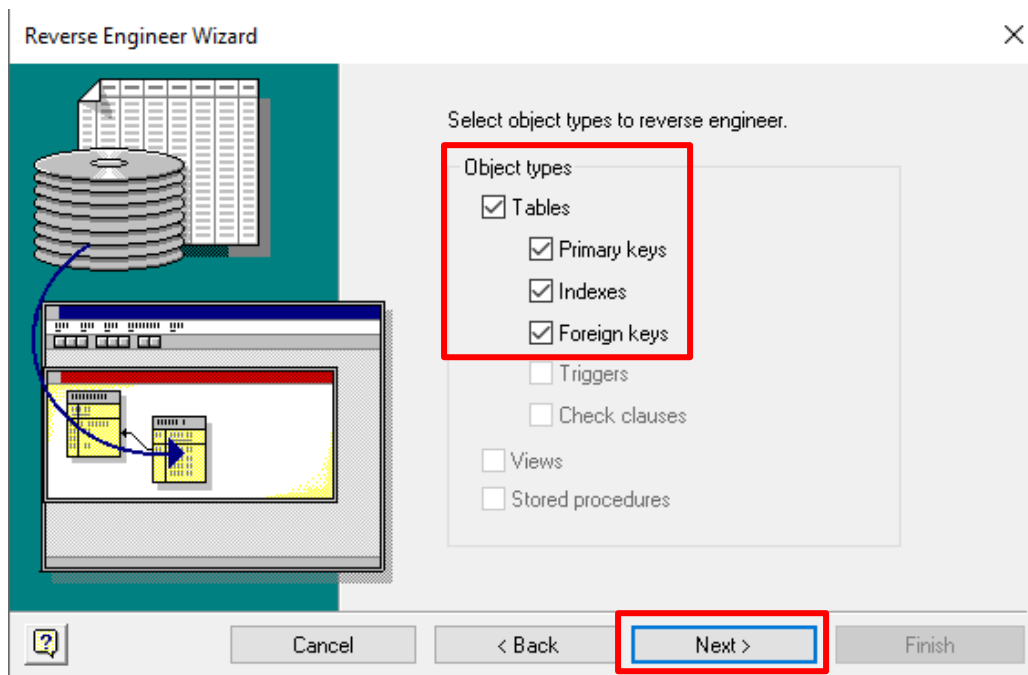
On the Connect Data Source window, click “OK”.



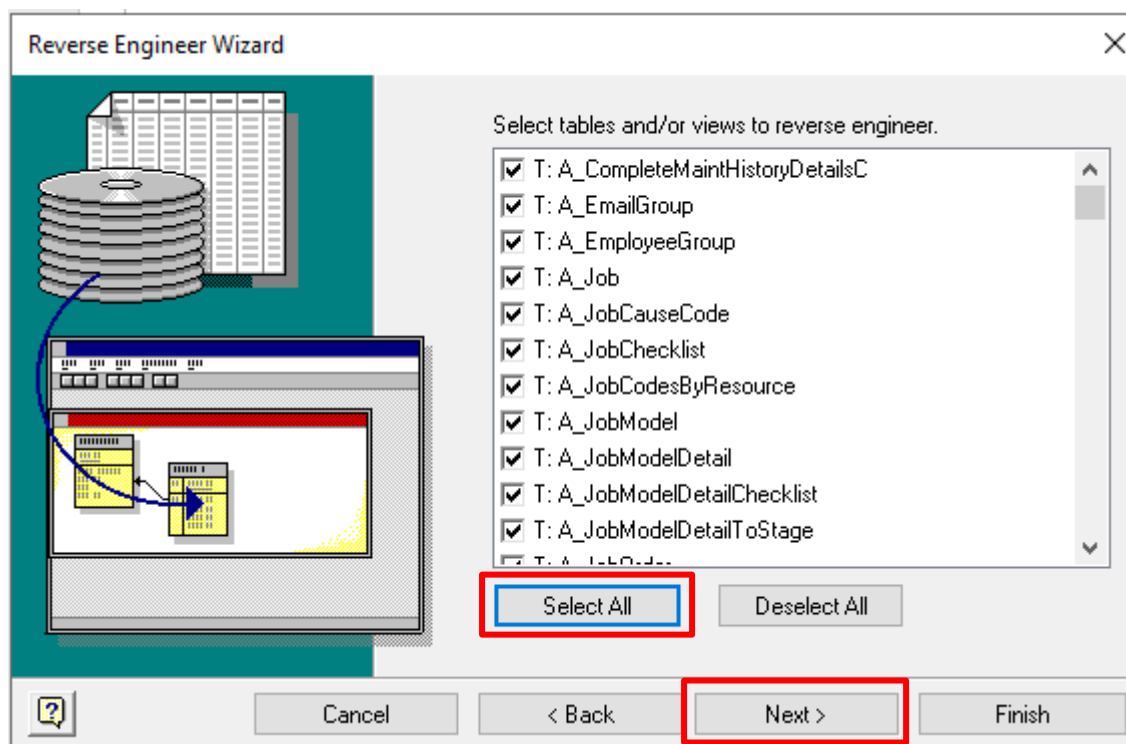
Click “OK” on the following warning window.



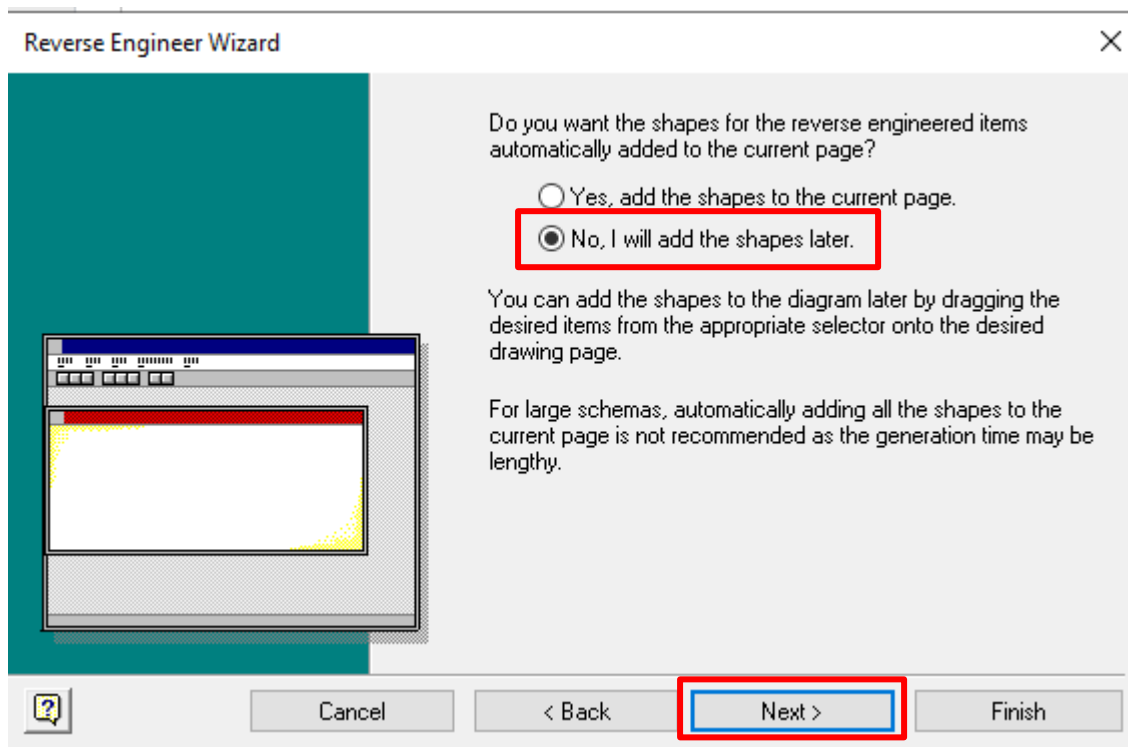
Click on “**Next >**”. Make sure in this window the **Tables**, **Primary Keys**, **Indexes** and **Foreign Keys** are **checked**, else something must be wrong during the Data Source Setup section.



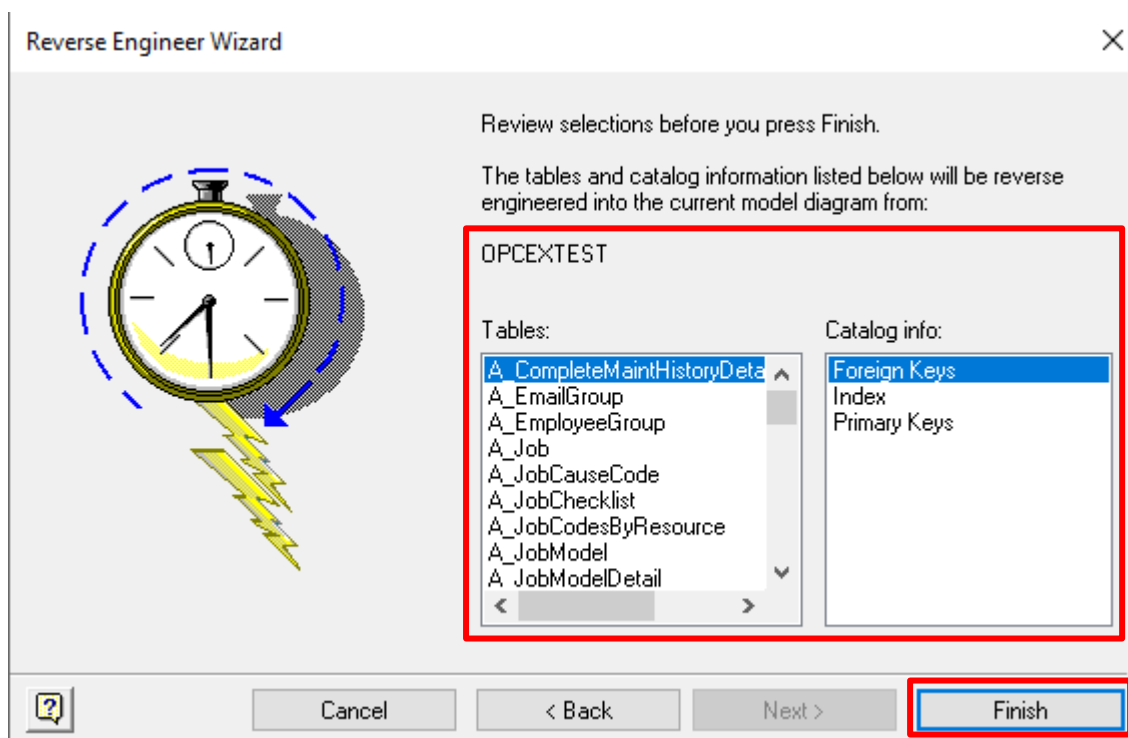
Then, select the tables or views to display the Entity Relationship Diagram (ERD) and click “**Next >**”.



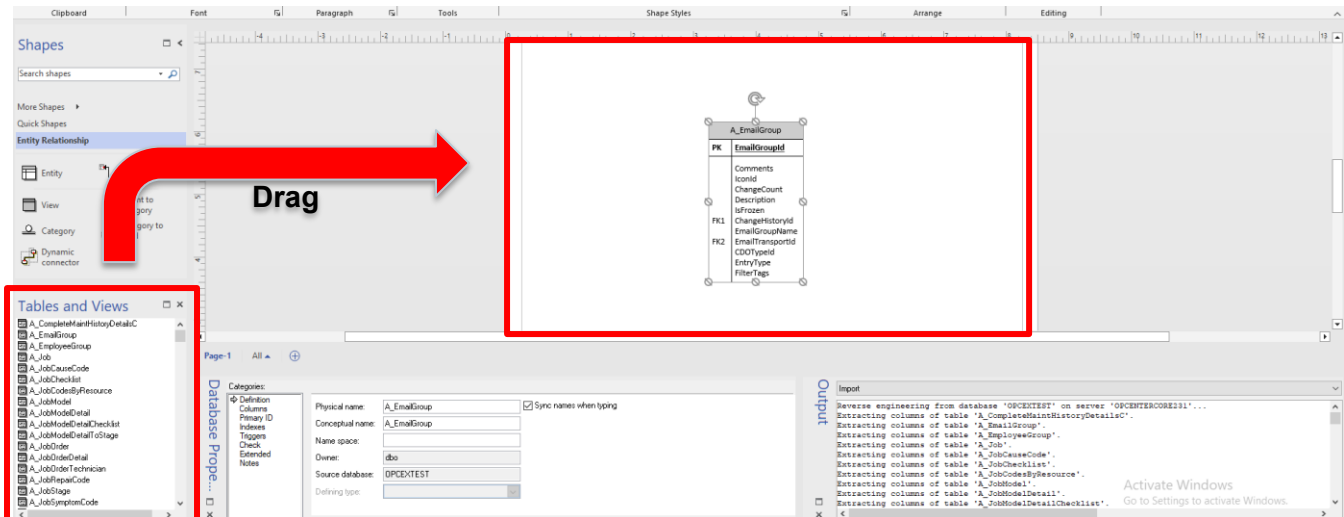
Select the option **“No, I will add the shapes later”** to display it manually, else select **“Yes, add the shapes to the current page.”** to generate the complete ERD once the wizard is completed, then click **“Next >”**.



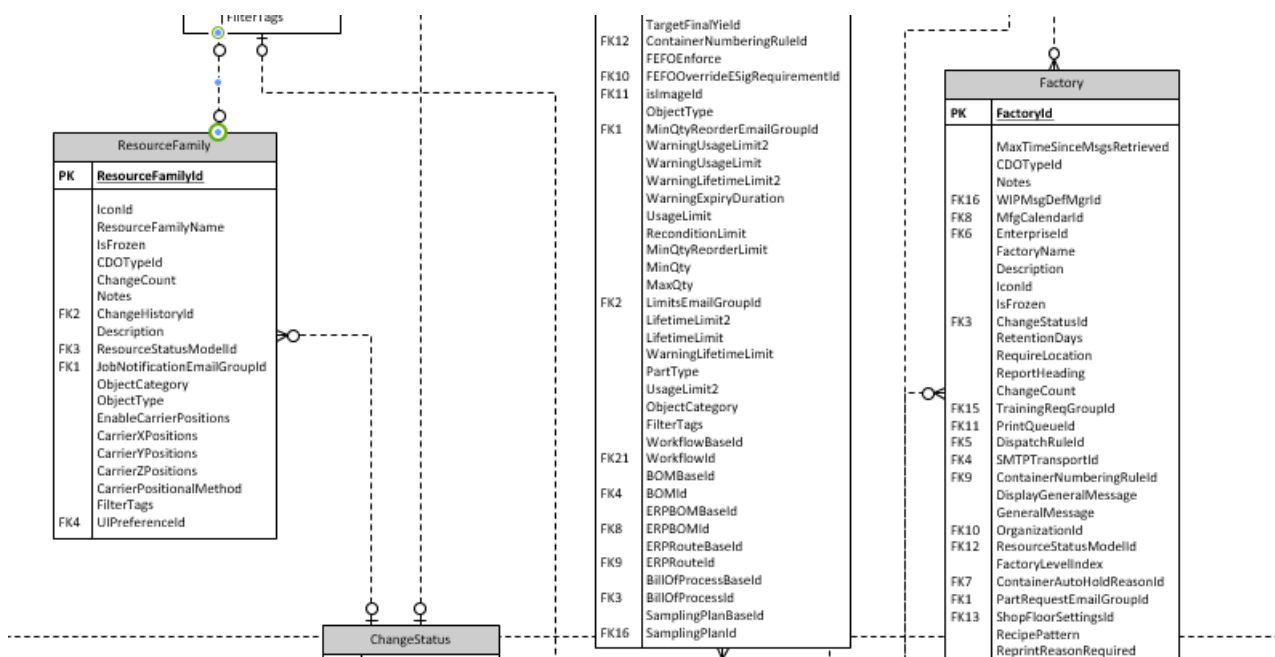
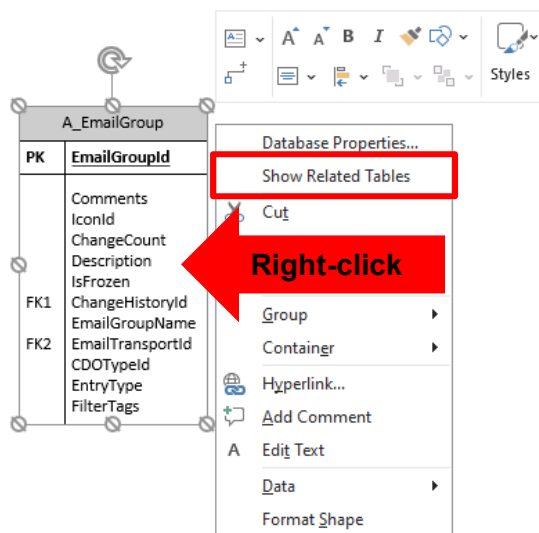
Then, the Reverse Engineer Wizard will display a summary of the selections. Click on **“Finish”**.



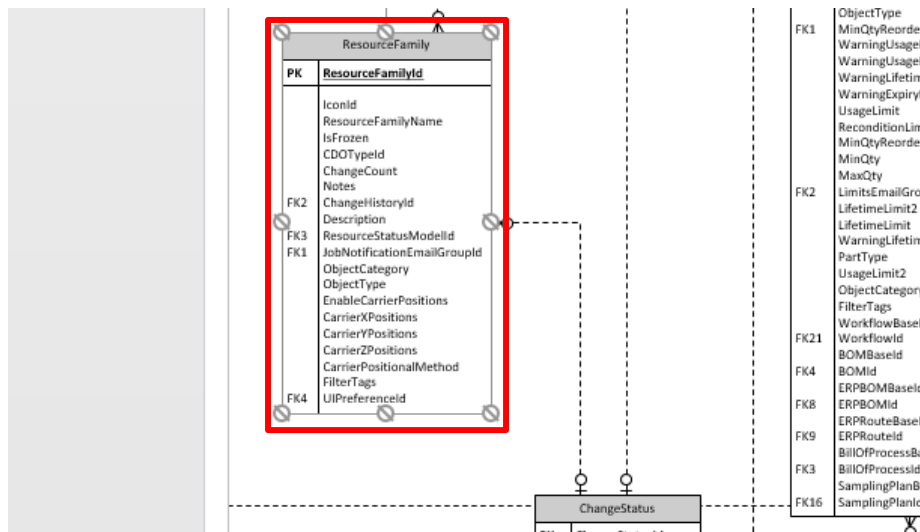
To view the ERD of the table, simply **drag** the table from “**Tables and Views**” to the space in the middle.



Right click on the table and select “**Show Related Tables**” to display its associated entities and relationships.



Users can click on the table or the dotted line to view its **Database Properties** as shown.



Page-1 All +

Database Properties

Categories:

- Definition
- Columns
- Primary ID
- Indexes
- Triggers
- Check
- Extended
- Notes

Physical name: ResourceFamily ☒ Sync names when typing

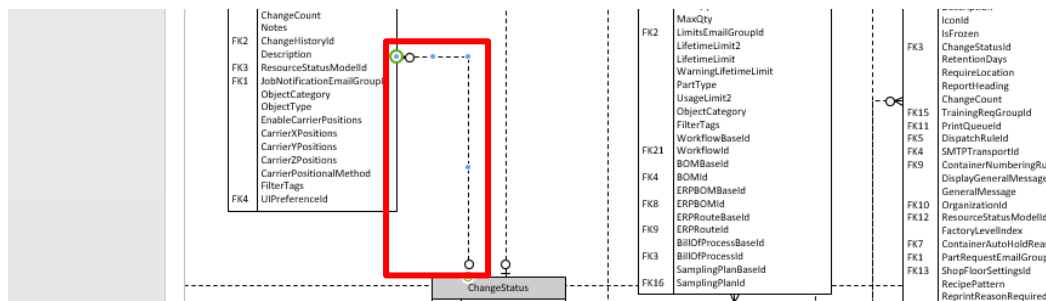
Conceptual name: ResourceFamily

Name space:

Owner: dbo

Source database: OPCEXTEST

Defining type:



Page-1 All +

Database Properties

Categories:

- Definition
- Name
- Miscellaneous
- Referential Action

Parent: ChangeStatus

Child: ResourceFamily

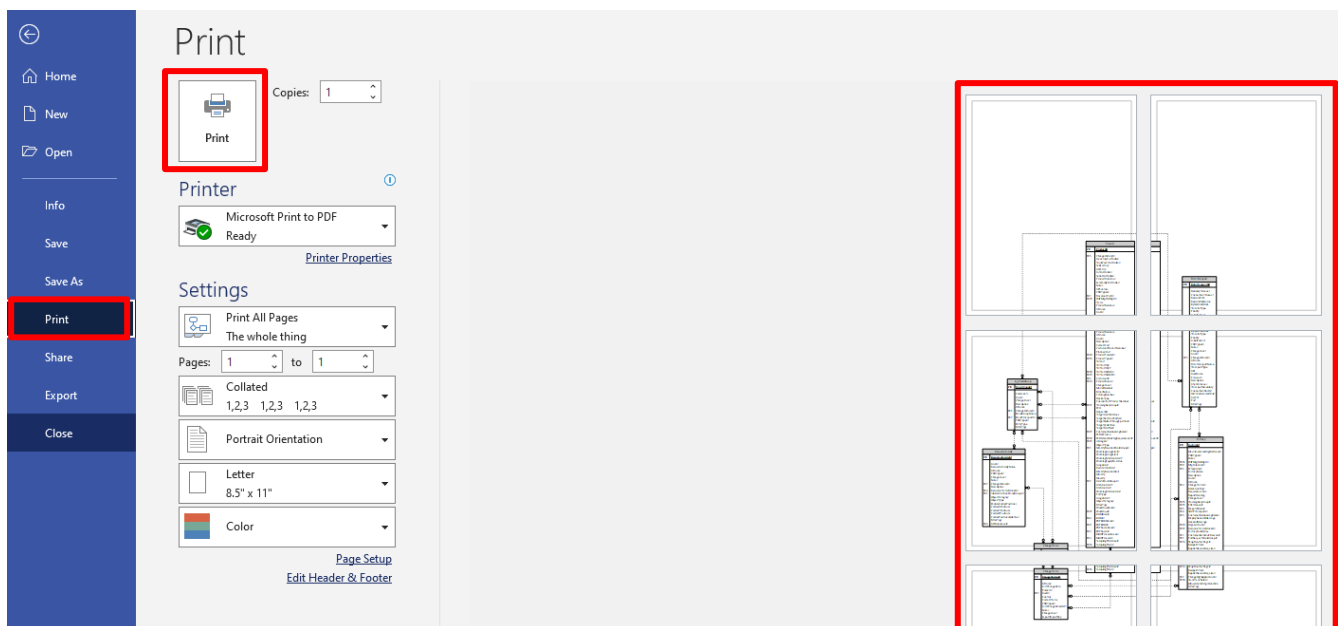
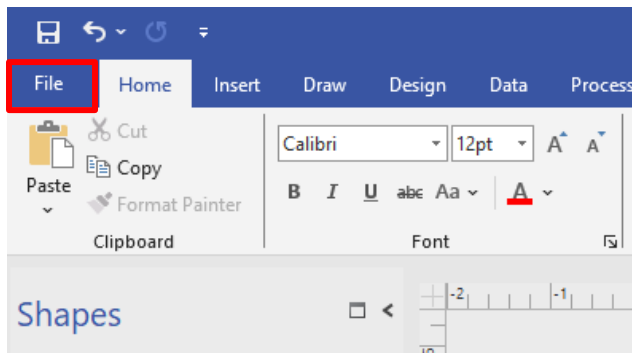
Foreign key role name:

IsFrozen, LastChangeDate, ChangeStatusId, ParentId, UserId, Control, ChangeStatus

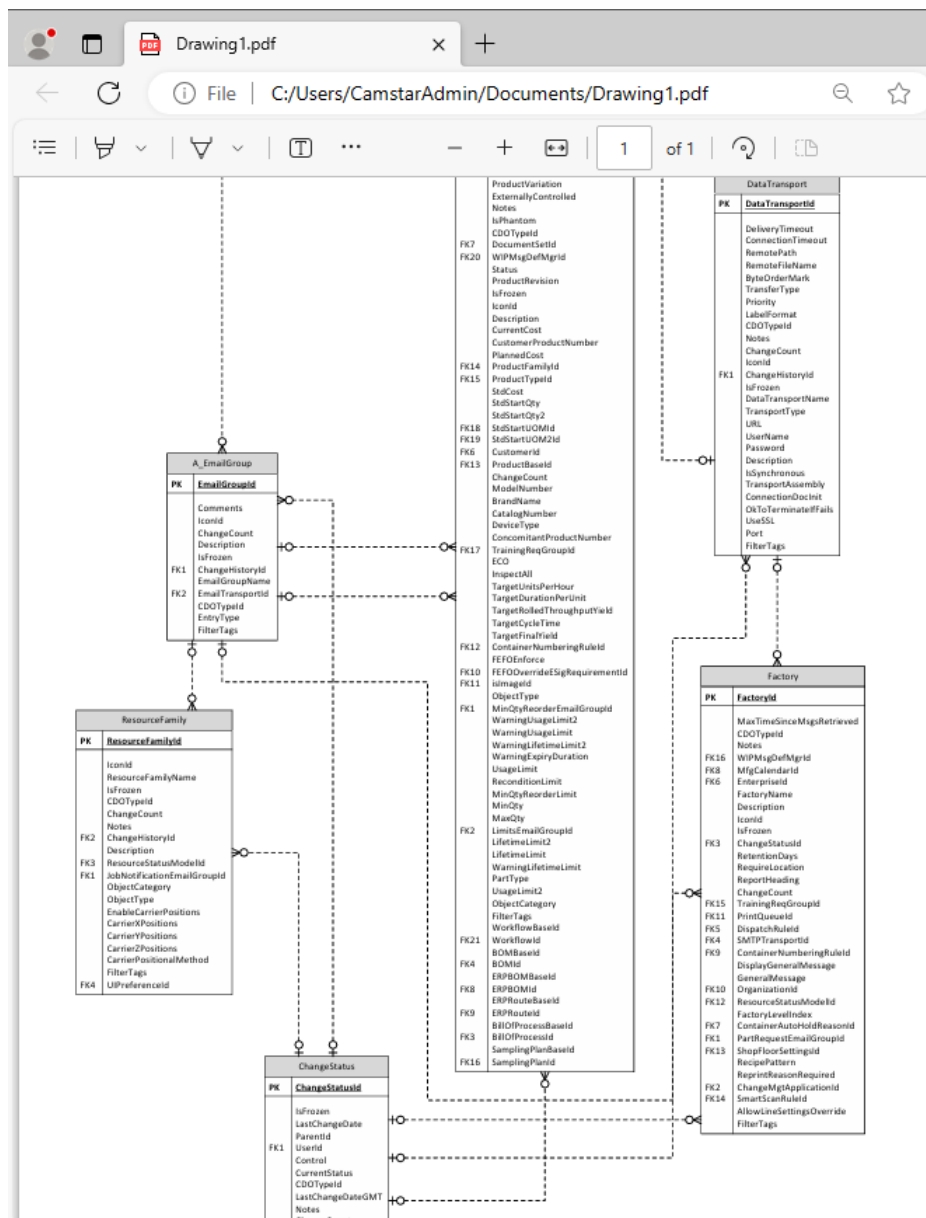
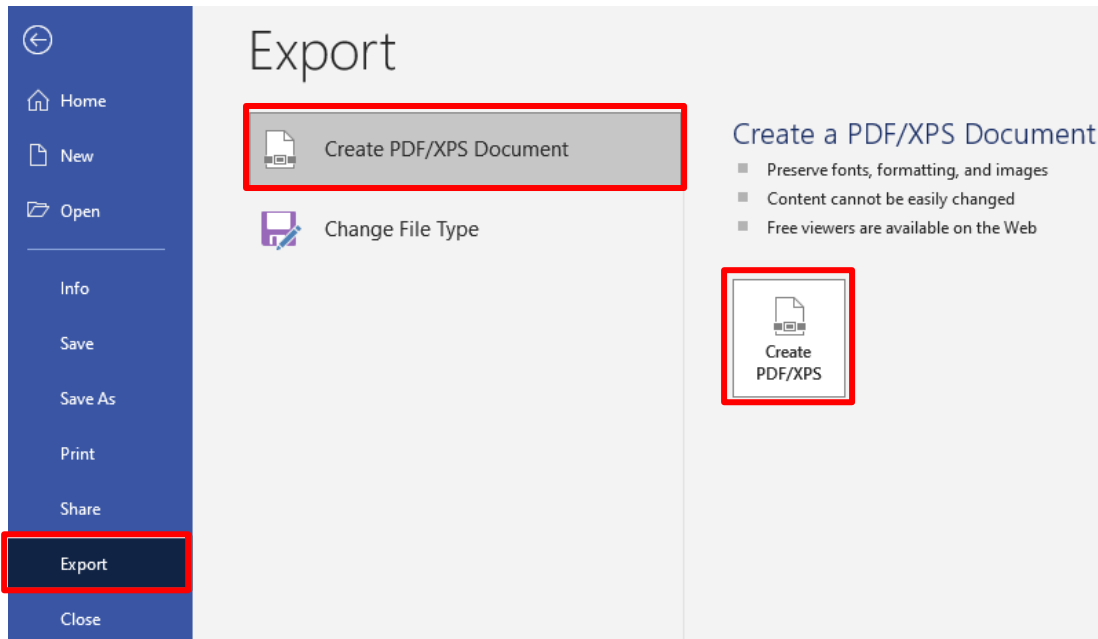
IconId, ResourceFamilyName, IsFrozen, ResourceFamilyId, CDOTypeld, ChangeCount, Notes, ChangeHistoryId, Description, ResourceStatusModelId

4.2.2.3 Export

To print the Entity Relationship Diagram (ERD), click on **“File”** on the menu, then select **“Print”** for the print preview and printing.



To export the Entity Relationship Diagram (ERD) to PDF, click on **Export**, then select **Create PDF/XPS Document** and select **Create PDF/XPS** to save the PDF document.



4.3 Visio with Microsoft Access

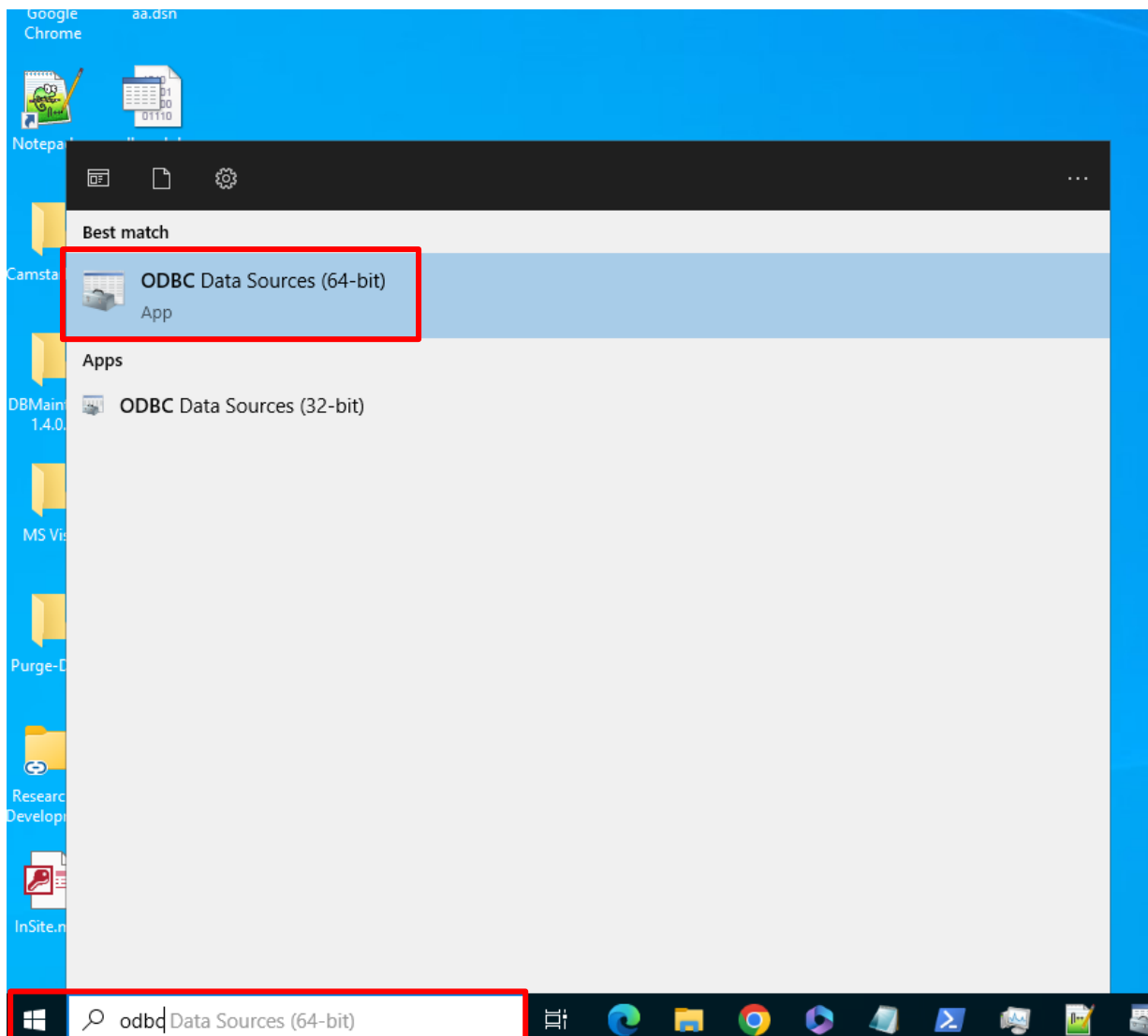
4.3.1 Prerequisites

- Microsoft Visio Professional 2019
- Microsoft Access Driver

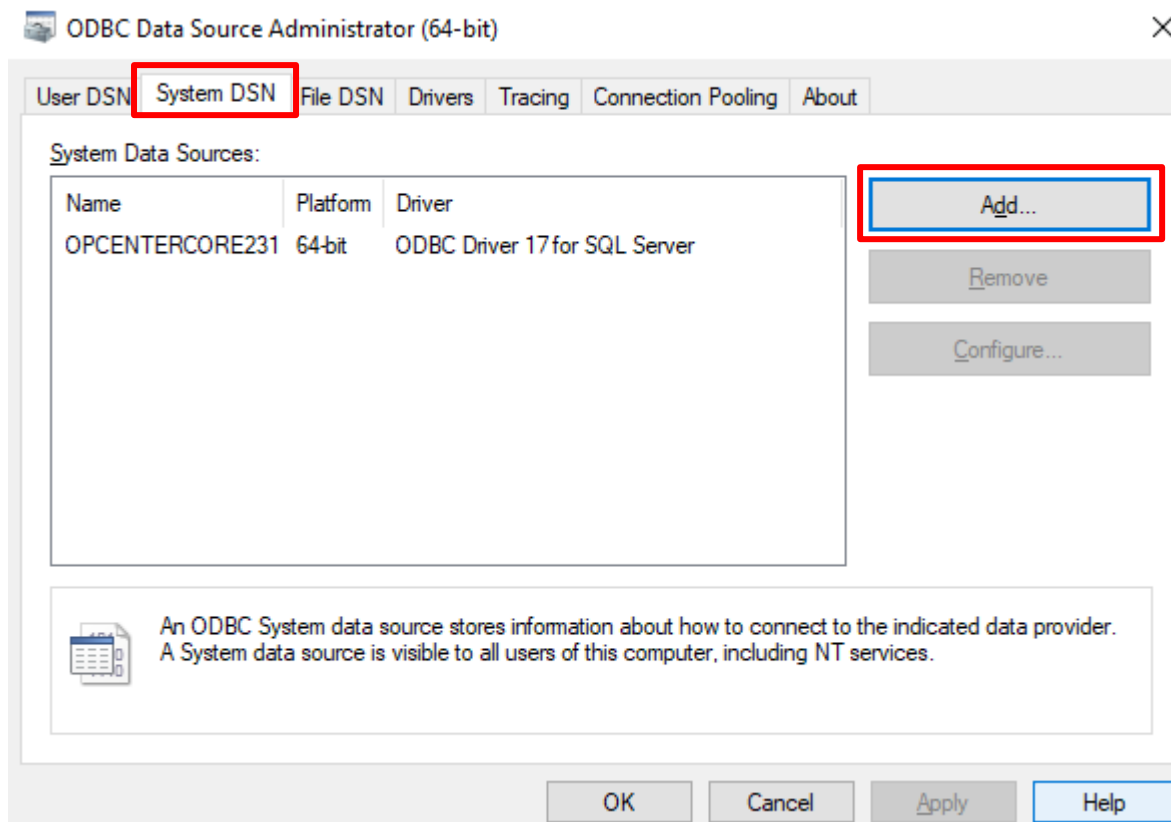
4.3.2 Demonstration

4.3.2.1 Data Source Setup

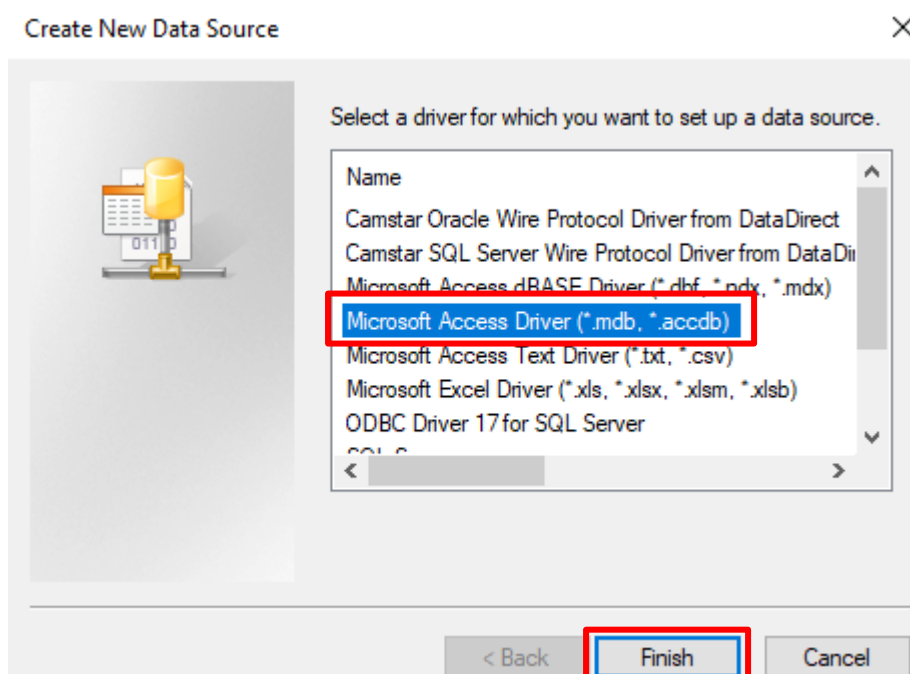
Go to the **Windows Search Bar**, and search “**ODBC**”, then select “**ODBC Data Sources (64-bit)**”.



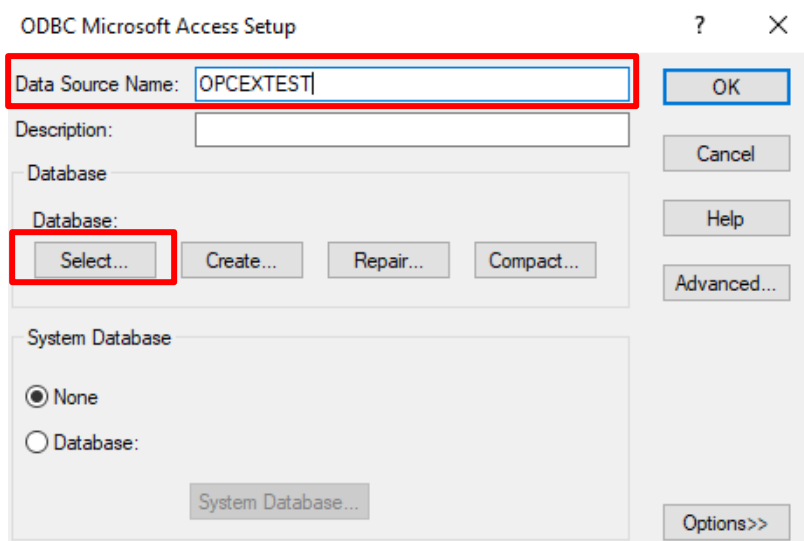
Select **"System DSN"** and select **"Add..."**.



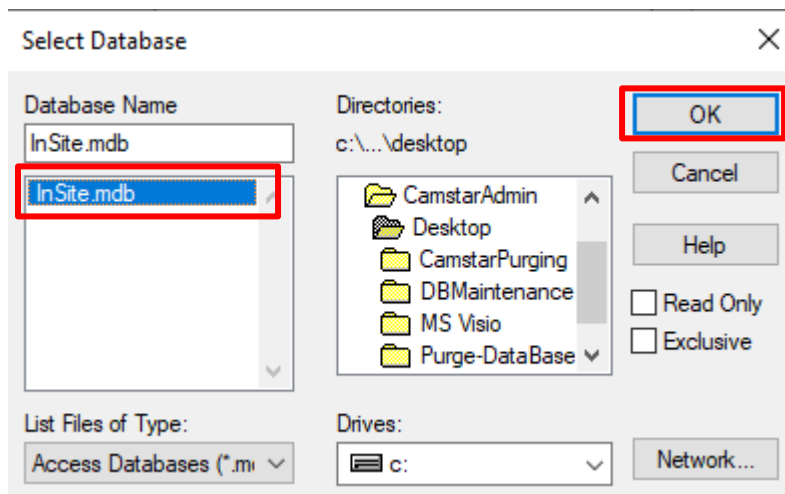
Select **"Microsoft Access Driver (*.mdb, *.accdb)"** and click **"Finish"**.



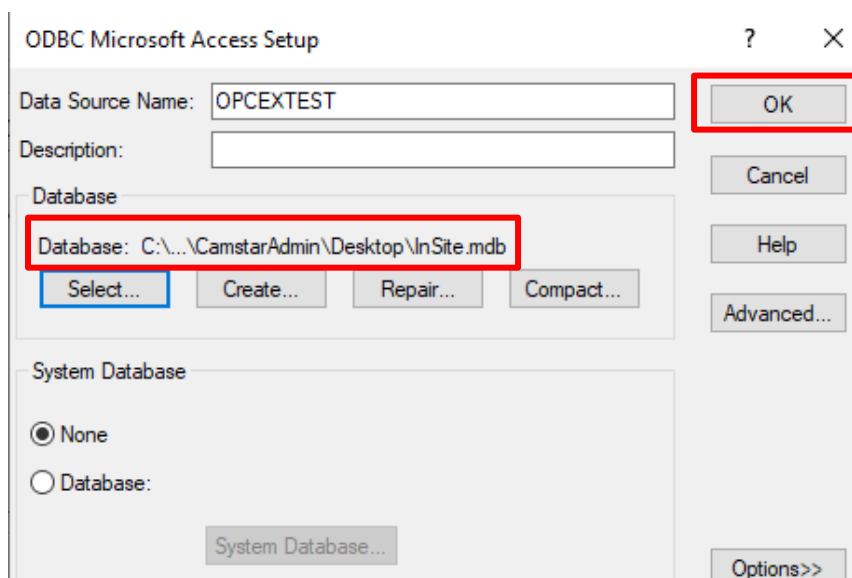
The following pop-up window will appear. Create a new **Data Source Name** (i.e. **OPCEXTEST**) and click on “**Select...**” under the **Database** section.



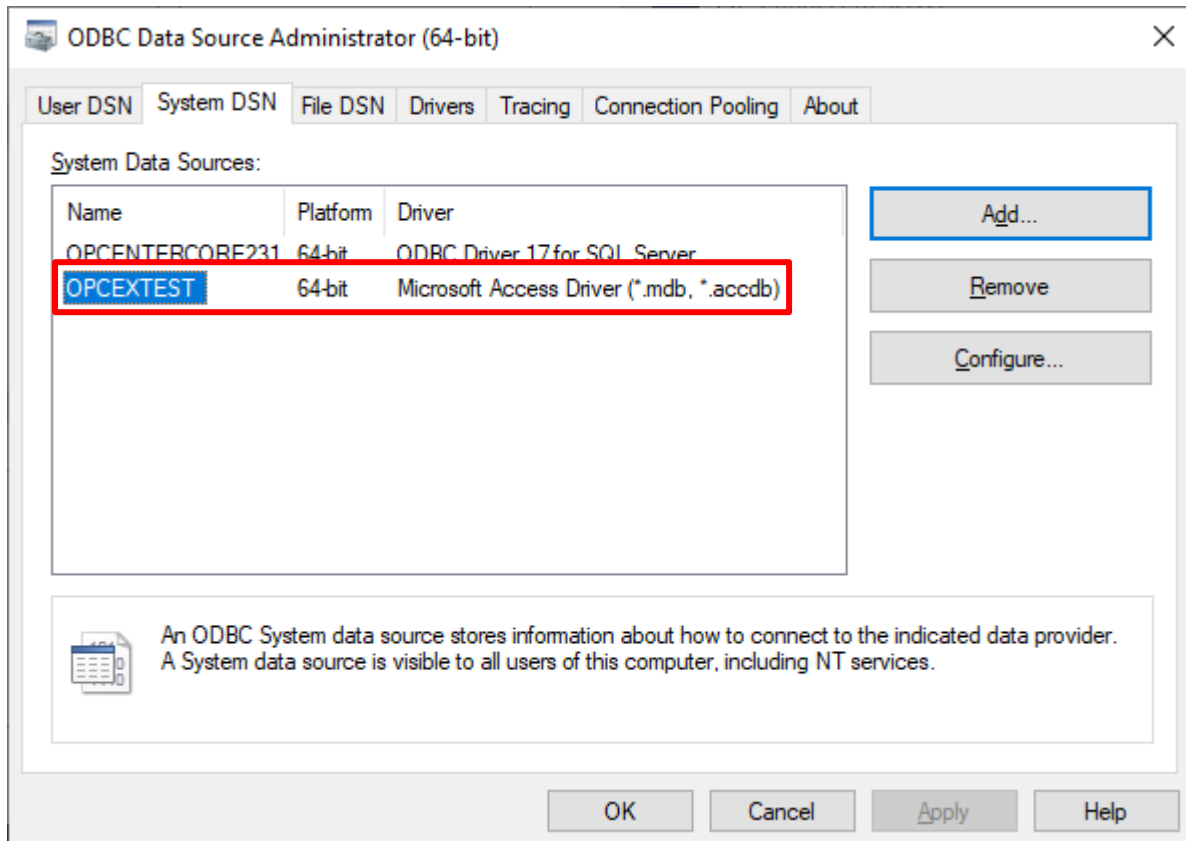
Browse and select the Microsoft Access supported extension (*.mdb, *.accdb) and click “**OK**”. (In this example, **InSite.mdb** is the file.)



The Database will display the directory path of the database file. Then, click “**OK**”.

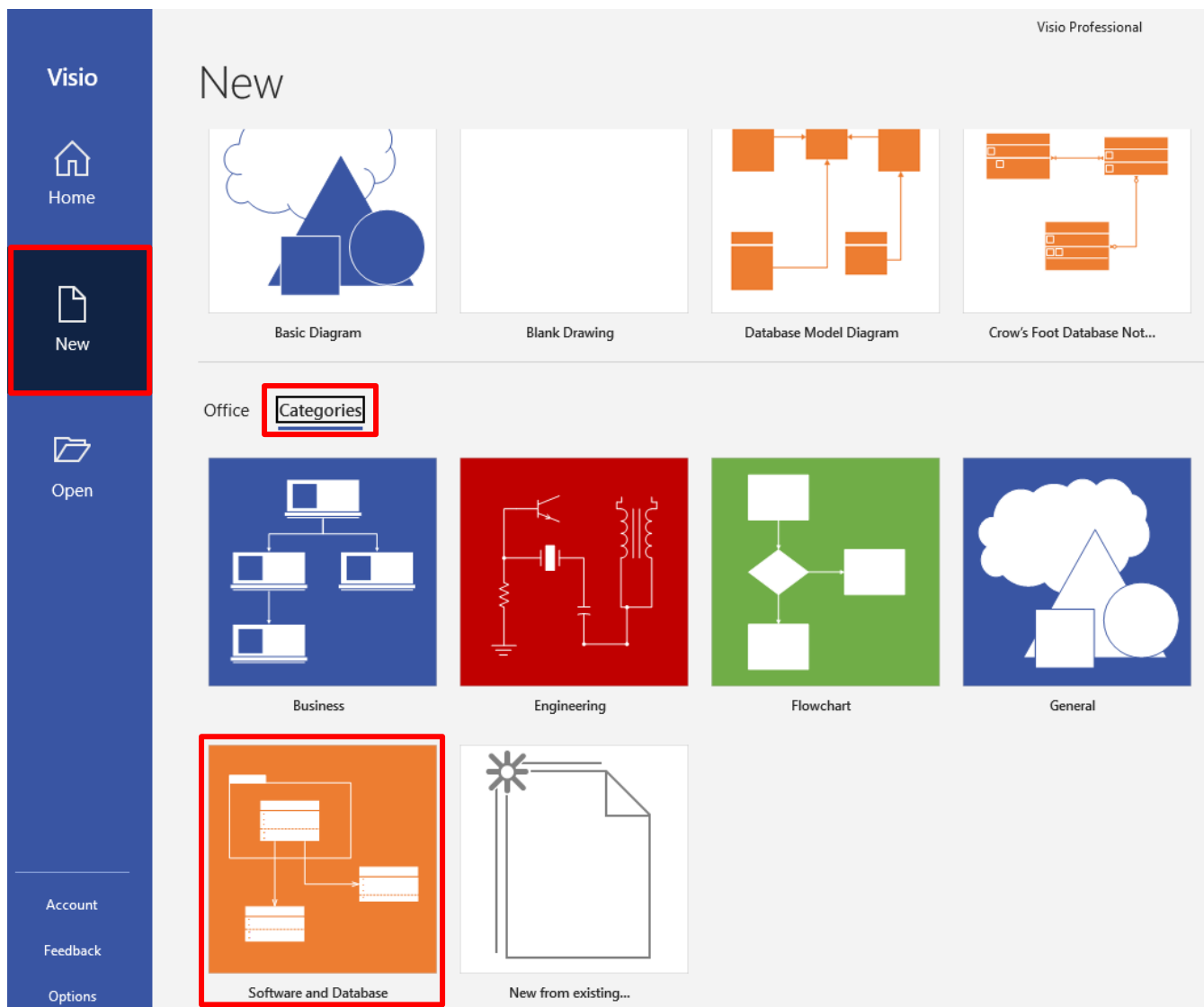


A new System Data Source will be created and displayed as shown in the following windows.

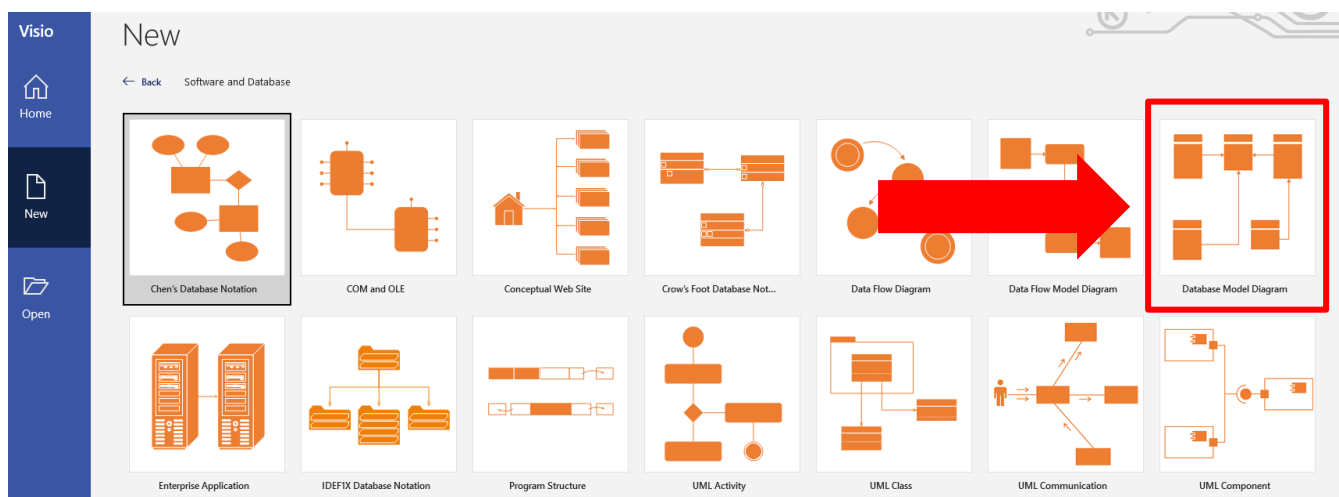


4.3.2.2 Reverse Engineering Setup

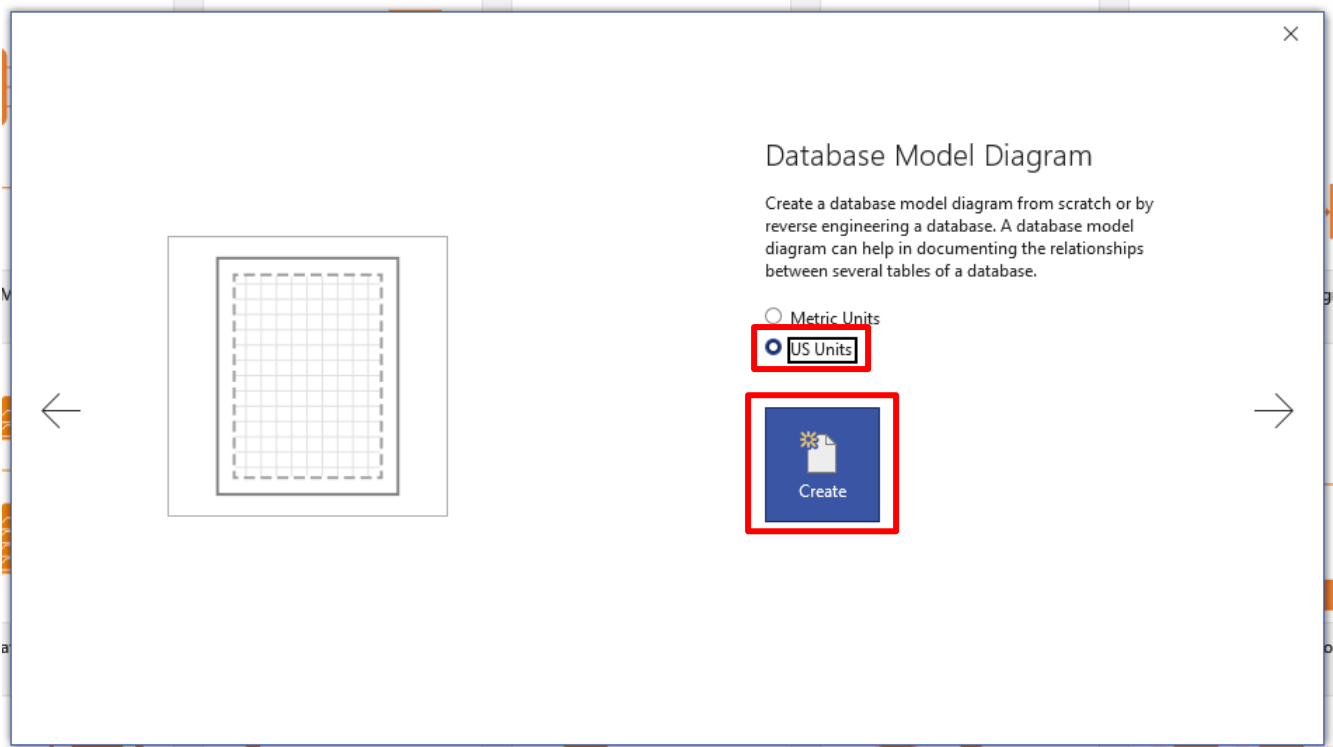
Open Microsoft Visio and select “**New**”, then select “**Categories**” and select “**Software and Database**”.



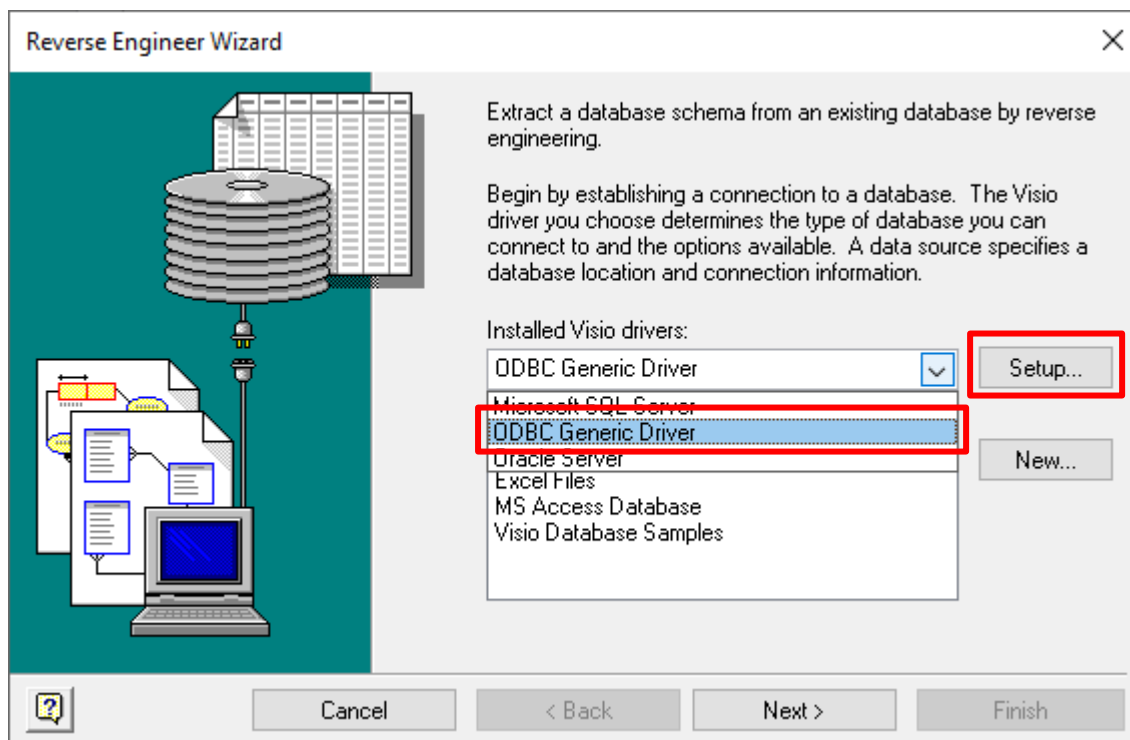
There are many Software and Database diagrams available in Microsoft Visio. To generate an Entity Relationship Diagram (ERD), select “Database Model Diagram”.



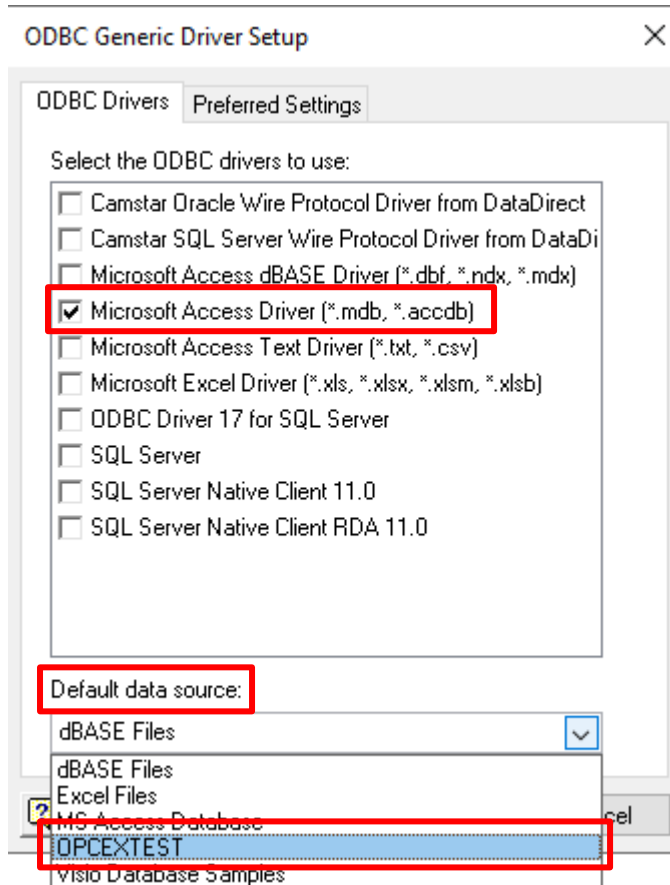
On the following pop-up window, select the “**US Units**” option and click “**Create**”.



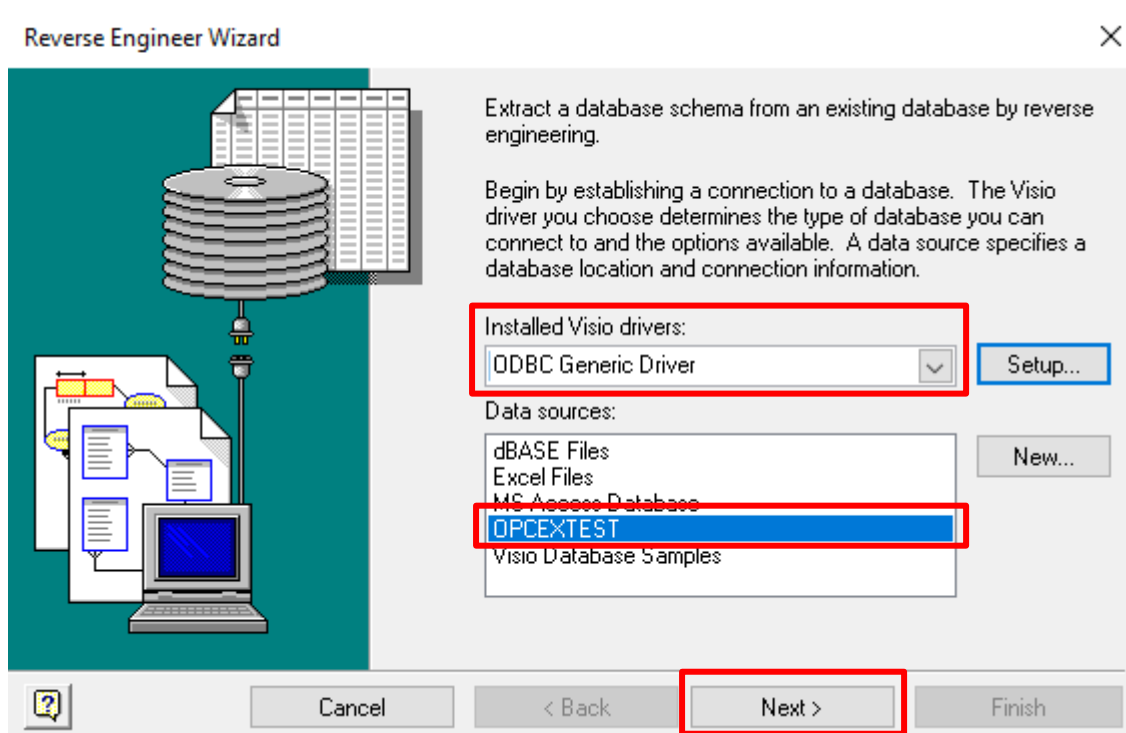
The **Reverse Engineer Wizard** window will appear. Select “**ODBC Generic Driver**” from the dropdown list and click “**Setup...**”



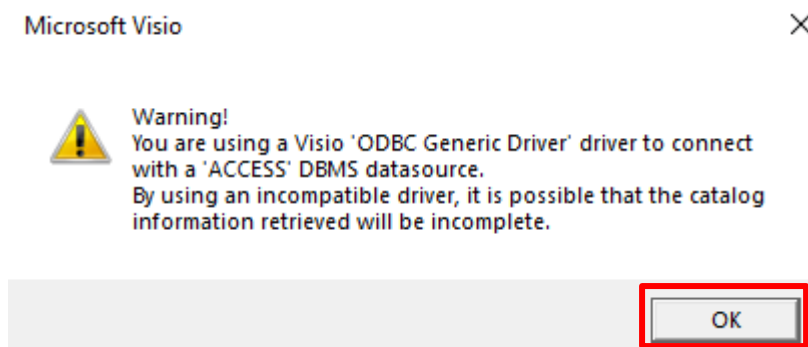
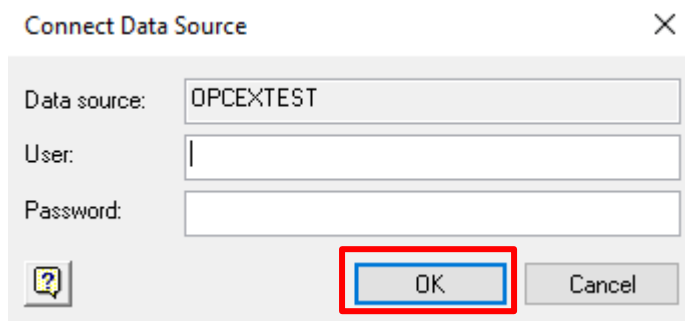
Uncheck all other drivers and only check “Microsoft Access Driver (*.mdb, *.accdb)”, and select the data source created on 3.3.2.1 (i.e. **OPCEXTEST**) as the **Default Data Source**. Then, click “OK”.



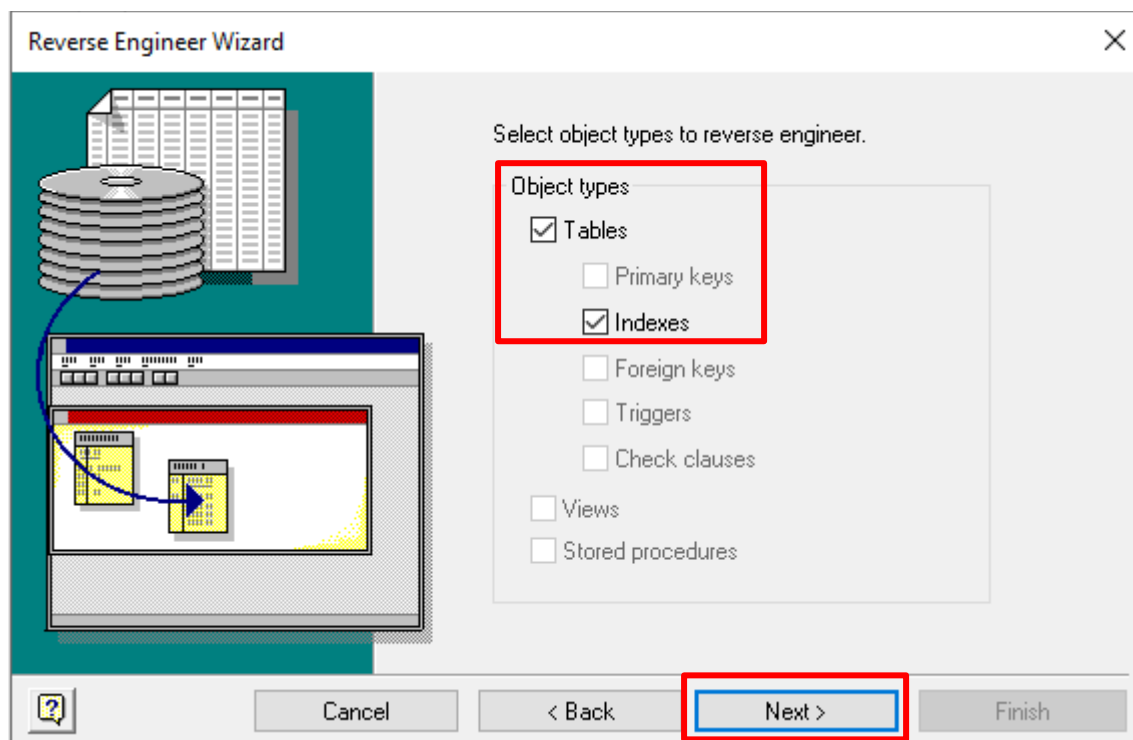
The Reverse Engineer Wizard will reflect the latest changes, then select “**Next >**”.



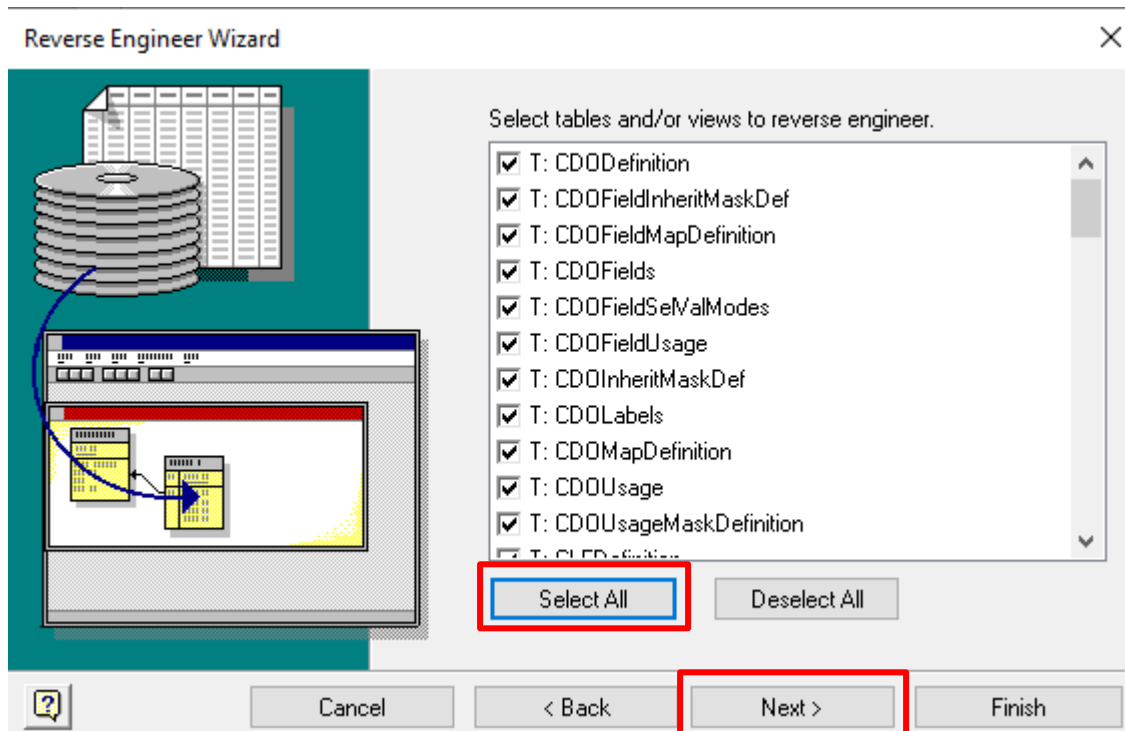
Click “OK” on the following pop-up windows.



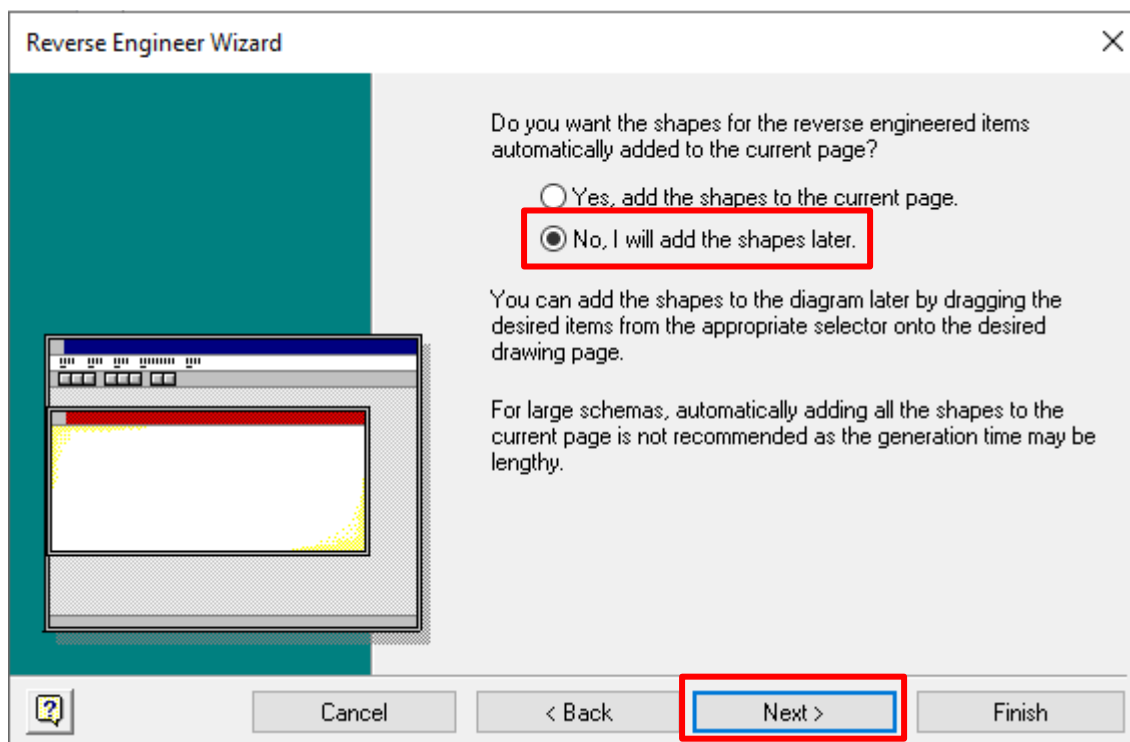
On Microsoft Visio Professional 2019, generating ERD through Microsoft Access only supports displaying the Tables and Indexes of the database. Then, click “Next >”.



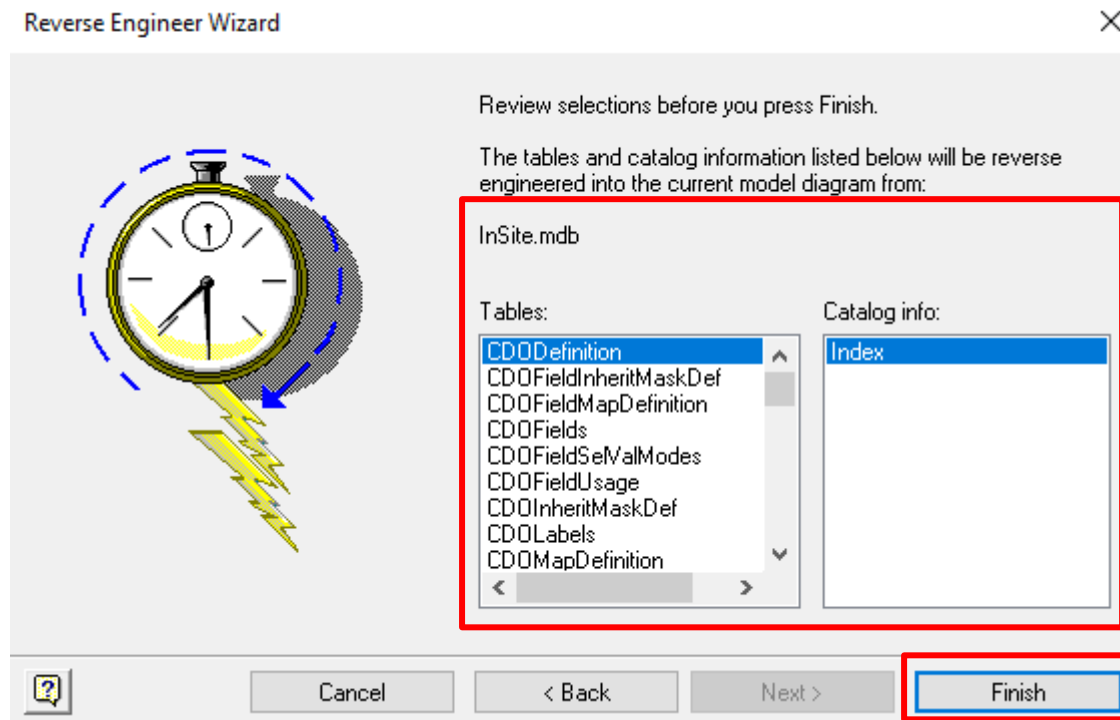
Click on **"Select All"** to check all the tables and views. Then click **"Next >"**.



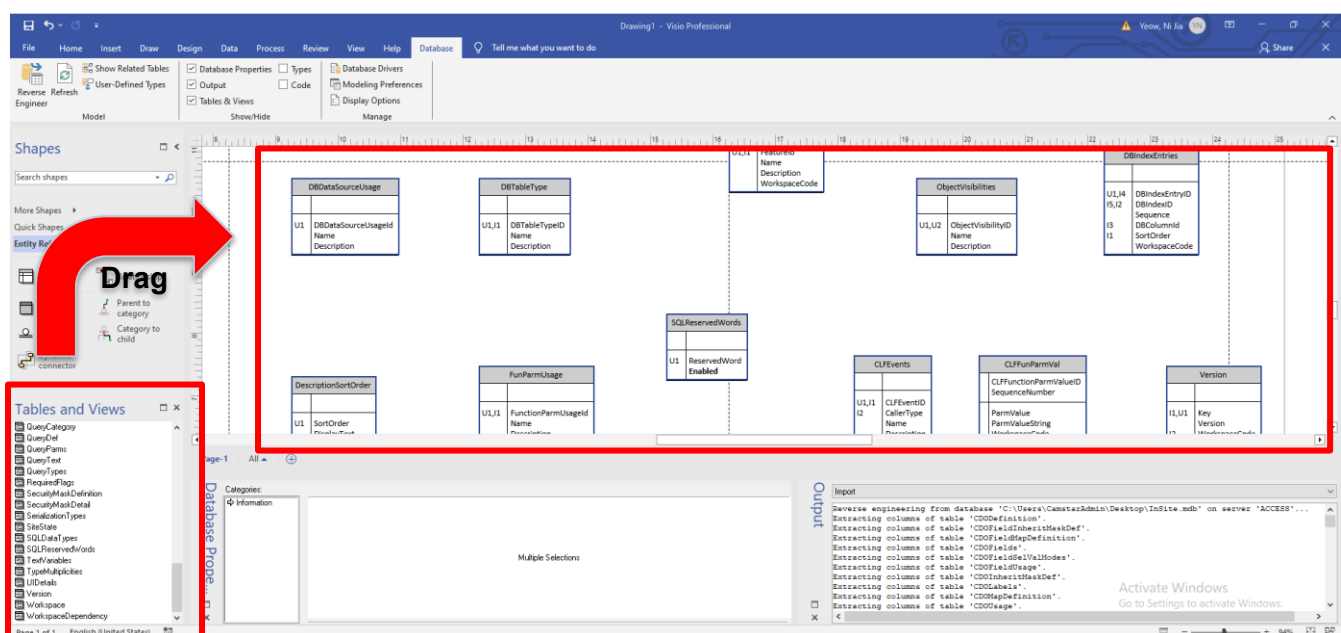
Select the option **"No, I will add the shapes later"** to display it manually, else select **"Yes, add the shapes to the current page."** to generate the complete ERD once the wizard is completed, then click **"Next >"**.



Lastly, the Wizard will show the summary of the information extracted from the database to generate the ERD. Then, click on **“Finish”**.

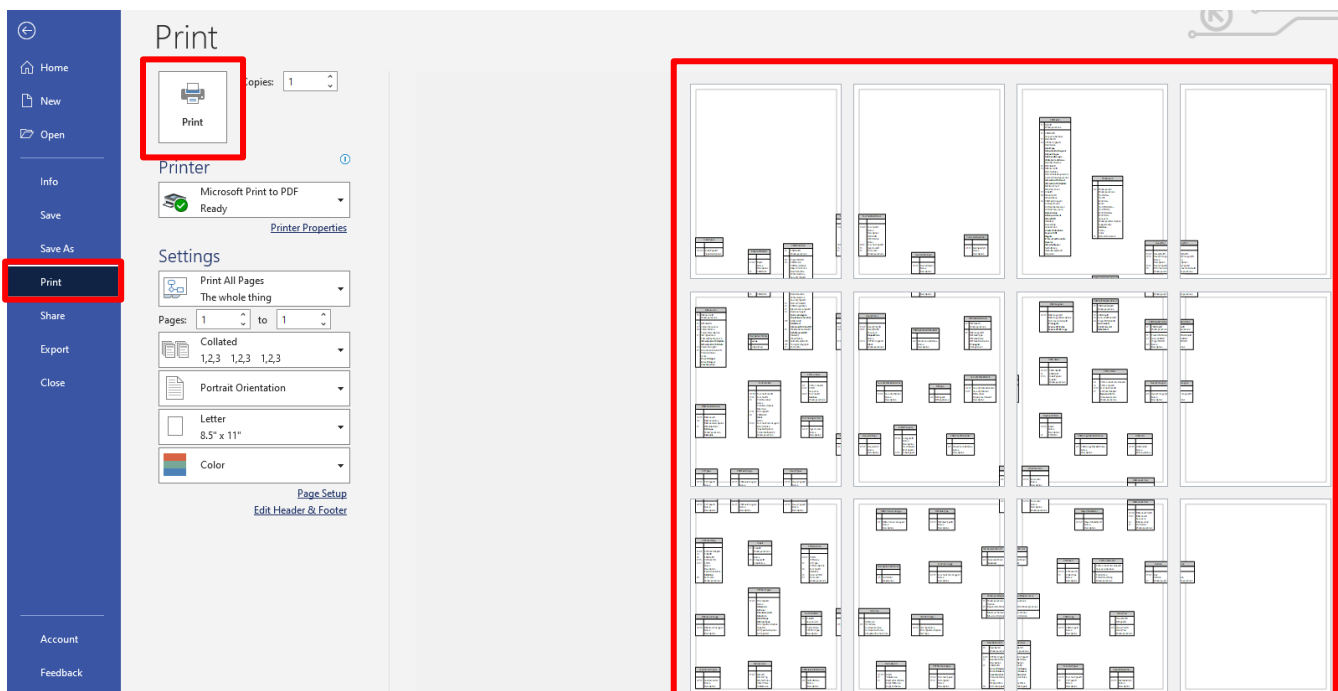
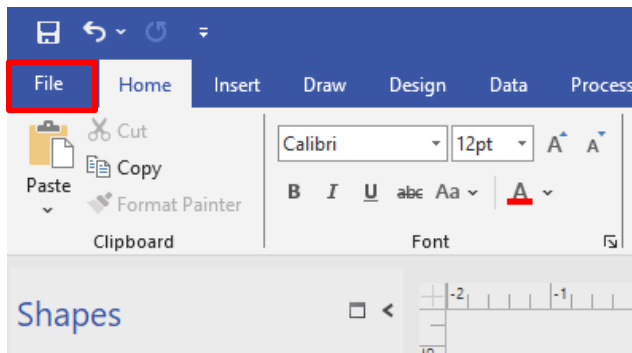


To view the ERD of the table, simply **drag** the table from **“Tables and Views”** to the space in the middle.

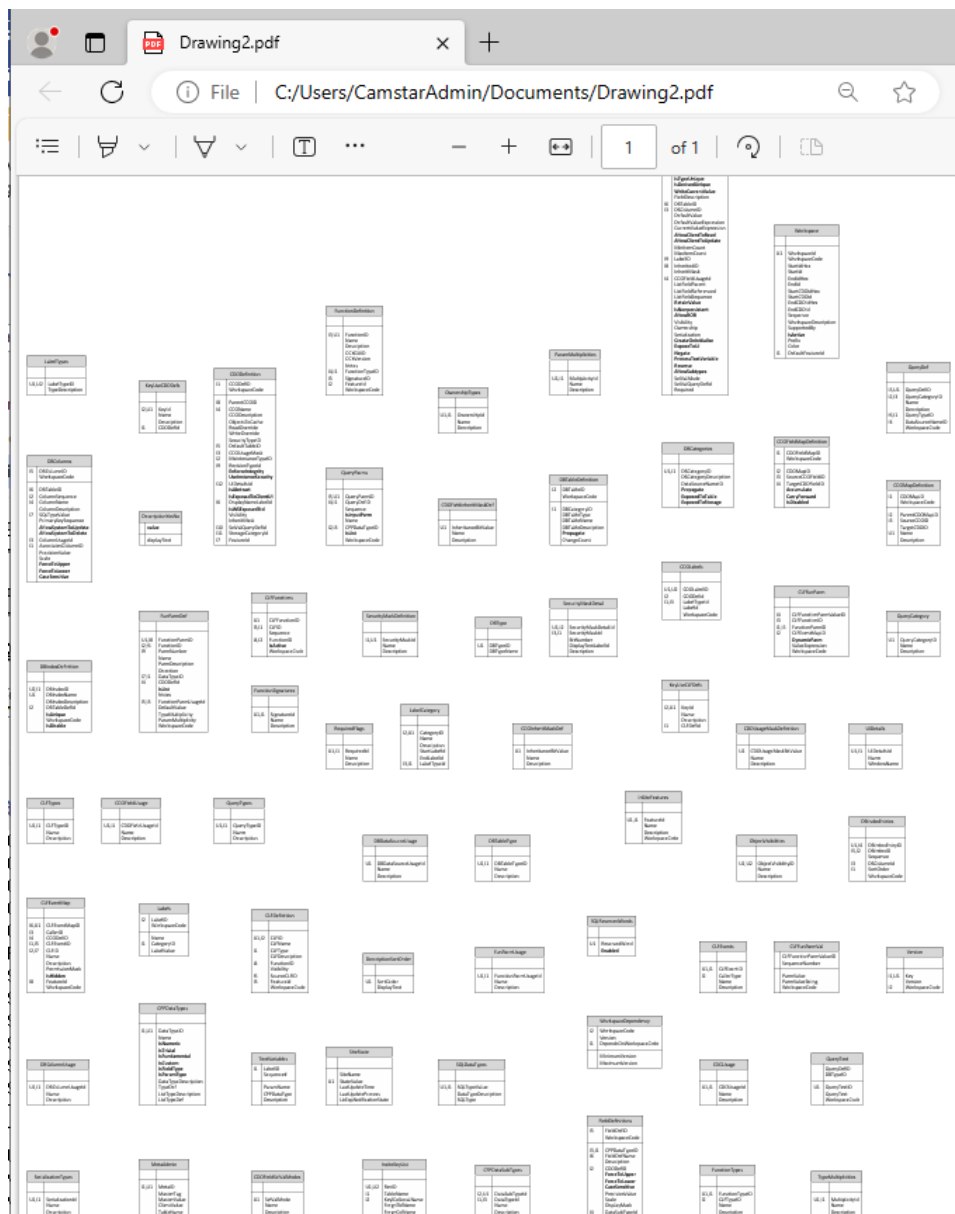
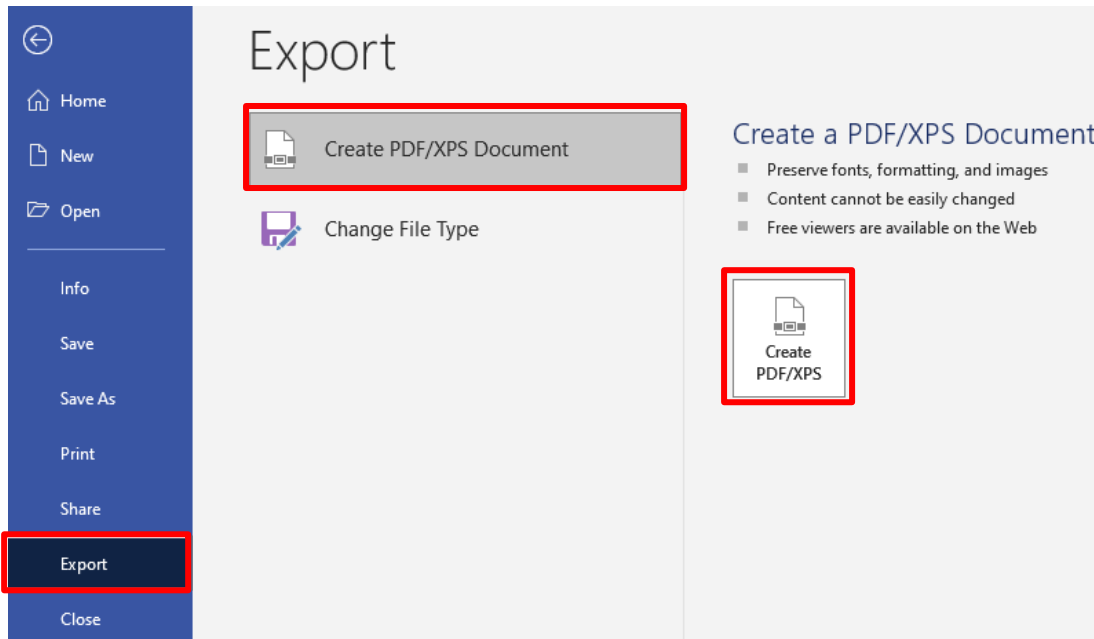


4.3.2.3 Export

To print the Entity Relationship Diagram (ERD), click on “**File**” on the menu, then select “**Print**” for the print preview and printing.

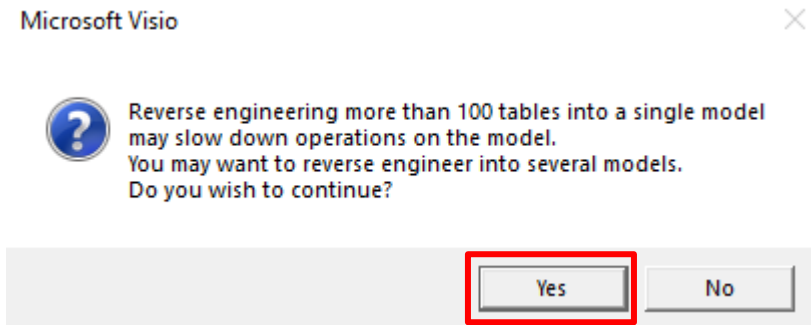


To export the Entity Relationship Diagram (ERD) to PDF, click on **Export**, then select **Create PDF/XPS Document** and select **Create PDF/XPS** to save the PDF document.

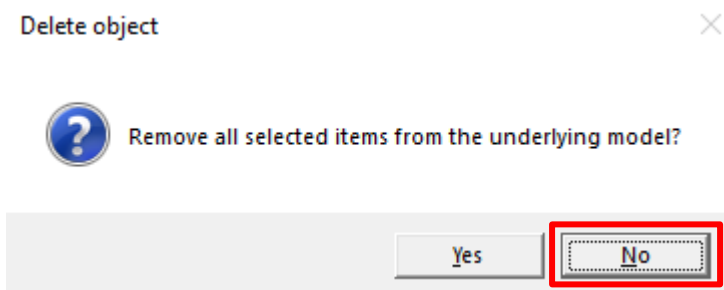


4.4 Disclaimers

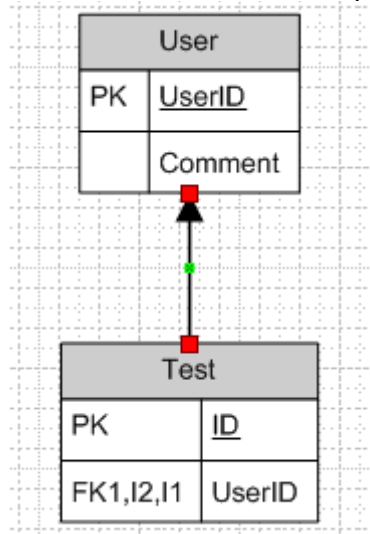
- The Reverse Engineering Wizard will prompt a warning that if the targeted database has more than 100 tables, it might take more time to generate the Entity Relationship Diagram (ERD) model.



- If the user want to remove/delete the table from the ERD, it would generate the following pop-up window. **Select “No”**, else it would be removed from the database model and user would need to reload the database again to retrieve the entity back.



- The Entity Relationship Diagram (ERD) generated from Visio through Microsoft Access could not generate the Primary Keys (PK), Foreign Keys (FK) and Entities Relationships due to unsupported driver on Microsoft Visio 2019. (refer to this [link](#) for more info.)
- If wish to generate ERD with its entities and relationships through Microsoft Access, use Microsoft Visio 2007. Sample of the ERD is shown below.



- Changing the entity name/columns in Microsoft Visio will not reflect changes** to the SQL Server and Microsoft Access database.

5 Generate ERD by Oracle

5.1 Introduction

Oracle SQL Developer is an integrated development environment (IDE) provided by Oracle Corporation for working with Oracle databases. It is a powerful tool designed to enhance productivity for developers, database administrators (DBAs) and data analysts.

In this user guide, we would demonstrate how to generate Entity Relationship Diagram (ERD) on Oracle Database through Oracle SQL Developer.

5.2 Prerequisites

- Oracle Database 19c
- Oracle SQL Developer 21.4.1

5.3 Demonstration

5.3.1 Oracle Database Schema Setup

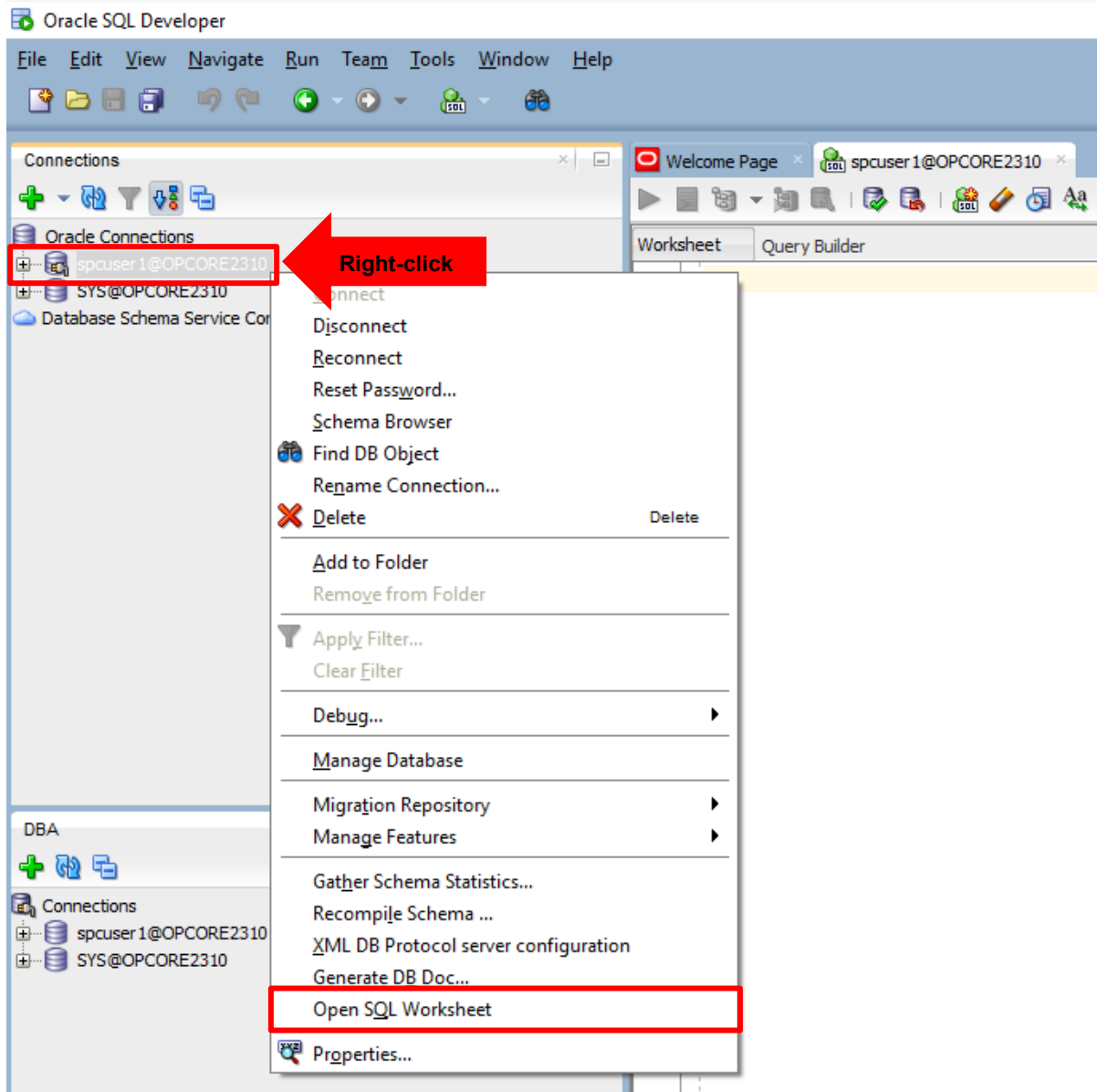
This is an **optional** step to create a target schema, if it doesn't already exist.

Connect and authenticate as SYSDBA in Oracle SQL Developer, **right-click** the database connection and select **"Open SQL Worksheet"** to create a new SQL script.

Copy the **"CreateDB.or.sql"** script from the path "C:\Program Files (x86)\Camstar\InSite Administration\OpMetaLink\scripts\Oracle" into the **"SQL Worksheet"**, then execute it to create the database with the relationship association named <Oracle Username>. Schema name and username will be the same.

5.3.2 Generate ERD

Launch Oracle SQL Developer, connect to the <Oracle Username>. database connection, **right-click** the database connection and select **"Open SQL Worksheet"** to create a new SQL script.



Copy the Opcenter Meta Link-generated Oracle output SQL script from the output folder, then click the 'Run Script' icon to execute and build the table.

Sample Opcenter Meta Link-generated Oracle output file format:

- Oracle_DDL_Cdo_YYYY-MM-DD-HH-MM-SS
- Oracle_DDL_Database_YYYY-MM-DD-HH-MM-SS

Welcome Page x spcuser1@OPCORE2310 x

Click to generate database 85.78700256 seconds

Worksheet Query Builder

```

ALTER TABLE ExecuteChecklistHistoryDtl
ADD (CONSTRAINT FK_ExecuteChecklistHistoryDtl_ChecklistEntry_ChecklistEntryId FOREIGN KEY (ChecklistEntryId) REFERENCES ChecklistEntry (ChecklistEntryId))
CONSTRAINT FK_ExecuteChecklistHistoryDtl_ResponseSet_ResponseSetId FOREIGN KEY (ResponseSetId) REFERENCES ResponseSet (ResponseSetId)
CONSTRAINT FK_ExecuteChecklistHistoryDtl_ExecuteChecklistHistory_ExecuteChecklistHistoryId FOREIGN KEY (ExecuteChecklistHistoryId) REFERENCES ExecuteChecklistHistory (ExecuteChecklistHistoryId)
ALTER TABLE ExecuteChklistResponseHistDtl
ADD (CONSTRAINT FK_ExecuteChklistResponseHistDtl_ExecuteChecklistHistoryDtl_ExecuteChecklistHistoryDtlId FOREIGN KEY (ExecuteChecklistHistoryDtlId) REFERENCES ExecuteChecklistHistoryDtl (ExecuteChecklistHistoryDtlId))
ALTER TABLE ExecuteChecklistHistory
ADD (CONSTRAINT FK_ExecuteChecklistHistory_Checklist_ChecklistId FOREIGN KEY (ChecklistId) REFERENCES Checklist (ChecklistId))
CONSTRAINT FK_ExecuteChecklistHistory_HistoryMainline_HistoryMainlineId FOREIGN KEY (HistoryMainlineId) REFERENCES HistoryMainline (HistoryMainlineId))
ALTER TABLE DataPointCollectionGroup
ADD (CONSTRAINT FK_DataPointCollectionGroup_DataPointCollection_DataPointCollectionId FOREIGN KEY (DataPointCollectionId) REFERENCES DataPointCollection (DataPointCollectionId))
CONSTRAINT FK_DataPointCollectionGroup_Employee_LastEnteredById FOREIGN KEY (LastEnteredById) REFERENCES Employee (EmployeeId))
CONSTRAINT FK_DataPointCollectionGroup_RoleDef_LastEnteredByRoleId FOREIGN KEY (LastEnteredByRoleId) REFERENCES RoleDefinition (RoleDefinitionId))
ALTER TABLE ChecklistEntry
ADD (CONSTRAINT FK_ChecklistEntry_ResponseSet_ResponseSetId FOREIGN KEY (ResponseSetId) REFERENCES ResponseSet (ResponseSetId))
CONSTRAINT FK_ChecklistEntry_Employee_LastCompletedById FOREIGN KEY (LastCompletedById) REFERENCES Employee (EmployeeId))
CONSTRAINT FK_ChecklistEntry_RoleDef_LastCompletedByRoleId FOREIGN KEY (LastCompletedByRoleId) REFERENCES RoleDefinition (RoleDefinitionId))
ALTER TABLE ChecklistTemplate
ADD (CONSTRAINT FK_ChecklistTemplate_ChecklistTemplateBase_ChecklistTemplateBaseId FOREIGN KEY (ChecklistTemplateBaseId) REFERENCES ChecklistTemplateBase (ChecklistTemplateBaseId))

```

Script Output x

Task completed in 85.787 seconds

Table CHECKLISTTEMPLATE altered.

Table CHECKLISTTEMPLATEBASE altered.

Table RESPONSESET altered.

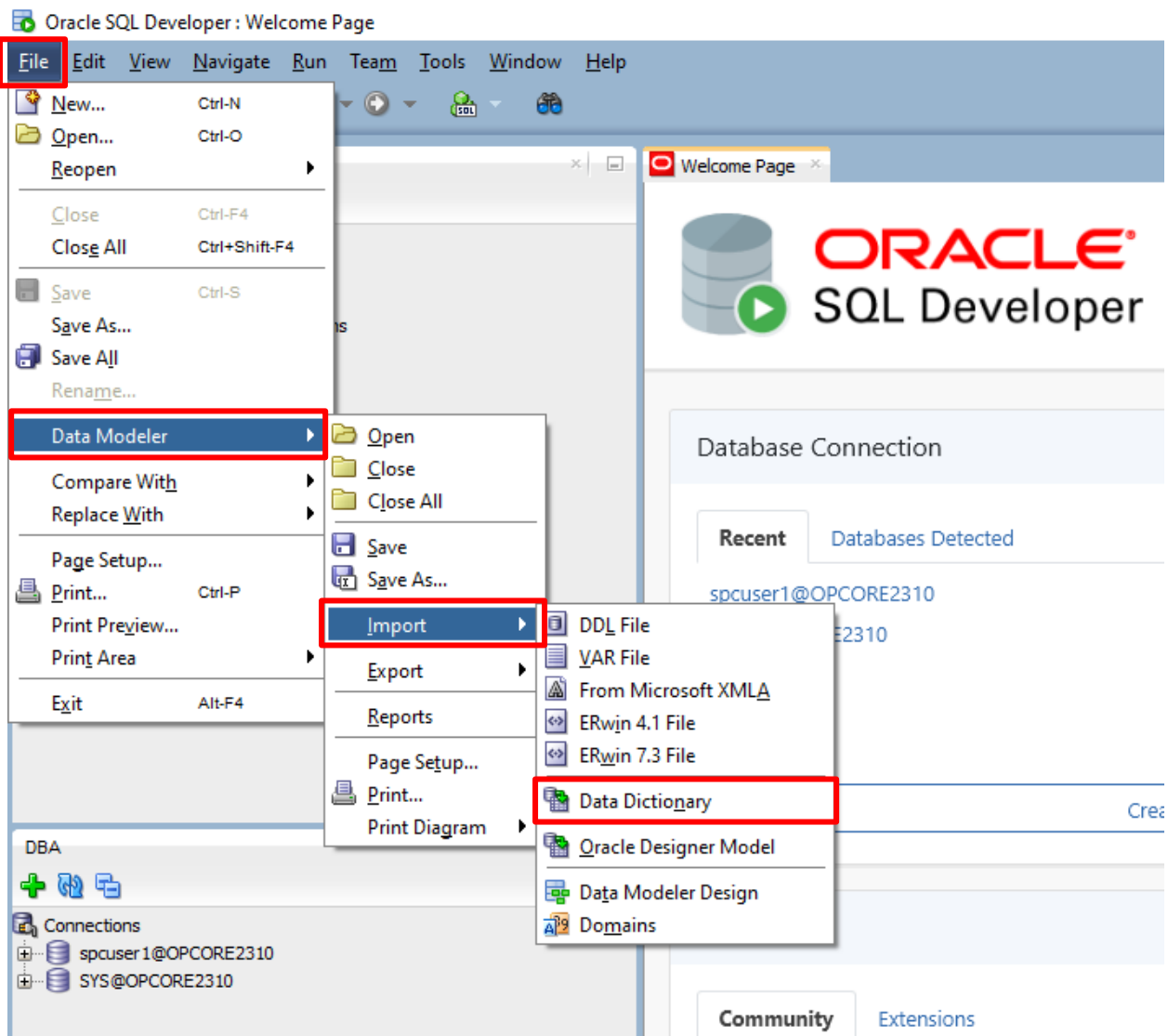
Table RESPONSEITEM altered.

Table HISTORYMAINLINEADDITIONALESIGS altered.

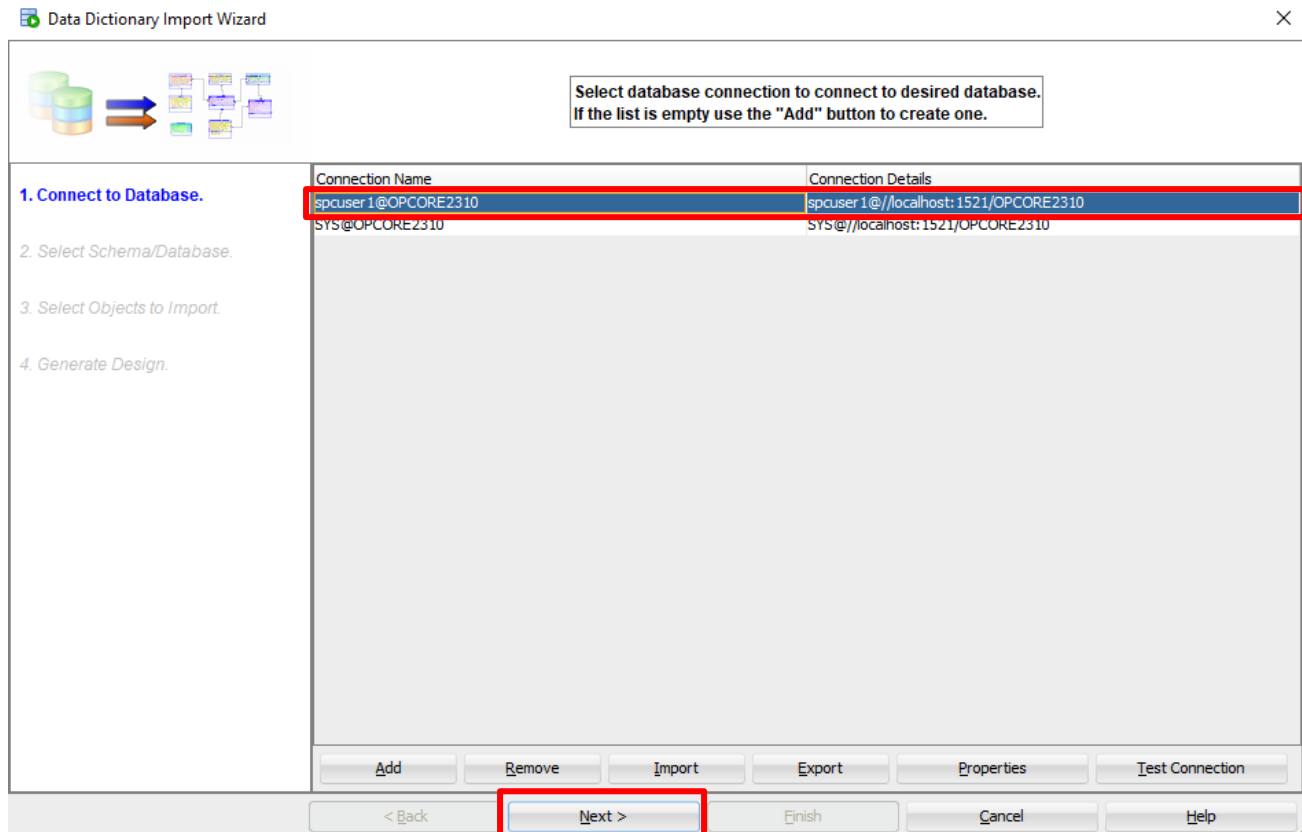
Table COLLECTDATAPOINTSHISTORY altered.

5.3.3 Data Dictionary Import Wizard

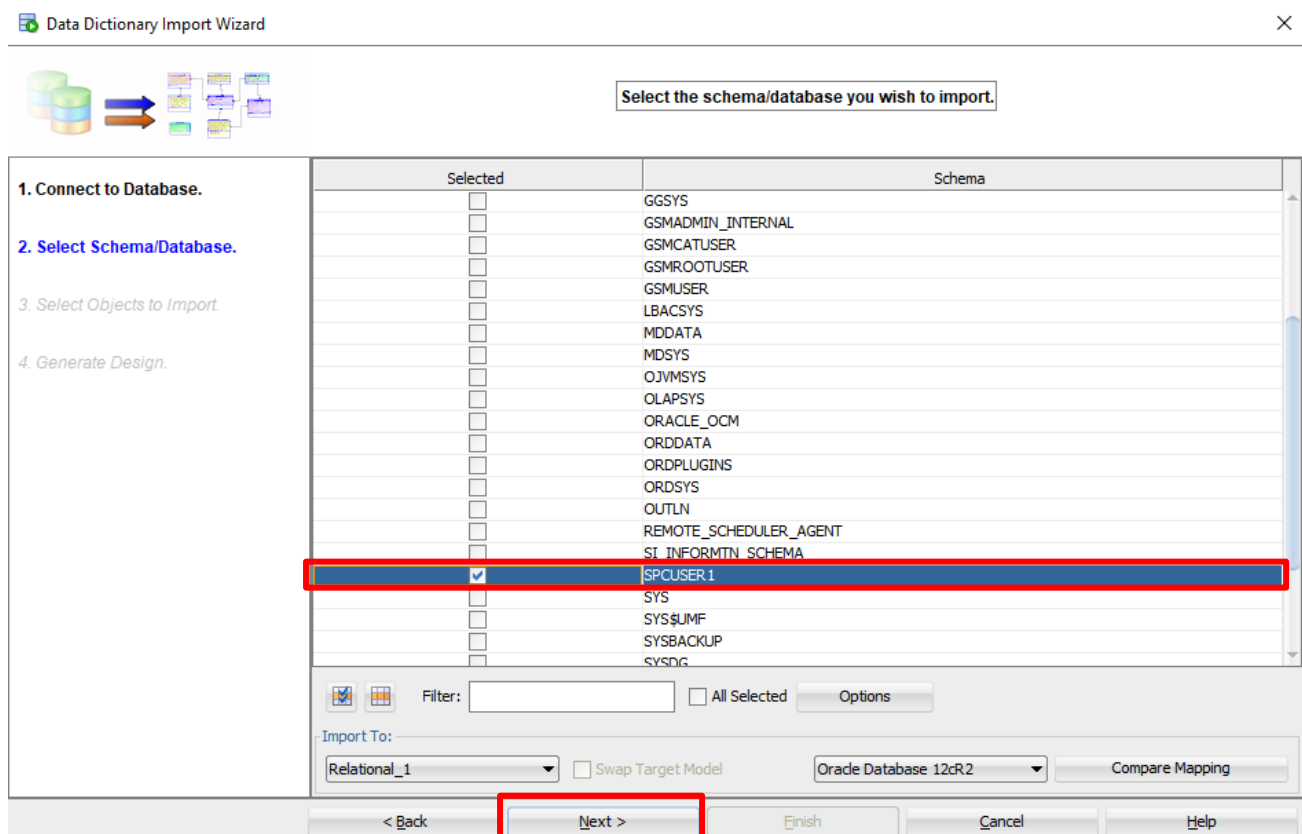
Open Oracle SQL Developer, then **select “File” → “Data Modeler” → “Import” → “Data Dictionary”**.



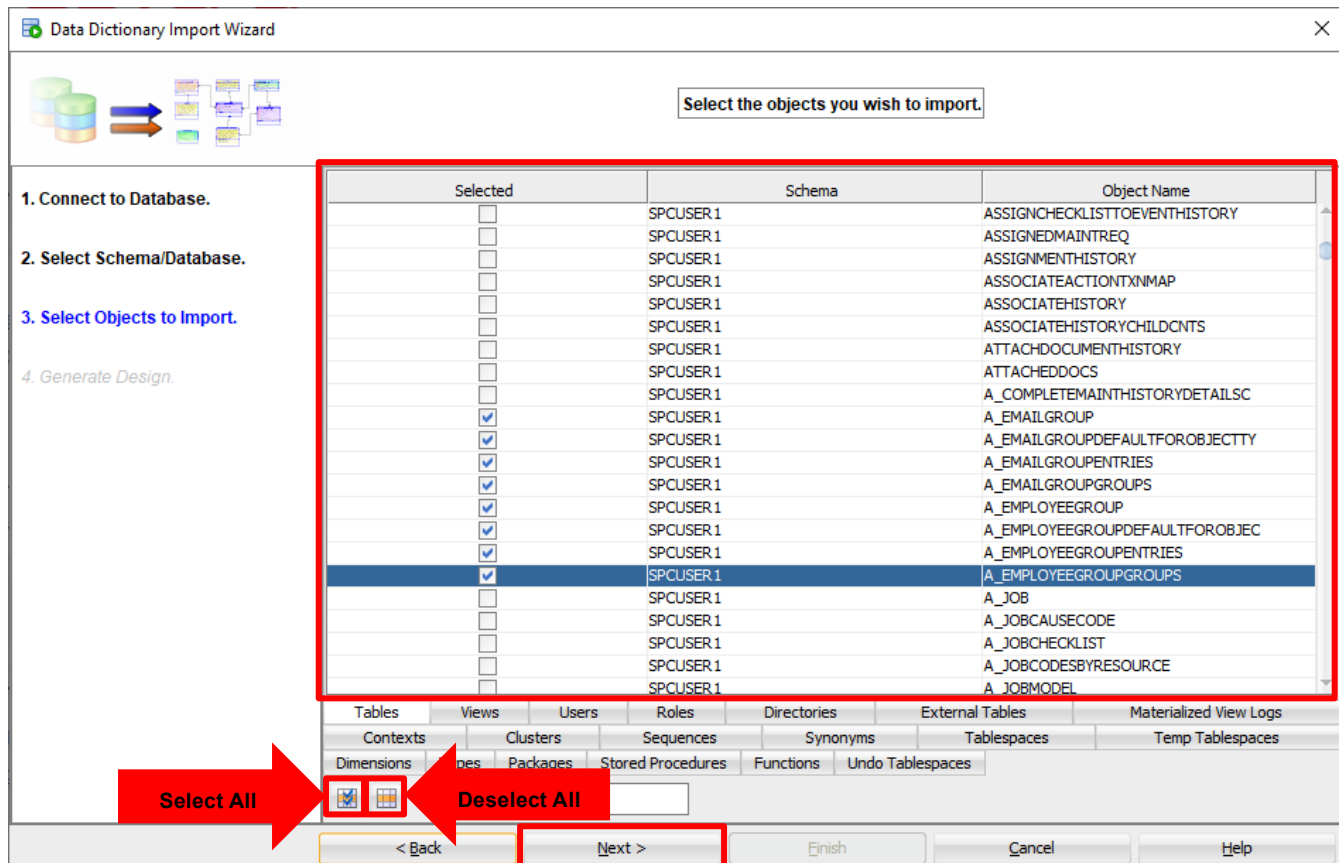
Select the connection to the Oracle Database, then select “**Next >**”. (In this example, **spcuser1@OPCORE2310** is the connection.)



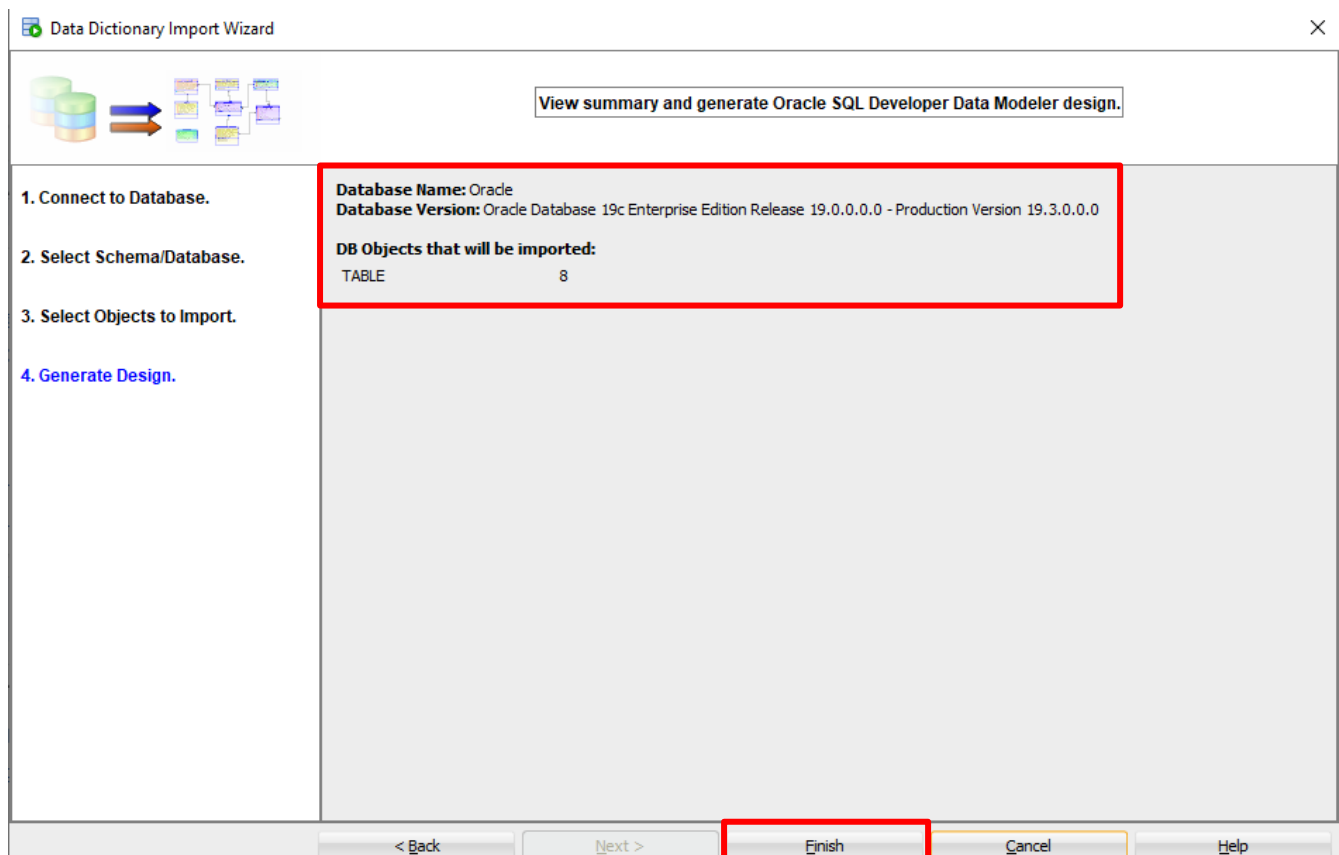
Select the Schema/Database to generate the ERD, then select “**Next >**”. (In this example, **SPCUSER1** is the schema/database.)



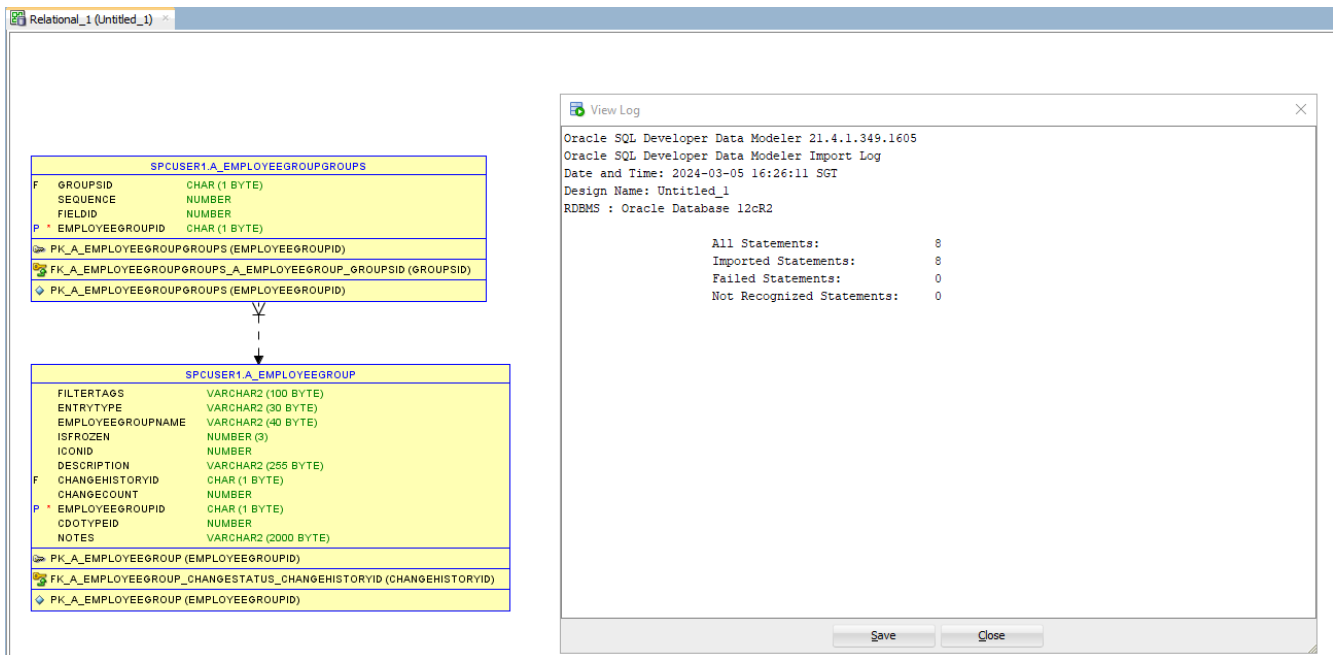
Select the objects/entities of the database to generate the ERD, then select “**Next >**”.



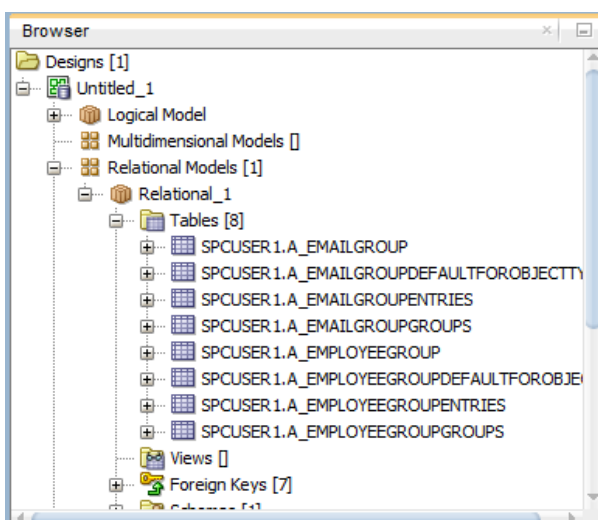
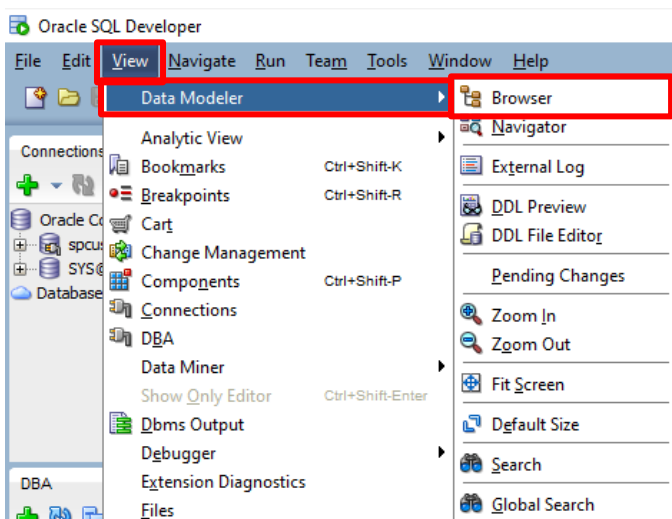
The Wizard will display the number of objects/entities to generate the ERD. Then, select “**Finish**” to begin generating the ERD.



After the process is complete, the Oracle SQL Developer will generate the **ERD** and **View Log** as follows.

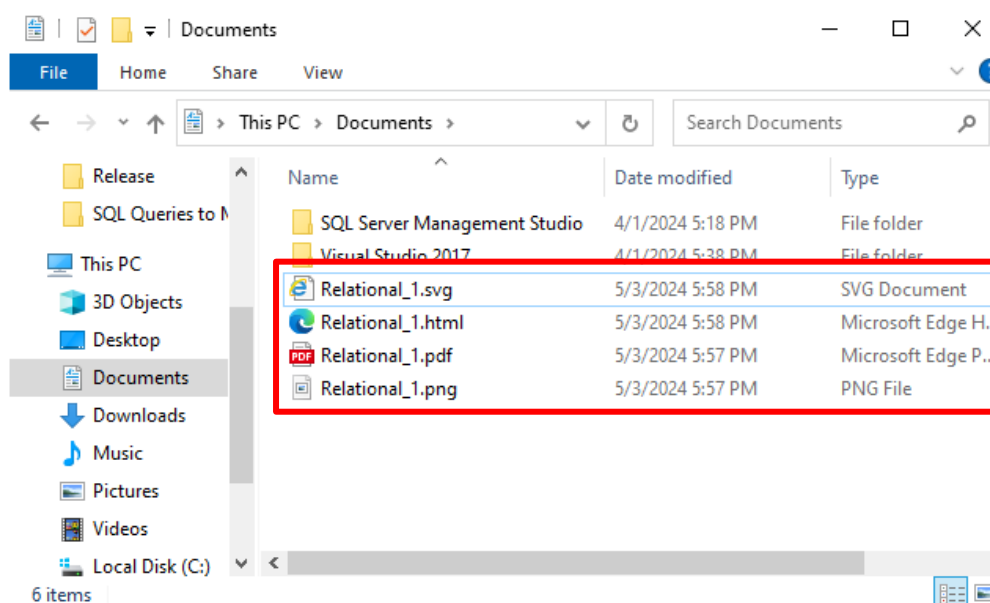
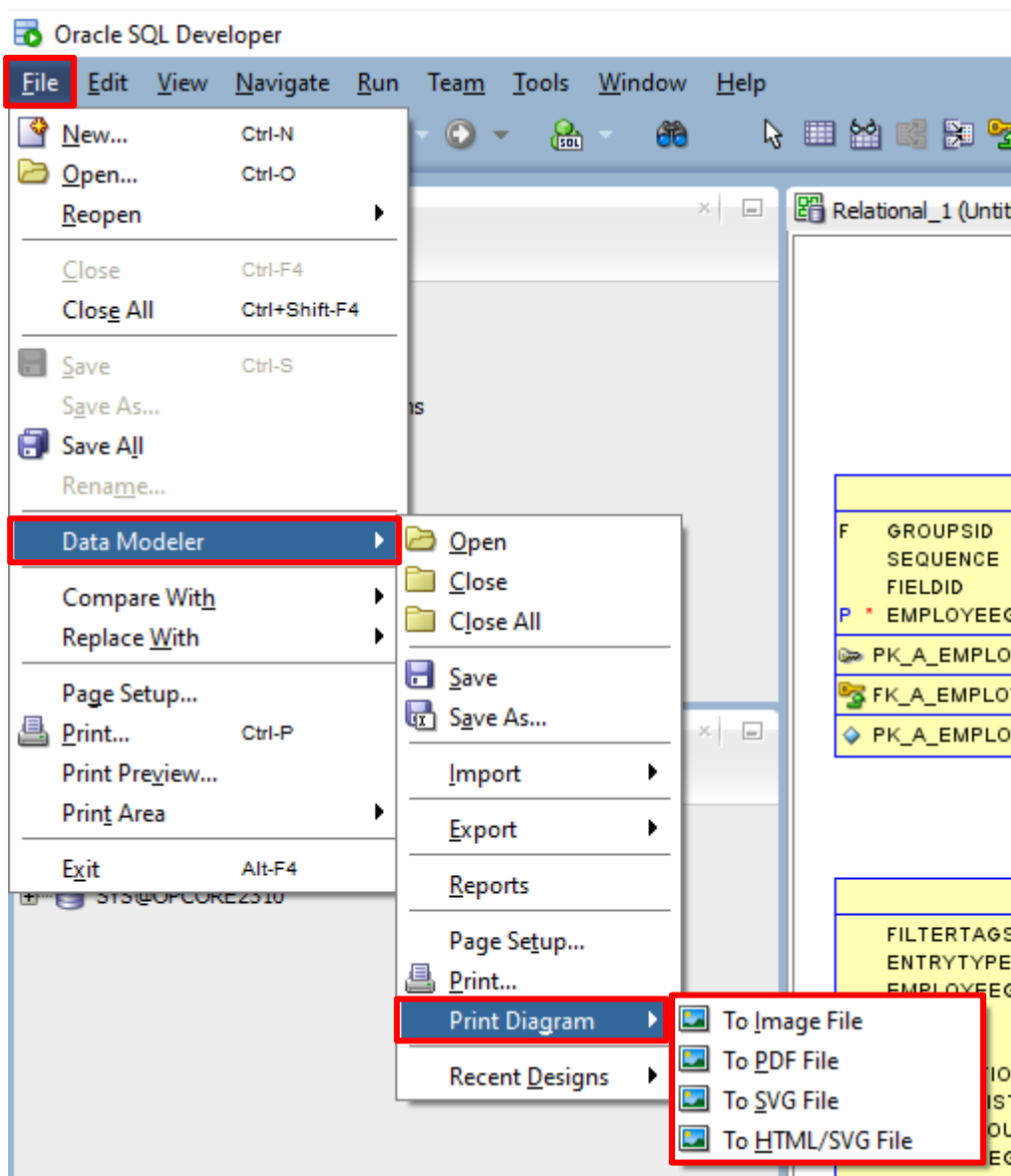


To view the menu for the Relational Model Diagram, select “**View**” → “**Data Modeler**” → “**Browser**”. Then, the **Browser** menu will show up along with the Connections and DBA menus as follow.



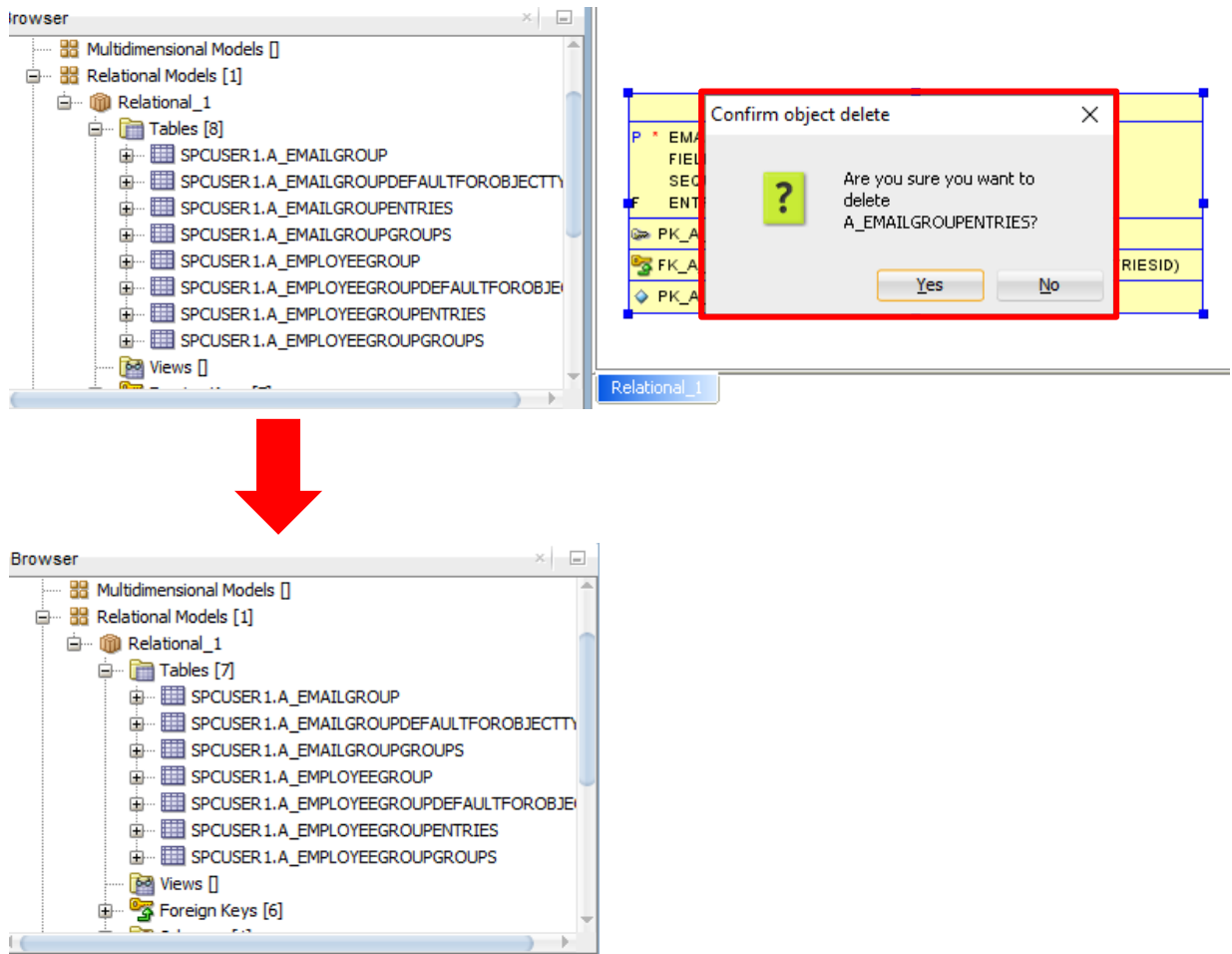
5.3.4 Export

To print the ERD, select **File** → **Data Modeler** → **Print Diagram** → **To Image/PDF/SVG/HTML File**.



5.4 Disclaimers

- Deleting the entity ERD would result in deleting it from Relational Models. If you want to display again the entity's ERD, you need to re-import from the **Data Modeler**.



- If import >100 tables from the schema, it will impact the performance of the Oracle SQL Developer.

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Usage Disclaim

This tool and document are the results of a collaboration between Technical Enablement and Global Practice teams.

Excluding this manual and the documentation available on the **CoP Solution & Product Consulting Community**, No other support is provided.

Anyway, the objective is to continuously improve the existing practice and create new ones.

If you have a suggestion for improvement or new practice please contact the MoM Global Practice team.

All the 3rd party tools suggested are not supported by Siemens and should not be installed on the Customer

